

A Geometric and Category-Theoretic Theory of Viability: How Sequential Adaptations Induce Path-Dependent Risk

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Abstract. Adaptive systems survive fluctuating environments by adjusting internal strategies. Sequences of individually harmless adjustments can push a system into failure when environmental shifts occur in a non-commuting order. We develop the geometric and category-theoretic framework defining and quantifying this phenomenon: *viability-weighted curvature*, the viability loss a closed loop accumulates through policy holonomy. The theory has four pillars. (i) The *SAVGS architecture* unifies control base, Fisher–Rao policy bundle, viability margin, maintenance graph, and 2-categorical boundary span into one stratified bundle. (ii) The 2-category $\mathbf{StCon}(B)$ of stratified connections carries a lax-functorial gluing theorem and a piecewise-holonomy formula, with boundary resets at their true order for transversal wall-crossing loops in *pairs*. Each stratum carries the Fisher-minimal transport law (KKT projection) as connection, and the small-loop theorem bounds endpoint erosion by the viability-weighted curvature. (iii) A single composition theorem types seven bridges as optics, with per-optic Lipschitz constants and a Banach contraction of the Krasnoselskii–Mann-averaged update; the projected CPTP contraction settles the Zeno self-reference. (iv) The filtered-colimit construction of RAF sets is proved at Set level, verified at scale; the ∞ -categorical extension in homotopy type theory carries marked proof-sketch status. The validation battery is robust across six axes: carbon, oxygen, nitrogen, phosphate, and iron supply plus non-medium maintenance stress leave labels invariant, re-stratifying only at regime switches and nitrogen-source substitution. The association is invariant under canonical flux selection, the declared tie-break closing its near-degeneracy boundary. The application paper develops the atomic curvature measure and its genome-scale validation.

Keywords: applied category theory; optic category; stratified connection; homotopy type theory; holonomy; viability theory

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1 Introduction

The problem. Living and engineered adaptive systems must continuously adjust their internal controls to remain functional under shifting external conditions. In classical viability theory, survival is treated primarily as a set-membership problem: a system is viable as long as its state stays within a predefined safe region [6]. This static, set-theoretic perspective obscures a subtle and ubiquitous failure mode: *path-dependent vulnerability*. An organism or machine can successfully survive a change in temperature, and it can separately survive a change in nutrient availability. Yet, if these two environmental changes are applied sequentially in a closed cycle — first shifting temperature, then nutrients, then reversing temperature, and finally reversing nutrients — the system may return to its original environment with its internal reserves severely degraded. The order matters, not only the magnitudes: this failure occurs because optimal control strategies frequently do not *commute* — applying adaptation A followed by adaptation B produces a different internal state than applying B followed by A . In differential geometry, the net discrepancy accumulated around a closed loop is called *holonomy*, and its intensity per unit area is called *curvature*. For orientation: a *policy* is the optimizer’s chosen response at each state; a *connection* is the rule for comparing policies at nearby states; and a policy loop’s *holonomy* is the drift the closed cycle accumulates — the geometric signature of non-commutativity.

The obstruction. Why has this geometric insight not been systematically applied to adaptive systems? The difficulty lies in *active-set switches*. In any realistic system, adaptations are governed by active physical bottlenecks — maximum enzyme capacities, nutrient uptake limits, thermal boundaries. Inside a regime where the active bottlenecks do not change, the optimal strategy varies smoothly. But when an environmental shift pushes the system across a constraint boundary — releasing an old bottleneck and activating a new one — the system’s sensitivity changes abruptly. At these boundary interfaces, the connection 1-form describing strategic adjustments undergoes a discontinuous jump, horizontal subspaces cease to exist, classical smooth calculus and standard curvature integrals break down, and traditional holonomy cannot be evaluated across the interface — exactly where the question is most interesting. Classical viability theory supplies set-valued dynamics and tangential conditions, but no geometric object that measures how a non-commuting sequence of individually manageable changes accumulates into a viability threat.

The solution. This paper resolves the challenge by constructing a rigorous, stratified geometric framework for adaptive systems. Rather than treating constraint switches as troublesome discontinuities, we show that the boundaries between operational regimes carry their own well-defined, discrete geometric data. The missing object is constructed categorically: a stratified connection with a boundary gluing law, a viability-weighted curvature that contracts the curvature 2-form with survival covectors, and a compositional semantics (optics — a map paired with its feedback channel) under which heterogeneous domain bridges compose into one typed endomorphism whose fixed point is well posed and, for an explicit instantiation, contractive.

Contributions. Our core contributions are organized as follows:

1. **The SAVGS architecture** (Section 3): the *SAVGS object* (Definition 3.1) — control base, policy bundle with Fisher–Rao geometry, viability margin, maintenance graph, and a 2-categorical boundary span, assembled into one stratified G_C -reduced bundle.
2. **Stratified connections and boundary holonomy** (Section 3): the 2-category $\mathbf{StCon}(B)$ of stratified G_C -connections (Definition 3.3), the 2-categorical gluing theorem (Theorem 3.4), and the piecewise holonomy formula (Theorem 3.6), with the small-loop expansion stated for closed loops crossing a switching wall transversally in pairs and the boundary-reset contribution

at its true order, in agreement with the machine-verified numerics in five loop geometries and three structure-group regimes.

3. **Viability-weighted curvature and vulnerability bounds** (Section 3): the *Fisher-minimal transport law* on constant-active-set strata in closed KKT form (Proposition 3.14) and the *small-loop viability-holonomy theorem* (Theorem 3.16): endpoint erosion of a viability margin is bounded, to leading order, by the viability-weighted curvature (Definition 2.11).
4. **Conservation laws via information geometry** (Section 4): the Bregman–Hessian Noether correspondence (Proposition 4.1) with its falsifiable precondition check, giving the gauge-covariance statement for the curvature.
5. **Unified composition via optics** (Sections 6 and 7): the *single composition theorem* (Theorem 6.4) — seven domain bridges typed as optics, the realized update on a complete metric state space (Definition 7.2), per-optic analytic Lipschitz constants (Lemma 7.1), the Banach contraction of the Krasnoselskii–Mann-averaged update for the chosen seven-map instantiation (Theorem 7.5), and the projected CPTP channel contraction (Theorem 7.7) settling the Zeno self-reference resolution.
6. **Operational closure at scale** (Section 10): the filtered-colimit construction of RAF sets, proved at the Set level with the adapter-level optic statement (Theorem 10.3) and verified at scale (Proposition 10.8).
7. **The homotopy-theoretic extension** (Section 12): the ∞ -categorical extension via homotopy type theory (Theorem 12.2, Corollary 12.3), with proof-sketch status explicitly marked (Remark 12.4), and the three-phase autopoiesis closure test built on it (Definition 12.5).

Relation to the application paper. In a companion manuscript [1], this geometric framework is applied to genome-scale metabolic models of *Escherichia coli*: there the viability-weighted-curvature idea is realized measure-theoretically as an atomic curvature measure on the active-set complex of parametric flux balance analysis, a gene-level sensitivity metric κ^μ is derived from it, and the metric is validated at genome scale against carbon-depletion transcriptional response ($r = +0.395$, $n = 424$ genes), with an explicit protein-layer null ($r = -0.083$ on matched quantitative proteomics, $n = 366$) and selection-rule robustness checks. The application paper cites this paper for the constructions, machine verifications, and proofs at the status marked per result, recording the division of labor in one Discussion paragraph — complete proofs for the gluing formula, the Fisher-minimal transport law, the composition and contraction layers, and the terminal-coalgebra theorem; machine-verified statements for the numerics; citation-level reduction for the stratified descent; proof-sketch status, marked as such, for the homotopy-type-theoretic extension. The present paper develops the full categorical machinery and cites the application paper where the empirical line is concerned. The division of labor is deliberate: the empirical claims of the application paper are statistical, the categorical claims of this paper are mathematical, and neither paper depends on the other’s claims.

Status conventions. Theorem-level claims are labeled by proof status throughout: complete proofs; proofs that reduce to cited descent or coherence theory (marked in remarks); proof sketches (marked); and machine-verified statements, whose scripts and artifacts are committed in the project repository and regenerated by them. Open problems are collected in Section 15.

Plan of the paper. Section 2 states the preliminaries one definition at a time, none used before it is stated. Sections 3–4 develop the core geometric theory: the SAVGS architecture, the 2-category $\mathbf{StCon}(B)$ with its gluing theorem and piecewise-holonomy formula, the Fisher-minimal transport law, the viability-weighted curvature, and the Bregman–Noether correspondence. Section 5 organizes the framework’s predictions into a seven-claim falsification hierarchy.

Sections 6–7 develop the optic-composition semantics: seven domain bridges typed as optics, per-optic Lipschitz constants, and the Banach contraction of the composed update; Section 9 then constructs the smooth envelope of the algorithmic rate-distortion surrogates that its observables require. Section 8 executes the falsification battery computationally — experiments appear in this theory paper because each claim is quantitative, so its refutation is a terminating computation. Sections 10 and 11 construct autocatalytic (RAF) sets by filtered colimits and characterize the maximal one as a terminal coalgebra. Section 12 extends the composition to ∞ -categories in homotopy type theory and defines the three-phase closure test; Section 13 benchmarks that test on biochemical networks, in the same computational spirit. Section 14 states the four load-bearing conclusions, and Section 15 collects the open problems. Two distinct sevens appear in the paper: the seven forward maps typed as optics in the composition theorem, and the seven claims A–G of the falsification hierarchy.

2 Preliminaries

To ensure precision while remaining accessible across disciplines, we first define the core mathematical components of the framework, one definition at a time: the viability depth functional (and its distinction from curvature), the structure group of the policy bundle, the optic category, and the RAF, CPTP, Bregman, and algorithmic rate-distortion notions, up to the viability-weighted curvature itself. Each is used where cited; none is used before it is stated.

Definition 2.1 (Viability depth functional D_V ; distinct from curvature). Let M be a smooth state manifold, $V : M \rightarrow \mathbb{R}$ a smooth viability function with maximum $V_{\max} = \max_M V$, and $\gamma_a : [0, 1] \rightarrow M$ a smooth policy loop of amplitude a , parametrized at unit speed so that $|\gamma_a| = \int_0^1 |\dot{\gamma}_a(t)| dt$ is the loop length. The *viability depth functional* is

$$D_V(\gamma_a) = \frac{1}{|\gamma_a|} \int_{\gamma_a} (V_{\max} - V) ds = \frac{1}{|\gamma_a|} \int_0^1 (V_{\max} - V \circ \gamma_a(t)) |\dot{\gamma}_a(t)| dt. \quad (1)$$

For the radially symmetric viability $V(x_1, \dots, x_{n-1}) = 1 - \sum_i x_i^2$ and the circular loop $\gamma_a(t) = (a \cos 2\pi t, a \sin 2\pi t, 0, \dots, 0)$ at unit speed, $D_V(a) = a^2$. The viability depth is a loop-averaged deficit; it is *not* a curvature 2-form: it carries no orientation, no dependence on path ordering, and no commutator structure. The curvature-based viability-weighted curvature κ_V (Definition 2.11 below, contracting the curvature 2-form F with viability covectors) is a distinct object; the equality $D_V(a) = \kappa_V(a) = a^2$ in the radial prototype is a model-specific relation, not an identity.

Definition 2.2 (Structure group: $O(r)$, $SO(r)$, or $CO(r)$ with Weyl scale). The Fisher–Rao metric on a policy fiber of dimension r admits an orthonormal frame bundle with structure group $O(r)$ ([4, 35]). If the policy frame is oriented, the structure group reduces to $SO(r)$. A reduction to the conformal group $CO(r) = \mathbb{R}_+ \times SO(r)$ requires an additional *scale* (Weyl) *structure* on the policy bundle: a positive scale bundle $L \rightarrow E$ together with a conformal class $[g^F]$ of the Fisher metric and a Weyl connection preserving the conformal class. The three cases are distinct modeling commitments; Chentsov’s uniqueness theorem [20] fixes the Fisher metric under statistical-map invariance but does *not* by itself select among $O(r)$, $SO(r)$, or $CO(r)$. For the framework’s prototypes we use: $G_C = SO(2)$ for the abelian $n = 3$ prototype; $G_C = SO(3)$ for the non-abelian $n = 4$ prototype; and $CO(r)$ only when an endogenous scale variable (effective precision, temperature, sample size, or Weyl factor) is explicitly modeled (Remark 2.3).

Remark 2.3 (Fisher–Weyl policy structure, when $CO(r)$ is justified). A Fisher–Weyl policy structure consists of: (i) a policy bundle $E \rightarrow B$; (ii) a vertical Fisher metric g^F ; (iii) a positive scale bundle $L \rightarrow E$; (iv) a conformal class $[g^F]$; (v) a Weyl connection preserving the conformal class. The conformal frame bundle then has structure group $CO(r)$. If the cost functional depends only on the conformal class of g^F and is invariant under positive rescalings of orthonormal frames, its

stabilizer contains $\text{CO}(r)$; if scale is fixed, the stabilizer reduces to $\text{SO}(r)$ or $O(r)$. This preserves the $\text{CO}(r)$ ambition with mathematical precision: $\text{CO}(r)$ is an *added modeling structure*, not a consequence of Chentsov’s theorem alone.

Remark 2.4 (Physical meaning of the $n = 3 \rightarrow n = 4$ transition). At $n = 3$ the policy simplex is two-dimensional, the prototype’s structure group is $\text{SO}(2)$ (Definition 2.2; the full Fisher–Rao point stabilizer on the simplex is $O(n - 1) \supseteq \text{SO}(n - 1)$, and $\text{CO}(2) = \mathbb{R}_+ \ltimes \text{SO}(2)$ arises only under the Weyl structure of Remark 2.3), and the Lie algebra $\mathfrak{so}(2)$ is one-dimensional, generated by the single infinitesimal rotation about the origin. Two rotations about the same origin necessarily commute (they lie in the same one-parameter subgroup), so the holonomy of a sequence of loops is path-independent up to homotopy, and the path-ordering in the parallel-transport exponential collapses to ordinary commutative multiplication. The non-abelian signature (Claim F) is then unobservable: the commutator $[R_1, R_2] = 0$ identically, regardless of the rotation amplitudes. At $n = 4$ the policy simplex is three-dimensional, the prototype’s structure group is $\text{SO}(3)$ (Definition 2.2; $\text{CO}(3)$ again only under the Weyl structure), and $\mathfrak{so}(3)$ is three-dimensional with three independent generators $\{\mathfrak{l}_x, \mathfrak{l}_y, \mathfrak{l}_z\}$ satisfying $[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y$ etc. Two rotations about *distinct* axes (e.g. $R_z(\alpha_1)R_x(\alpha_2)$ vs $R_x(\alpha_2)R_z(\alpha_1)$) no longer commute, and the small-angle commutator is $[R_1, R_2] \approx \alpha_1\alpha_2[\mathfrak{l}_i, \mathfrak{l}_j]$ with Frobenius norm $\sqrt{2}\alpha_1\alpha_2$ (each $\mathfrak{so}(3)$ basis element has Frobenius norm $\sqrt{2}$). The non-abelian signature is therefore a quantitative, falsifiable prediction specific to $n \geq 4$; it cannot be tested at $n = 3$ by any experimental design.

Definition 2.5 (Optic category). Following Riley [48], for a monoidal category $(\mathcal{C}, \otimes, I)$ with finite limits the *optic category* $\mathbf{Optic}(\mathcal{C})$ has as *objects* pairs (M, C) of a forward carrier M and a backward carrier C , and as *morphisms* from (M, C) to (M', C') the equivalence classes $[R, f, g]$ of triples consisting of a residual object R , a forward map $f : M \otimes R \rightarrow M'$, and a backward map $g : C' \rightarrow C \otimes R$, quotiented by the coend relation: $[R, f, g] = [R', f', g']$ whenever there exists $h : R \rightarrow R'$ with $f' \circ (\text{id}_M \otimes h) = f$ and $(\text{id}_C \otimes h) \circ g' = g$. Composition is Riley’s Proposition 2.3 [48]: the composite of $[R, f, g]$ followed by $[R', f', g']$ is $[R \otimes R', f' \circ (f \otimes \text{id}_{R'}), (g \otimes \text{id}_{R'}) \circ g']$, with unit $[I, \rho, \eta]$ given by the left/right unitors of $\mathcal{C}$. The adapter subcategory is the case $R = I$ (trivial residual).

In words: an optic pairs a forward map, which turns an internal carrier state together with a residual (memory) into an external action, with a backward map, which turns external feedback together with that residual into an internal adjustment — a map paired with its feedback channel, in the terminology of the Introduction.

Remark 2.6 (Strict feedback representatives). The operational sections of this paper (Construction 6.1, Table 1, Definition 7.2, and Section 10) work with *strict feedback representatives* over $\mathcal{C} = \mathbf{Set}$ (cartesian monoidal): a morphism from (M, C) to (M', C') is a triple (R, f, g) with $f : M \times R \rightarrow M'$ and $g : R \times C' \rightarrow C$ — the representative-level shape in which the residual, consumed by the forward pass, is reused as the backward pass’s input (the feedback instantiation appropriate when the forward pass transports the residual, as in Definition 7.2). Strict feedback morphisms compose by

$$(R_2, f_2, g_2) \circ (R_1, f_1, g_1) = (R_1 \times R_2, f_2 \circ (f_1 \times \text{id}_{R_2}), g_1 \circ (\text{id}_{R_1} \times g_2)), \quad (2)$$

with unit $(1, \pi, \text{co}\pi)$ on the (co)product projections; associativity and unitality follow directly from function composition and the associativity of finite products, and the composite’s residual $R_1 \times R_2$ is the representative-level form of Riley’s $R \otimes R'$. On the adapter subcategory ($R = 1$) the strict feedback form and the coend form of Definition 2.5 coincide. The general translation between the strict feedback form and the coend-quotient form is recorded as an open problem (Section 15); nothing below depends on it: every statement of this paper is typed in the strict form and every composition law is verified at the representative level.

Definition 2.7 (Hordijk–Steel closure and RAF set). Following Hordijk and Steel [33, 53], given a catalytic reaction network $(X, \mathcal{R}, F, c)$ over a molecule set X with food set $F \subset X$ and catalysis

assignment $c : \mathcal{R} \rightarrow 2^X$, define for each $A \subseteq \mathcal{R}$ the *Hordijk–Steel closure* $\text{cl}_A(F)$ of the food set under A as the least set of molecules containing F and closed under the reactions of A :

$$\text{cl}_A(F) := \bigcup_{n \geq 0} S_n, \quad S_0 = F, \quad S_{n+1} = S_n \cup \{\text{react}(r) \cup \text{prod}(r) : r \in A, \text{react}(r) \subseteq S_n\}. \quad (3)$$

A subset $R' \subseteq \mathcal{R}$ is a *reflexively autocatalytic food-generated (RAF) set* if (i) every $r \in R'$ is catalyzed by some element of $\text{cl}_{R'}(F)$, and (ii) every reactant of every $r \in R'$ lies in $\text{cl}_{R'}(F)$. (Both conditions use the closure: catalysts must be food-reachable through R' , not merely present as reactants or products of R' .)

Definition 2.8 (CPTP channel). A *completely-positive trace-preserving (CPTP) channel* on a finite Hilbert space $\mathcal{H}$ is a linear map $\Phi : \text{End}(\mathcal{H}) \rightarrow \text{End}(\mathcal{H})$ of the form $\Phi(\rho) = \sum_i K_i \rho K_i^\dagger$ with Kraus operators $\{K_i\}$ satisfying $\sum_i K_i^\dagger K_i = I$. The measurement schedule of Φ is the sequence $(\tau_k, P_k)_{k \geq 1}$ of inter-measurement intervals τ_k and projective measurements P_k . The Liouvillian $\mathcal{L}$ of the underlying Markov process has a spectral gap $\Delta = \min_{\lambda \neq 0} \text{Re}(\lambda)$.

Definition 2.9 (Bregman divergence). A *Bregman divergence* [4, 19] generated by a strictly convex differentiable function ϕ is

$$D_\phi(p, q) = \phi(p) - \phi(q) - \langle \nabla \phi(q), p - q \rangle. \quad (4)$$

The divergence is not symmetric in general. The *dual affine coordinates* are p (the primal) and $\nabla \phi(p)$ (the dual). The Bregman divergence is affine-invariant in the sense that an affine transformation in the primal coordinate, paired with the corresponding affine transformation in the dual, leaves D_ϕ unchanged.

Definition 2.10 (Algorithmic rate-distortion distance). Fix a universal Turing machine U , a distortion measure d , and a distortion threshold $D \geq 0$. The *algorithmic rate-distortion distance* of x at threshold D is

$$\text{dist}_D(x) = \min\{|p| : U(p) \text{ outputs } \hat{x}, d(x, \hat{x}) \leq D\}. \quad (5)$$

It is monotone non-increasing in D , bounded above by the Kolmogorov complexity $K(x)$, and upper-semicomputable by dovetailing over all programs. It is a single-string, intrinsically deterministic quantity: no set-average rate and no random-coding gap arise. The construction is the algorithmic rate-distortion distance of Vereshchagin and Vitányi [56], stated here in its single-string form; the set-level rate-distortion theory of Shannon is the coarse-grained counterpart.

Definition 2.11 (Viability-weighted curvature κ_V). Let $(B, E, \mathcal{P}, \varepsilon, \Gamma)$ be a SAVGS (Definition 3.1) with a G_C -connection on the policy bundle whose curvature 2-form on a stratum is F . Let

$$h_\alpha(\theta, x) = D_\phi(\text{dist}_D(x), \text{dist}_D(x_0)) \quad (6)$$

be a viability margin built from a Bregman divergence (Definition 2.9) evaluated at the algorithmic rate-distortion distance (Definition 2.10) of the current state relative to a reference x_0 , with $h_\alpha > 0$ on the open feasibility set. The *per-pair viability-weighted curvature* is

$$\kappa_\alpha(\theta, x; u, v) = \frac{[-D_{F(u,v)} h_\alpha]^+(\theta, x)}{h_\alpha(\theta, x)}, \quad (7)$$

where $[\cdot]^+$ is the positive part: only viability-eroding curvature contributes. The *viability-weighted curvature* is the worst case over unit-area bivectors and active viability covectors,

$$\kappa_V(\theta, x) = \sup_{\substack{u, v \in T_\theta B \\ \|u \wedge v\|_{g_F} = 1}} \max_{\alpha : h_\alpha(\theta, x) > 0} \kappa_\alpha(\theta, x; u, v). \quad (8)$$

The quantity $\kappa_V(\theta, x)$ is the worst fractional loss of viability margin per unit oriented environmental-loop area, to leading order; it contracts the geometric defect F with the survival covectors Dh_α instead of reporting $\|F\|$ alone. In intuitive terms, κ_V ignores harmless policy rotations and responds only to directional shifts that consume the system’s remaining viability buffer. Gauge invariance of κ_V follows from the square-root embedding (Proposition 3.19) together with the Bregman–Noether correspondence (Proposition 4.1); the wedge-norm constraint makes κ_V a per-area quantity, matching the leading-order expansion of Theorem 3.16. The instantiation (6) is one admissible choice: any C^2 viability function with the stated positivity works. The application paper [1] realizes the measure-theoretic *analogue* of this object: its atomic crease measure stands in for the curvature 2-form and its value–flux coupling supplies analogue survival covectors, with the per-event mass corresponding, at active atoms, to the positive part of the margin change at the crossed facet normalized by the local margin — a stated correspondence at active atoms, not a proved domination; the ratio-form comparison is recorded as open (Section 15).

3 The SAVGS Framework

We now assemble the components into a single mathematical structure: the Stratified Autopoietic Viability Geometric System (SAVGS) — the minimal object on which the viability-weighted curvature can be stated with its type structure, gauge behavior, and premises explicit. Five components assemble into one stratified object: a control base, a policy bundle, a viability margin, a maintenance graph, and a boundary span.

Definition 3.1 (SAVGS). The SAVGS is the tuple $(B, E, \mathcal{P}, \varepsilon, \Gamma)$:

1. A continuous control base manifold $B \subset \mathbb{R}^d$, with coordinates including food scarcity, danger intensity, and sensor noise.
2. A policy bundle $\pi : E \rightarrow B$ with total space E and policy fiber $\mathcal{P} = \pi^{-1}(\theta)$, where $\mathcal{P}$ is the open simplex $\Delta_\circ^{m-1}$ (for m actions), the circle S^1 (abelian $n = 3$ prototype), the rotation group $SO(3)$ (non-abelian $n = 4$ prototype), or another declared policy manifold. The bundle projection $\pi : E \rightarrow B$ is the canonical projection onto the base manifold; on the trivial bundle $E = B \times \mathcal{P}$ this is $\pi(\theta, p) = \theta$.
3. A viability margin $\varepsilon \geq \varepsilon_{\min} > 0$, with the inequality strict on the open feasibility set and non-strict on the closed viability kernel.
4. An endogenous maintenance graph $\Gamma = (M, R, E_\Gamma)$ of maintenance operations M , regeneration rules R , and maintenance edges E_Γ (renamed to avoid clash with total space E).
5. A 2-categorical span $\text{Stab}_1 \leftarrow \text{Boundary} \rightarrow \text{Stab}_2$ that resolves the constraint-switching boundary discontinuity between strata (Theorem 3.4 for the full gluing theorem).

The five components compose into a single object whose geometry is a stratified G_C -reduced principal bundle over the control manifold, with policy fibers over each stratum and viability kernels defined by non-strict inequalities on each stratum. The viability-weighted curvature of Definition 2.11 is computed on this object.

Remark 3.2 (Why the 2-categorical span is needed). At a constraint-switching boundary of the viability margin ε the smooth connection breaks down in the manner described in §1, and the holonomy of a boundary-crossing loop is defined only after the boundary behavior itself is specified. The fifth SAVGS component supplies exactly that specification: a 2-categorical span $\text{Stab}_1 \leftarrow \text{Boundary} \rightarrow \text{Stab}_2$ that glues the two smooth strata via a boundary 2-cell; the connection is defined piecewise on each stratum, and a matching condition on the boundary supplies the 2-categorical compatibility. The formal development — the lax-functorial gluing and the boundary

2-cell's coherence with the connection 1-forms — is given in Section 3.1 below, and the homotopy-theoretic well-definedness of the resulting holonomy is treated in Section 12. The operational requirement on the abelian and non-abelian prototypes is that the active set remains constant along each loop, which places the loop entirely within a single smooth stratum.

3.1 2-categorical gluing theorem for stratified connections

We now close the 2-categorical gluing agenda sketched in Remark 3.2 by constructing the 2-category $\mathbf{StCon}(B)$ of stratified G_C -connections on a stratified base B , proving the gluing theorem, and deriving the piecewise holonomy formula. Throughout, $G_C \in \{O(r), SO(r), \mathrm{CO}(r)\}$ is the structure group selected by the policy-fiber geometry (Definition 2.2).

Definition 3.3 (2-category $\mathbf{StCon}(B)$ of stratified connections). Let B be a stratified manifold with strata $\{S_i\}_{i \in I}$ ($B = \bigsqcup_i S_i \cup \bigcup_{ij} B_{ij}$ where $B_{ij} = S_i \cap \bar{S}_j$ are boundary hypersurfaces, and $T_{ijk} = S_i \cap S_j \cap S_k$ are triple intersections). The 2-category $\mathbf{StCon}(B)$ of stratified G_C -connections is:

- *Objects*: stratified principal G_C -bundles $P \rightarrow B$ with a connection $A = \{A_i \text{ on } S_i, g_{ij} : B_{ij} \rightarrow G_C\}$ satisfying:

(O1) A_i is a smooth connection 1-form on stratum S_i ;

(O2) $g_{ij} : B_{ij} \rightarrow G_C$ is a smooth transition function;

(O3) the matching condition on B_{ij} : with the convention that the transition g_{ij} acts on frames as $e_j = e_i \cdot g_{ij}$ (so g_{ij} carries stratum S_i 's frame to stratum S_j 's frame), the connection forms are related by the standard gauge transformation

$$A_j = g_{ij}^{-1} A_i g_{ij} + g_{ij}^{-1} dg_{ij}, \quad (9)$$

whose infinitesimal form ($g_{ij} = \exp u_{ij}$ near the identity) is $A_j = A_i + du_{ij} + [A_i, u_{ij}] + O(u_{ij}^2)$. For abelian $G_C = U(1)$ this reduces to $A_j - A_i = d\alpha_{ij}$ where $g_{ij} = e^{i\alpha_{ij}}$; the reverse-direction reading ($g_{ji} = g_{ij}^{-1}$ carrying A_j to A_i) is equivalent and is the one used in the holonomy formula below;

(O4) the Čech cocycle on T_{ijk} : $g_{ij} \cdot g_{jk} = g_{ik}$ (multiplication in G_C).

- *1-Morphisms*: connection-preserving equivariant bundle maps $\phi : P \rightarrow P'$, smooth on each stratum, with $\phi \circ g_{ij} = g'_{ij} \circ \phi$ on each boundary B_{ij} .
- *2-Morphisms*: gauge transformations $\eta : \phi \Rightarrow \phi'$ (vertical automorphisms of P' that conjugate ϕ to ϕ'), smooth on each stratum, with the coherence 2-cell condition at triple intersections (η commutes with g_{ij} on T_{ijk} in the sense of strict 2-functorial equality).

Theorem 3.4 (2-categorical gluing theorem). *Given a stratified base B with strata $\{S_i\}$, boundary hypersurfaces $\{B_{ij}\}$, a smooth Čech cocycle $\{g_{ij} : B_{ij} \rightarrow G_C\}$ satisfying $g_{ij} \cdot g_{jk} = g_{ik}$ on triple intersections T_{ijk} , and connection 1-forms $\{A_i \text{ on } S_i\}$ satisfying the matching condition (O3) of Definition 3.3 on each B_{ij} ,*

$$A_j|_{B_{ij}} = g_{ij}^{-1} A_i g_{ij} + g_{ij}^{-1} dg_{ij},$$

there exists a unique (up to 2-isomorphism) object (P, A) in $\mathbf{StCon}(B)$ whose restriction to each S_i is A_i and whose transition on each B_{ij} is g_{ij} .

Proof. The proof is by 2-descent in the 2-stack of stratified G_C -bundles with connection. The 2-stack property (objects glue uniquely up to 2-isomorphism on overlaps; 1-morphisms glue uniquely up to 2-isomorphism; 2-morphisms glue strictly) follows from the Giraud axioms for the underlying G_C -bundles extended to stratified spaces:

- *Object gluing*: the underlying G_C -bundle P is constructed by the standard clutching construction on each stratum, glued along the boundary via the transition g_{ij} . The Čech cocycle (O4) ensures consistency at triple intersections. The result is unique up to bundle isomorphism.
- *Connection gluing*: the connection 1-forms $\{A_i\}$ glue via the matching condition (O3): on B_{ij} , g_{ij} is a gauge transformation carrying A_j to A_i , so A_i on S_i and A_j on S_j define a single stratified connection on P . The glued connection is unique up to gauge transformation (2-isomorphism).
- *1-morphism gluing*: given two stratified connections (P, A) and (P', A') and a 1-morphism $\phi_i : P|_{S_i} \rightarrow P'|_{S_i}$ on each stratum that commutes with g_{ij} on boundaries, the $\{\phi_i\}$ glue to a unique 1-morphism $\phi : P \rightarrow P'$, with the boundary compatibility $\phi \circ g_{ij} = g'_{ij} \circ \phi$ supplying the gluing data.
- *2-morphism gluing*: gauge transformations are determined locally; on overlaps they agree by the strict 2-functorial coherence condition, so 2-morphisms glue strictly.

Hence $\mathbf{StCon}(B)$ is a 2-stack (a 2-sheaf of groupoids), and the gluing theorem holds by 2-descent [10, 17, 18, 27]. $\square$

Remark 3.5 (Proof status of Theorem 3.4). The proof reduces the gluing to the 2-stack descent property for stratified G_C -bundles: the Giraud-type axioms for the underlying bundles are cited [18, 27], not re-proved, and the extension from smooth spaces to stratified spaces (stratum-by-stratum clutching with boundary matching) is verified here at the level of construction, with the coherence of the boundary 2-cells checked on triple intersections. The piecewise holonomy consequences (Theorem 3.6), by contrast, are machine-verified in five loop geometries and all three structure-group regimes (Remarks 3.8–3.12).

Theorem 3.6 (Piecewise holonomy formula). *Let $(P, A) \in \mathbf{StCon}(B)$ be a stratified connection, and let $\gamma : [0, 1] \rightarrow B$ be a smooth loop that crosses finitely many boundaries transversally at points $\{p_k\}_{k=1}^n$ in cyclic order ($p_1 < p_2 < \dots < p_n$). Let γ be decomposed as $\gamma = \gamma_1 \sqcup \gamma_2 \sqcup \dots \sqcup \gamma_n$ where each piece γ_k lies in stratum S_{i_k} . Then the stratified holonomy is*

$$\mathrm{Hol}^{\mathrm{strat}}(\gamma) = \mathrm{Hol}_{S_{i_n}}(\gamma_n) \cdot g_{i_n i_1}(p_n) \cdot \mathrm{Hol}_{S_{i_{n-1}}}(\gamma_{n-1}) \cdot \dots \cdot g_{i_1 i_2}(p_1) \cdot \mathrm{Hol}_{S_{i_1}}(\gamma_1), \quad (10)$$

a product of stratum-holonomies with boundary-transition maps, the order fixed by the chronological (right-to-left) traversal of the loop. For a closed loop the number of transversal crossings of any single wall is necessarily even: a small loop of area ε^2 centered on a wall B_{+-} crosses it twice, at points $p_+ \neq p_-$ with $|p_+ - p_-| = O(\varepsilon)$. For such a loop the piecewise curvature formula is

$$\mathrm{Hol}^{\mathrm{strat}}(\gamma_\varepsilon) = \exp\left(-\varepsilon^2 \left[\iint_{\Sigma \cap S_+} F_+ dA + \iint_{\Sigma \cap S_-} F_- dA \right] - [\log g_{+-}(p_+) - \log g_{+-}(p_-)] + O(\varepsilon^2) \right), \quad (11)$$

where $F_\pm$ are the curvature 2-forms on strata $S_\pm$ and Σ is the small enclosed disk. For affine wall transitions the terms neglected beyond the displayed $O(\varepsilon)$ and $O(\varepsilon^2)$ contributions are $O(\varepsilon^3)$ — the regime the numerics of Remark 3.8 verify; for general transitions, second-order Taylor terms of $\log g_{+-}$ along the wall and commutators of the two resets enter at $O(\varepsilon^2)$ (Open Problem 1). The two boundary resets cancel at leading order by the cocycle condition ($g_{-+} = g_{+-}^{-1}$ on the wall): the residual boundary contribution $\log g_{+-}(p_+) - \log g_{+-}(p_-)$ is the tangential variation of the transition function between the two crossing points, of order $O(\varepsilon)$; it vanishes identically when the transition function is constant along the wall, and it is linear in the wall-coordinate difference in the machine-verified regime of Remark 3.8. An $O(1)$ reset term $\log g_{+-}(p_)$ arises only for an open arc that crosses the wall once and is not closed by the reverse transition.*

Proof. By Theorem 3.4, the stratified connection A exists and is unique up to 2-isomorphism, so the holonomy is well-defined. Decompose γ into pieces $\{\gamma_k\}$ on strata $\{S_{i_k}\}$ separated by boundary crossings $\{p_k\}$. The parallel transport of A along γ factorizes as the product of stratum-parallel transports on each γ_k (using $A|_{S_{i_k}} = A_{i_k}$ and Stokes' theorem on each stratum) interleaved with the boundary transitions $g_{i_k i_{k+1}}(p_k)$ applied at each crossing. The ordering (right-to-left in (10)) follows the chronological traversal of the loop; the composition order is genuinely order-dependent for non-abelian G_C (Remark 3.9). For an ε -small *closed* loop crossing the wall twice, the stratum-holonomies reduce by Stokes to $\varepsilon^2 \iint_{\Sigma \cap S} F_S dA$ on each stratum (where Σ is the small enclosed disk), while the two boundary transitions contribute $\log g_{+-}(p_+) - \log g_{+-}(p_-)$ in the Lie algebra: the $O(1)$ parts of the reset pair cancel by the cocycle condition, and the residual is the tangential variation of the transition function between the crossing points, of order $O(\varepsilon)$. For abelian G_C with the matching condition (9) the cancellation is exact at leading order: along the short segment of γ inside S_- , the jump $A_- - A_+ = d\alpha_{+-}$ integrates to $\alpha(p_-) - \alpha(p_+)$, which cancels the reset variation identically, so the holonomy is governed by the curvature terms at ε^2 plus the stated $O(\varepsilon)$ residual. This is the regime verified numerically in Remark 3.8 (boundary reset linear in the wall coordinate with slope ε , the loop width). $\square$

Remark 3.7 (Regime delineation with the application paper). The two holonomy-scaling regimes are complementary, and the pair of papers states both. Loops contained in a single constant-active-set stratum accumulate curvature at $O(\varepsilon^2)$ per loop — the smooth arm, where the companion's quadratic law $\kappa(a) = a^2$ and the Fisher-minimal transport law live. Loops that cross a constraint-switching wall accumulate the boundary contribution at $O(\varepsilon)$ per crossing (Theorem 3.6) — the wall-crossing arm, which the application paper's regime dichotomy measures as the piecewise-affine slope 1.00 against the smooth arm's 2.000.

Remark 3.8 (Numerical verification of the piecewise holonomy formula). The piecewise holonomy formula — including the two-crossing small-loop form of (11), with the boundary contribution $\alpha(p_+) - \alpha(p_-)$ at its true order $O(\varepsilon)$ — is numerically verified on a two-stratum base $B = \mathbb{R}^2$ with the x -axis as the boundary, $G_C = U(1)$ (abelian, simplifying the matching condition), and constant-curvature connections $A_{\pm} = (c_{\pm}/2)(x dy - y dx)$ on the upper (S_+) and lower (S_-) half-planes. The transition function $g_{+-}(x, 0) = e^{i\alpha(x)}$ with $\alpha(x) = a_1 x$ is the boundary reset map. The verification: (a) computes the analytic holonomy $H_{\text{an}}(\gamma) = e^{-i(F\varepsilon^2 + \alpha(p_+) - \alpha(p_-))}$ for a rectangular loop of side ε centered at $(x_c, 0)$ crossing the boundary twice; (b) computes the numerical holonomy by parallel transport along the loop with the connection 1-form A and the boundary transitions; (c) confirms that $|H_{\text{an}}| = |H_{\text{num}}|$ to within 10^{-13} (machine precision) for all tested $\varepsilon \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$, with the boundary crossings correctly detected ($n_{\text{cross}} = 2$ per loop); (d) verifies the boundary reset term is linear in a_1 with slope ε (the loop width), as predicted by $\alpha(p_+) - \alpha(p_-) = a_1 \cdot \varepsilon$. Outputs: `two_cat_gluing_stratified.{png, csv, txt}`.

Remark 3.9 (Non-abelian extension to $G_C = SO(3)$, the $n = 4$ prototype's policy fiber). The abelian verification of Remark 3.8 is extended to the *non-abelian* $G_C = SO(3)$ setting of the $n = 4$ prototype's policy fiber (Definition 2.2); the deposited verification artifacts cover both settings. The $\mathfrak{so}(3)$ -valued connection on each stratum is $A_{\pm} = (F/2)(x dy - y dx) T_z$ where T_z is the $\mathfrak{so}(3)$ generator of rotations about the z -axis, and the boundary transition $g_{+-}(x, 0) = \exp(\alpha(x) T_y)$ is a rotation about the y -axis by $\alpha(x) = a_1 x$. The commutator $[T_y, T_z] = T_x \neq 0$ makes the matrix product in the piecewise holonomy formula (10) genuinely order-dependent: the boundary reset $R_b = \alpha(p_*) T_y$ lies in a different $\mathfrak{so}(3)$ direction than the stratum curvature FT_z . The verification: (a) computes the analytic piecewise holonomy $H_{\text{an}} \in SO(3)$ as the ordered matrix product of stratum-holonomies $\exp(-(F/2) \int (x dy - y dx) T_z ds)$ and boundary transitions $\exp(\pm \alpha(p_k) T_y)$ in the traversal order; (b) computes the numerical $SO(3)$ holonomy by segment-by-segment matrix-exponential parallel transport $H_{\text{num}} = \prod_k \exp(-A_{\text{mid}} ds) \cdot \prod_k g(p_k)^{\pm 1}$; (c) confirms $\|H_{\text{num}} - H_{\text{an}}\|_F < 10^{-10}$ for all tested $\varepsilon \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$, with $\det(H_{\text{num}}) = +1$ and $H_{\text{num}}^T H_{\text{num}} = I_3$ (genuine special-orthogonal matrices); (d) recovers the abelian limit $\alpha = 0 \Rightarrow H = \exp(-F\varepsilon^2 T_z)$ (single z -rotation by $F\varepsilon^2$); (e) detects the non-abelian mixing

via the off-diagonal entry $H_{\text{num}}[0, 2]$ (the $x \rightarrow z$ block), which is zero at $a_1 = 0$ and grows linearly in a_1 (slope $\approx \varepsilon \cdot F\varepsilon^2/2$ in the small-loop regime), demonstrating that the boundary y -rotation rotates the holonomy matrix out of the stratum-curvature T_z direction. The non-abelian verification thus confirms Theorem 3.6 in the regime matching the $n = 4$ prototype. Outputs: `two_cat_gluingso3.{png, csv, txt}`.

Remark 3.10 (Third verification regime: non-abelian $G_C = CO(3)$ for endogenous reversibility). The piecewise holonomy formula is further verified in the *third* structure-group regime of Definition 2.2, $G_C = CO(3) = \mathbb{R}_+ \ltimes SO(3)$, corresponding to endogenous reversibility in the policy-fiber hierarchy. The Lie algebra $\mathfrak{co}(3) = \mathbb{R}S \oplus \mathfrak{so}(3)$ decomposes as a direct sum of the *central* scaling direction $S = I_3$ ($[S, \cdot] = 0$) and the non-abelian rotation part $\mathfrak{so}(3)$. The two-stratum base and stratum connection match the $SO(3)$ setup, but the boundary transition is now a $CO(3)$ -valued map

$$g_{+-}(x, 0) = \exp(\alpha(x) T_y + \beta(x) S) = R_y(\alpha(x)) \cdot \Lambda(\beta(x)), \quad (12)$$

where $\Lambda(\beta) = e^\beta I_3$ is uniform scaling by e^β , $\alpha(x) = a_1 x$ is the rotation amplitude (as in the $SO(3)$ test), and $\beta(x) = b_1 x$ is the scaling amplitude (new in $CO(3)$). The boundary reset thus combines a T_y -direction rotation with an S -direction scaling; since S is central, the scaling factors compose multiplicatively (abelian), while the rotations compose non-commutatively (as in the $SO(3)$ regime). The $CO(3)$ holonomy is consequently a product of non-abelian rotation matrices and abelian scaling factors, with $\det(H) = \lambda_{\text{total}}^3 > 0$ and $H^\top H = \lambda_{\text{total}}^2 \cdot I_3$ (conformal orthogonality: $CO(3)$ preserves angles up to uniform scaling). The numerical verification confirms: (a) $\|H_{\text{num}} - H_{\text{an}}\|_F < 10^{-10}$ for all tested $\varepsilon \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$; (b) $\det(H_{\text{num}}) = \lambda_{\text{total}}^3$ to within 10^{-9} ; (c) $\|H_{\text{num}}^\top H_{\text{num}} - \lambda_{\text{total}}^2 I_3\|_F < 10^{-9}$; (d) the abelian limit ($\alpha = \beta = 0$) recovers $H = \exp(-F\varepsilon^2 T_z)$ (same as the $SO(3)$ abelian limit, since the scaling direction contributes nothing when the boundary transition vanishes); (e) the pure-scaling limit ($\alpha = 0, \beta \neq 0$) recovers $H = \exp(-F\varepsilon^2 T_z) \cdot \Lambda(b_1 \varepsilon)$ to machine precision (rotation holonomy composed with uniform scaling). The $CO(3)$ verification completes the confirmation of Theorem 3.6 in *all three* structure-group regimes: abelian $U(1)$, non-abelian $SO(3)$, and endogenous-reversibility $CO(3)$. Outputs: `two_cat_gluingsco3.{png, csv, txt}`.

Remark 3.11 (Topologically distinct loop: triangular geometry). To verify that the piecewise holonomy formula is *geometry-independent* (and not an artifact of the rectangular loop shape), the $SO(3)$ verification is repeated on a *triangular* loop with vertices $V_1 = (0, +\varepsilon)$, $V_2 = (+\varepsilon, -\varepsilon)$, $V_3 = (-\varepsilon, -\varepsilon)$ (traversed counterclockwise). The triangle is topologically distinct from the rectangle: 3 vertices and 3 edges instead of 4 and 4; boundary crossings at *non-perpendicular* angles (the triangle's edges cross the x -axis at oblique angles, not vertically); an *asymmetric* distribution of stratum pieces (3 pieces in S_- , 2 in S_+ ; the rectangular loop splits its 4 pieces symmetrically with 2 per stratum). The triangle has signed area $2\varepsilon^2$ (twice the rectangular loop's ε^2 at the same scale). The boundary crossings occur at $p_1 = (\varepsilon/2, 0)$ on edge $V_1 \rightarrow V_2$ and $p_3 = (-\varepsilon/2, 0)$ on edge $V_3 \rightarrow V_1$, giving 5 stratum pieces in total: edge 1a ($V_1 \rightarrow p_1$ in S_+), edge 1b ($p_1 \rightarrow V_2$ in S_-), edge 2 ($V_2 \rightarrow V_3$ entirely in S_-), edge 3a ($V_3 \rightarrow p_3$ in S_-), edge 3b ($p_3 \rightarrow V_1$ in S_+). The piecewise formula gives $H = \text{Hol}_{S_+}(3b) \cdot g(p_3)^{-1} \cdot \text{Hol}_{S_-}(3a) \cdot \text{Hol}_{S_-}(2) \cdot \text{Hol}_{S_-}(1b) \cdot g(p_1) \cdot \text{Hol}_{S_+}(1a)$, with each Hol a T_z -rotation by the line integral of $A = (F/2)(x dy - y dx)$. Verification: (a) $\|H_{\text{num}} - H_{\text{an}}\|_F < 10^{-10}$ for all $\varepsilon \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$ (machine precision); (b) $\det(H_{\text{num}}) = +1$ and $H_{\text{num}}^\top H_{\text{num}} = I$ (genuine $SO(3)$); (c) the abelian limit ($\alpha = 0$) recovers $H = \exp(-F \cdot \text{Area}(\triangle) T_z) = \exp(-2F\varepsilon^2 T_z)$ (the triangle's larger area produces $2\times$ the rectangular loop's rotation angle, exactly matching the area formula $F \cdot \text{Area}$); (d) the non-abelian feature is detected ($H_{\text{num}}[0, 2] = 0$ at $a_1 = 0$, grows linearly in a_1). The triangular-loop verification confirms that the piecewise holonomy formula holds for arbitrary loop geometry, not just axis-aligned rectangles. Outputs: `two_cat_gluingso3_triangular.{png, csv, txt}`.

Remark 3.12 (Pentagon, circle, and ellipse loop topologies). To further validate geometry-independence, the $SO(3)$ verification is repeated on three additional loop topologies that are topologically and geometrically distinct from both the rectangular and triangular loops:

- *Pentagon* (5-vertex regular polygon, inscribed in a circle of radius ε): a polygonal loop with higher vertex count than the triangle (3) or rectangle (4). Pentagon area: $(5/2)\varepsilon^2 \sin(2\pi/5) \approx 2.38\varepsilon^2$ (at $\varepsilon = 1$). Pentagon boundary crossings: 2 (the polygon spans both strata), but with 5 edges contributing to the piecewise decomposition.
- *Circle* (smooth non-polygonal curve, 200 equal-angle sample points starting at angle π/n to avoid placing a sample exactly on the boundary): a smooth curve, the analytic limit of polygonal refinement. Circle area: $\pi\varepsilon^2$.
- *Ellipse* (smooth anisotropic curve, 200 equal-angle sample points with semi-major axis $a = \varepsilon$ and semi-minor axis $b = \varepsilon/2$): a loop with broken rotational symmetry, testing that the piecewise formula extends to anisotropic loop geometry. Ellipse area: $\pi\varepsilon \cdot (\varepsilon/2) = \pi\varepsilon^2/2$.

For each topology, the verification is run on the same two-stratum base as the rectangular ($U(1)$ and $SO(3)$) and triangular ($SO(3)$) verifications: $F_+ = F_- = 2$, $\alpha(x) = 1.5x$, $\varepsilon \in \{0.05, 0.1, 0.2, 0.4, 0.8\}$. The numerical holonomy is computed by the same segment-by-segment matrix-exponential parallel transport, with 2000 sample points (numerical) and 8000 sample points (analytic reference). For all three topologies: (a) $\|H_{\text{num}} - H_{\text{an}}\|_F < 10^{-10}$ for all ε (machine precision); (b) $\det(H_{\text{num}}) = +1$ and $H_{\text{num}}^\top H_{\text{num}} = I$ (genuine $SO(3)$); (c) $n_{\text{crossings}} = 2$ per loop (entry + exit); (d) the abelian limit ($\alpha = 0$) recovers $H = \exp(-F \cdot \text{Area}(\text{loop}) \cdot T_z)$ within machine precision for the corresponding area formula (pentagon $(5/2)\varepsilon^2 \sin(2\pi/5)$, circle $\pi\varepsilon^2$, ellipse $\pi\varepsilon^2/2$); (e) the non-abelian feature is detected ($H_{\text{num}}[0, 2] = 0$ at $a_1 = 0$, grows linearly in a_1 in all three topologies). Combined with the rectangular ($U(1)$, $SO(3)$), $CO(3)$, and triangular verifications, the piecewise holonomy formula (Theorem 3.6) is verified on *five* distinct loop shapes (rectangle, triangle, pentagon, circle, ellipse), confirming that the formula is *geometry-independent*: it holds for any loop shape, polygonal or smooth, isotropic or anisotropic. Outputs: `two_cat_gluing_so3_topology.{png, csv, txt}`.

Remark 3.13 (Pathwise viability versus endpoint viability). A small loop with bounded endpoint holonomy does not imply that intermediate points of the loop are viable: the endpoint bound is a weaker statement than pathwise viability. To control intermediate points, the calibration protocol of the abelian $n = 3$ prototype applies a Nagumo inward-pointing condition along the path, equivalently a viability tube of radius ε around the loop. A loop with bounded endpoint holonomy but intermediate viability violation is a false negative for the holonomy prediction; the tube check excludes it. The tube is operationalized by verifying that the entire ε -neighborhood of $\gamma_a([0, 1])$ lies inside the closed viability kernel $\{x : V(x) \geq V_{\max} - \varepsilon\}$; this is a binary check that produces a separate falsifiable verdict on the pathwise viability content of the holonomy prediction.

3.2 Fisher-minimal transport law on constant-active-set strata

On a constant-active-set stratum of SAVGS, the active viability constraints are represented locally by $c(\theta, p) = 0$ with $J_p = D_p c$ of full row rank and $J_\theta = D_\theta c$. For an environmental velocity $\dot{\theta}$, the Fisher-minimal policy velocity is the unique solution of $J_p v + J_\theta \dot{\theta} = 0$ minimizing $\frac{1}{2} v^\top G(p) v$, where $G(p)$ is the Fisher information metric on the open simplex Δ^{n-1} .

Proposition 3.14 (Fisher-minimal horizontal lift (KKT form)). *On a constant-active-set stratum where J_p has full row rank, the unique Fisher-minimal policy velocity is*

$$v^*(\theta, p, \dot{\theta}) = -G(p)^{-1} J_p^\top (J_p G(p)^{-1} J_p^\top)^{-1} J_\theta \dot{\theta}, \quad (13)$$

which defines the connection 1-form $A_i = G^{-1} J_p^\top (J_p G^{-1} J_p^\top)^{-1} \partial_i c$ on the stratum, satisfying $\dot{p} = -A_i(\theta, p) \dot{\theta}^i$. The horizontal lift $H_i = \partial_{\theta^i} - A_i$ is a genuine smooth connection on each constant-rank stratum, with curvature $F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j]$.

Proof. The constrained minimum of $\frac{1}{2}v^\top Gv$ subject to $J_p v + J_\theta \dot{\theta} = 0$ is found by the method of Lagrange multipliers. Setting $\nabla_v [\frac{1}{2}v^\top Gv + \mu^\top (J_p v + J_\theta \dot{\theta})] = Gv + J_p^\top \mu = 0$ gives $v = -G^{-1}J_p^\top \mu$. Substituting into the constraint: $-J_p G^{-1}J_p^\top \mu + J_\theta \dot{\theta} = 0$, so $\mu = (J_p G^{-1}J_p^\top)^{-1}J_\theta \dot{\theta}$. The full-row-rank hypothesis on J_p guarantees the inverse exists. The horizontal subspace $H_{(\theta,p)} = \ker(J_p) \cap T_p \Delta^{n-1}$ varies smoothly in (θ, p) on the stratum (constant-rank hypothesis), so the resulting 1-form A_i is C^2 ; the curvature F_{ij} is then the standard Ehresmann-curvature expression by direct computation [35]. $\square$

Remark 3.15 (Projected differential inclusion at constraint-switching boundaries). At a constraint-switching boundary—where an inequality enters or leaves the active set—the full-row-rank hypothesis of Proposition 3.14 fails: the rank of J_p jumps, and the smooth connection A_i is no longer defined. The correct replacement is the projected differential inclusion

$$\dot{p} \in \arg \min_{v \in T_{\mathcal{K}}(\theta, p; \dot{\theta})} \frac{1}{2} \|v - v_0\|_{F,p}^2, \quad (14)$$

where $T_{\mathcal{K}}(\theta, p; \dot{\theta})$ is the Bouligand contingent cone to the viability kernel $\mathcal{K}$ at (θ, p) in the direction $\dot{\theta}$ and v_0 is the unconstrained Fisher-minimal velocity. This is viability dynamics, not ordinary smooth-connection theory; the SAVGS 2-categorical span of Definition 3.1 glues the smooth-stratum connections across the boundary via the matching condition on the boundary 2-cell.

Theorem 3.16 (Small-loop viability–holonomy). *Let $(B, E, \mathcal{P}, \varepsilon, \Gamma)$ be a SAVGS as in Definition 3.1, on a constant-active-set stratum where Proposition 3.14 applies and the connection 1-form A_i is C^2 . Let $\gamma_\varepsilon : [0, 1] \rightarrow B$ be a square loop of side ε in the (θ^1, θ^2) -plane at θ_0 , contained entirely in a single stratum, and let $h_\alpha(\theta, x)$ be a C^2 viability function with $h_\alpha(\theta_0, x_0) > 0$. Then the parallel transport of the policy around γ_ε satisfies*

$$p_{\text{end}} - p_0 = \varepsilon^2 F_{12}(\theta_0, p_0) + O(\varepsilon^3), \quad (15)$$

up to the orientation convention for F_{12} . The induced change in the viability margin is

$$h_\alpha(x_{\text{end}}) - h_\alpha(x_0) = \varepsilon^2 D_p h_\alpha(F_{12}(\theta_0, p_0)) + O(\varepsilon^3) + \Delta_\alpha^{\text{other}}, \quad (16)$$

where $\Delta_\alpha^{\text{other}}$ collects all non-returning energy, integrity, repair, and environmental contributions along the loop. In particular, the endpoint remains viable whenever

$$h_\alpha(x_0) + \varepsilon^2 D_p h_\alpha(F_{12}) - C_\alpha \varepsilon^3 - |\Delta_\alpha^{\text{other}}| > 0 \quad \text{for every } \alpha, \quad (17)$$

where C_α depends on the C^2 bounds of h_α and A_i .

Proof. The expansion (15) is the standard non-abelian Stokes formula for the parallel transport around an infinitesimal loop (see, e.g., [35, 39]): writing the path-ordered exponential $\mathcal{P} \exp(-\oint A) = 1 - \iint_S F_{12} d\theta^1 d\theta^2 + O(\varepsilon^3)$ on a square S of side ε gives $p_{\text{end}} - p_0 = -\varepsilon^2 F_{12} + O(\varepsilon^3)$ (the sign convention is fixed by the orientation of A_i and is absorbed into F_{12}). Equation (16) follows by Taylor expansion of h_α at x_0 to first order in $p_{\text{end}} - p_0$ and second order in ε , with $\Delta_\alpha^{\text{other}}$ collecting the integral of $D_\theta h_\alpha \dot{\theta}$ along the loop plus the $O(\varepsilon^3)$ remainder. The sufficient condition (17) is then the requirement that the worst-case $O(\varepsilon^3)$ remainder plus the non-returning contribution not exceed the initial margin minus the leading-order curvature contribution. $\square$

Proposition 3.17 (Vulnerability bound). *Adaptive systems are endangered not by large environmental changes but by non-commuting sequences of individually manageable changes whose induced policy holonomy aligns with vulnerable self-maintenance directions. The upper bound on vulnerability is the viability-weighted curvature κ_V of Definition 2.11 on a G_C -structured stratified connection; the lower bound is 0, attained when $\kappa_V \equiv 0$ (flat connection).*

Proof. On a constant-active-set stratum, integrate Theorem 3.16 along the loop: the endpoint margin change is bounded above in magnitude by $\varepsilon^2 D_p h_\alpha(F_{12}) \leq \varepsilon^2 \kappa_V$ per unit loop area, with the positive part in Definition 2.11 discarding viability-restoring curvature; summing over the loop gives the upper bound. The flat connection $F \equiv 0$ gives $p_{\text{end}} = p_0$ to $O(\varepsilon^3)$, hence zero holonomy-induced vulnerability and the lower bound. $\square$

Remark 3.18 (What Theorem 3.16 does and does not claim). Theorem 3.16 bounds the *endpoint* viability margin by the leading-order curvature contribution plus an $O(\varepsilon^3)$ remainder. It does *not* bound the *pathwise* viability margin: a loop with bounded endpoint holonomy may transit non-viable regions. Pathwise viability requires the additional Nagumo inward-pointing condition of Remark 3.13—equivalently a viability tube of radius ε around $\gamma_a([0, 1])$ contained in the closed viability kernel. Curvature is therefore neither necessary nor sufficient for survival in isolation; the bidirectional “equivalence” is replaced by the unidirectional upper bound “vulnerability $\leq \kappa_V$ ” of Proposition 3.17, with the trivial lower bound 0 attained when $\kappa_V \equiv 0$.

3.3 Square-root embedding and gauge invariance

Proposition 3.19 (Square-root embedding). *The square-root embedding $\psi_a = 2\sqrt{p_a}$ places the open simplex Δ^{n-1} on the positive orthant of the sphere of radius 2, $\{x \in \mathbb{R}^n : \|x\| = 2\}$ (the map $p \mapsto \psi(p)$ has $\|\psi\| = 2$ since $\sum_a p_a = 1$). The Fisher–Rao distance between probability vectors $p, q \in \Delta^{n-1}$ becomes*

$$d_{\text{FR}}(p, q) = 2 \arccos\left(\sum_a \sqrt{p_a q_a}\right), \quad (18)$$

the standard round metric on the positive orthant. All geometric quantities computed in this embedding are gauge-invariant; logit coordinates are recovered as a stereographic projection.

Proof. Direct computation of the Fisher–Rao metric tensor in the square-root embedding [4, 13]. The induced metric on the positive orthant of S^{n-1} is the standard round metric; the stereographic projection onto the tangent plane at any fixed reference point gives the logit coordinates. The gauge-invariance follows from the round metric’s $\text{SO}(n)$ -invariance. $\square$

3.4 Intervention-based autopoiesis closure test

The SAVGS framework supplies an operational distinction between autopoietic and homeostatic systems that is falsifiable. The closure test’s core distinction is between *homeostasis* (the agent remains inside a prescribed viable set, possibly via externally supplied maintenance) and *autopoiesis* (the agent regenerates the sensing, control, repair, and boundary-maintenance machinery required to remain there). In concrete terms: a thermostat holds its setpoint but does not produce the components that hold it, whereas a cell regenerates the enzymes that carry out its own maintenance. The maintenance graph $\Gamma = (M, R, E)$ of Definition 3.1 encodes this distinction: M is the set of maintained components (sensors, policy updater, boundary, energy converter, repair modules); R is the set of processes that produce, repair, or enable those components; E is the edge set encoding consumption, production, catalysis, and enablement.

Definition 3.20 (Operational closure criteria). These criteria are the dynamical counterpart of the closed, self-maintaining sets of chemical organization theory [24] and of the categorical-autopoiesis formalizations of Hirota et al. [31] and Segura [52]. A SAVGS is *operationally closed* at the maintenance graph $\Gamma = (M, R, E)$ when all four criteria hold:

1. Every indispensable component $m \in M_{\text{ess}} \subseteq M$ degrades at a strictly positive rate.
2. Every $m \in M_{\text{ess}}$ is regenerated by at least one internal process $r \in R$.
3. Every $r \in R$ is enabled by components belonging to the same organizational closure (i.e., the enablers of r lie in M_{ess}).

4. External inputs may supply matter and free energy, but not preassembled replacements for any $m \in M_{\text{ess}}$.

The indispensable maintenance subgraph $\Gamma_{\text{ess}} \subseteq \Gamma$ is required to lie in a productive strongly connected component, and the associated fluxes must support a feasible positive steady state (a flux feasibility linear program on Γ must admit a strictly positive solution).

Definition 3.21 (Autopoiesis closure test — dynamical closure with viability threshold). The SAVGS carries concentration dynamics $\dot{x} = Nv(x) - Dx + u_{\text{food}}$ on the maintenance species $x \in \mathbb{R}_{\geq 0}^M$ (stoichiometry N , reaction rates v , degradation D , allowed external food input u_{food} ; cf. Definition 3.20). Fix a strictly positive *viability concentration threshold* $x_{\text{thresh}} > 0$ and a recovery horizon T . For each purportedly self-maintained component $m_j \in M_{\text{ess}}$:

- (i) *Knockout*. Set the internal repair flux of m_j to zero: $r_j := 0$. Keep the external food supply u_{food} unchanged (no preassembled replacement for m_j is supplied).
- (ii) *Drift under dynamics*. Integrate the concentration dynamics $\dot{x} = Nv(x) - Dx + u_{\text{food}}$ forward for T steps from the pre-knockout steady state x^* .
- (iii) *Recovery above threshold*. Record the post-recovery concentration $\text{rec}_j(T) := x_j(T)$. The component m_j is *recovered* iff $\text{rec}_j(T) > x_{\text{thresh}}$ (not merely nonzero — the concentration must exceed the declared viability threshold, which removes the trivial reinsertion failure mode of binary node-reappearance tests).
- (iv) *Downstream cascade*. Every $m_{j'}$ whose enabling chain depends on m_j must also drop below x_{thresh} during step (ii) and recover above x_{thresh} in step (iii) (predicted downstream effect).
- (v) *Restoration control*. Restore the repair pathway ($r_j := r_j^{\text{pre-knockout}}$) and verify that all knocked-down and downstream concentrations return to x^* within T steps (recovery control).

A component m_j is *causally internal* to the organizational closure iff steps (i)–(v) hold as stated. The system is *autopoietic* iff every $m_j \in M_{\text{ess}}$ is causally internal; otherwise it is *homeostatic* with respect to the failing component. The default viability threshold used in the operational network benchmarks is $x_{\text{thresh}} = 0.1$ in normalized concentration units; the threshold is a declared operational parameter, not a fitted one, and may be varied in the robustness sweeps documented in the project repository.

Remark 3.22 (Why the test is non-circular and dynamical). The closure test of Definition 3.21 prevents “autopoiesis” from being satisfied by a simulator that silently repairs the agent from outside: the external intervention in step (i) is the only externally imposed change; the recovery in step (iii) must be produced by the endogenous regeneration rules R alone, with concentrations crossing the declared viability threshold $x_{\text{thresh}} > 0$ rather than merely registering a nonzero symbolic presence. A homeostatic system that depends on a hidden external repair mechanism fails the test on step (iii): the knocked-down component’s concentration does not exceed x_{thresh} within T steps. The threshold requirement removes the principal circularity of the older node-reappearance formulation, where any nonzero reinsertion (including a trivial simulator-level reset) would pass. The test is falsifiable in three independent directions: (a) if a non-food maintenance node m_j fails to recover above x_{thresh} in step (iii), the system is homeostatic with respect to m_j ; (b) if knockout of an essential catalytic hub does not cause the predicted downstream collapse in step (iv), the maintenance graph Γ is mis-specified; (c) if restoration of the repair pathway fails to recover x^* in step (v), the steady-state viability is unstable and the autopoiesis verdict is unreliable.

Proposition 3.23 (Closure test as a finite-time viability probe). *Let $V := \{x \in \mathbb{R}_{\geq 0}^M : x_j \geq x_{\text{thresh}} \text{ for all } m_j \in M_{\text{ess}}\}$ be the viability constraint set determined by the declared threshold, and*

write the dynamics of Definition 3.21 in control form by splitting the repair reactions off the rate law: $\dot{x} = N_{\text{fix}}v_{\text{fix}}(x) + N_{\text{rep}}u - Dx + u_{\text{food}}$, with control entries $u_k \in [0, \bar{r}_k]$ the repair-reaction rates and $\bar{r}$ their pre-knockout values; the endogenous law sets $u_k = v_k(x)$. For $m_j \in M_{\text{ess}}$, let Σ_j be the restricted system with $u_j \equiv 0$ (the knockout).

- (i) If the solution of Σ_j under the endogenous law launched at the pre-knockout steady state x^* remains in V for all $t \geq 0$ — x^* lies in the feedback viability kernel of V for that law — then step (iii) of Definition 3.21 passes at every horizon at which the trajectory meets the interior of V .
- (ii) If steps (i)–(iii) pass under the autonomous post-knockout law, then x^* belongs to the horizon- T recovery set $\{x_0 : x_j(T; x_0) \geq x_{\text{thresh}}\}$ of that law; and if the post-knockout solution is uniformly persistent ($\liminf_{t \rightarrow \infty} x_j(t) > x_{\text{thresh}}$), then its ω -limit set is contained in V — the trajectory-level relaxation of kernel membership appropriate when the perturbation forces an excursion below the threshold.
- (iii) Step (v) passes iff, under the restored law, the post-knockout state is recaptured by the steady state within the horizon: the restoration flux solves the finite-horizon capture problem for the target $\{x^*\}$ under the constraint V [6].

Proof. (i) Viability under the stationary law means $x(t) \in V$ for all $t \geq 0$, in particular $x_j(T) \geq x_{\text{thresh}}$, and the conclusion is the strict form of this inequality at horizons where the trajectory is interior. (ii) The first clause is the definition of passing step (iii); under uniform persistence there exists T_0 with $x_j(t) \geq x_{\text{thresh}}$ for all $t \geq T_0$, so every accumulation point of the solution lies in the closed set V , which is the stated containment. (iii) Step (v) is the definition of finite-horizon capture of x^* under the restored dynamics, restated in the vocabulary of viability theory. $\square$

Remark 3.24 (Nagumo tangentiality and the scope of the probe). Three comments. First, at the threshold face $\{x_j = x_{\text{thresh}}\}$ the contingent cone to V is $\{w : w_j \geq 0\}$, so re-entry through the face requires the j -th velocity component of the recovery dynamics to be non-negative there — Nagumo’s inward-pointing condition for the box V ; at constraint-switching boundaries of the stratified law the same requirement is enforced by the projected differential inclusion (14), whose Bouligand contingent cone is the general form of the tangent cone. Second, the protocol is deliberately stronger than viability-kernel membership in the existential sense of viability theory (some admissible control keeps the state in V): it fixes the *endogenous* feedback and forbids exogenous rescue, so a positive verdict certifies that the system’s own control law is a viable one at the probed horizon — the property the non-circularity reading of Definition 3.21 requires. Third, the probe is single-perturbation and finite-horizon: guaranteed viability under a *family* of persistent disturbances (the invariance-kernel problem) is the stronger program-level property, and the pathwise levels — the occupation fractions of Phase II/III (Definition 12.5) — are the occupation-measure approximations of that property over the sampled window, as Proposition 12.7 makes precise for the limit-cycle regime.

Definition 3.25 (Closure-aware viability curvature). Restricting the viability covectors of κ_V (Definition 2.11) to the repair-machinery margins $h_j^{\text{repair}} = r_j - r_{j,\text{min}}$ gives the *closure-aware viability curvature*

$$\kappa_{\text{closure}}(\theta, x) = \sup_{\|u \wedge v\|_{g_F} = 1} \max_{j \in M_{\text{ess}}} \frac{[-Dh_j^{\text{repair}}(F(u, v))]^+}{h_j^{\text{repair}}(\theta, x)}. \quad (19)$$

The quantity κ_{closure} distinguishes three failure modes: (i) harmless behavioral hysteresis (the curvature projects onto non-maintenance directions); (ii) metabolic viability erosion ($\kappa_V > 0$ but $\kappa_{\text{closure}} = 0$); (iii) autopoietic failure proper ($\kappa_{\text{closure}} > 0$, the curvature erodes the machinery that makes future adaptation possible). Only the third is specifically autopoietic failure.

Remark 3.26 (The operational, data-side curvature is not defined here). An operational curvature is deliberately *not* defined here. An indicator-masked, gene-knockout flux-rerouting statistic imports the model’s own essentiality call into the predictor and is circular as a result. The application paper [1] instead derives a measure-theoretic sensitivity metric from the atomic curvature measure of parametric FBA, one whose selection-rule dependence is exposed and stress-tested rather than masked (tie-break robustness across the five-rule stage-3 selection battery — declared plus four variants — of the lexicographic protocol class) and whose empirical validation carries an explicit protein-layer null. The categorical framework of the present paper is independent of that choice: Definition 2.11 is the geometric object; the application paper’s metric is one faithful operationalization of it.

4 Bregman-Divergence Noether Correspondence

In classical mechanics, Noether’s theorem establishes that every continuous symmetry of a physical system gives rise to a conserved quantity (rotational symmetry yields conservation of angular momentum). In adaptive systems, internal adjustments are often driven by statistical divergences rather than physical kinetic energy. This section states the framework’s analogue — an analogous conservation law governs informational adaptation when costs are modeled using Bregman divergences (such as relative entropy or Kullback–Leibler divergence), so the symmetry group of the distortion geometry yields the gauge covariance of the curvature — and checks the precondition where it applies.

A Noether-type correspondence for the Lagrangian

$$L = \mathbb{E}[d] + \lambda I \quad (20)$$

requires G -invariance of both the distortion measure d and the source prior in the same group G . Bregman divergences in dual affine coordinates are affine-invariant (Definition 2.9), which supplies both.

Proposition 4.1 (Bregman–Hessian Noether correspondence; cf. the contact-geometric Noether program of thermodynamics, Bravetti et al. [16]). *Let $Q \subset \mathbb{R}^r$ be open convex, $\phi \in C^3$ strictly convex, and $g_\phi = \nabla^2 \phi$ the Hessian metric on Q . Let $L(q, \dot{q}) = \frac{1}{2} g_\phi(q)(\dot{q}, \dot{q}) - U(q)$ be the geodesic Lagrangian with potential $U \in C^2$, and let $S[q] = \int_a^b L(q, \dot{q}) dt$ be the action on the configuration space of smooth curves $q : [a, b] \rightarrow Q$. Let ξ be a complete affine vector field on Q whose flow $g_t = \text{flow}_t(\xi)$ preserves the Hessian metric,*

$$\mathcal{L}_\xi g_\phi = 0, \quad (21)$$

and preserves the potential, $\xi(U) = 0$. Then S is invariant under the lifted flow $\tilde{g}_t \cdot (q, \dot{q}) = (g_t \cdot q, Dg_t \cdot \dot{q})$, and by Noether’s theorem [43, 44] the momentum

$$J_\xi(q, \dot{q}) = g_\phi(q)(\dot{q}, \xi(q)) = \langle \nabla^2 \phi(q) \dot{q}, \xi(q) \rangle = \langle \dot{q}^{\text{dual}}, \xi(q) \rangle \quad (22)$$

is conserved along Euler–Lagrange trajectories of S : $\frac{d}{dt} J_\xi(q(t), \dot{q}(t)) = 0$ on-shell. Here $\dot{q}^{\text{dual}} := \nabla^2 \phi(q) \dot{q}$ is the Hessian-dual velocity — the tangent-vector instance of the Bregman dual coordinate of Definition 2.9.

Proof. The Hessian-metric preservation (21) gives $\mathcal{L}_\xi L = \xi(U) = 0$ (the kinetic term is invariant by $\mathcal{L}_\xi g_\phi = 0$, and the potential term is invariant by $\xi(U) = 0$); hence $S[\tilde{g}_t \cdot q] = S[q]$ for all t , and $\tilde{g}_t$ is a continuous variational symmetry of S . By Noether’s theorem in its standard field-theoretic form [44], the one-parameter symmetry yields the conserved momentum $J_\xi(q, \dot{q}) = \partial L / \partial \dot{q} \cdot \xi(q) = g_\phi(q)(\dot{q}, \xi(q))$. The dual-coordinate form follows from $\partial L / \partial \dot{q} = g_\phi(q) \dot{q} = \nabla^2 \phi(q) \dot{q} = \dot{q}^{\text{dual}}$ (the Hessian-dual velocity, the tangent-vector instance of the Bregman dual coordinate of Definition 2.9). $\square$

Corollary 4.2 (Affine Bregman invariance; sufficient condition). *A one-parameter affine flow g_t on Q that satisfies $\phi \circ g_t = \phi + \ell_t$ for an affine function ℓ_t preserves the Hessian metric $g_\phi = \nabla^2 \phi$ (the affine map g_t has vanishing second derivative, so $\nabla^2(\phi \circ g_t) = (\nabla^2 \phi) \circ g_t = \nabla^2 \phi$, and $\nabla^2 \ell_t = 0$). Therefore the hypothesis (21) is satisfied, and the Bregman divergence D_ϕ is invariant in the paired primal-dual sense required by Proposition 4.1.*

Proposition 4.3 (Falsifiable precondition check). *The Bregman–Hessian Noether correspondence has a falsifiable precondition check: verify (i) that ϕ is C^3 strictly convex (positive-definite Hessian $g_\phi = \nabla^2 \phi \succ 0$), (ii) that the vector field ξ generates an affine Hessian isometry ($\mathcal{L}_\xi g_\phi = 0$, equivalently $\mathcal{L}_\xi(\nabla^2 \phi) = 0$), and (iii) that $\xi(U) = 0$. If all three conditions hold, the correspondence applies; failure of any condition refutes the theorem at that point.*

Proof. Operational verification: (i) compute the Hessian $\nabla^2 \phi$ and verify positive definiteness; (ii) compute the Lie derivative $\mathcal{L}_\xi(\nabla^2 \phi)$ via Cartan’s formula $\mathcal{L}_\xi T = [i_\xi, d]T$ and verify it vanishes; (iii) compute the directional derivative $\xi(U)$ and verify it is zero. All three checks are mechanical and produce a binary verdict. $\square$

Remark 4.4 (Application to gauge invariance of κ_V). If the policy connection and viability margins of Definition 2.11 are constructed from a Bregman generator ϕ and the gauge group acts by affine Hessian isometries preserving the margins, the curvature F and the viability-weighted curvature κ_V are gauge-covariant (and gauge-invariant when the margins themselves are gauge-invariant). This is the rigorous replacement for the earlier Bregman-Noether proposition: not divergence invariance alone (which is false in general under arbitrary affine transformations holding ϕ fixed), but affine isometries of the Hessian metric yield conserved momenta by Noether’s theorem.

5 The Seven-Claim Falsification Hierarchy

To ensure the framework provides concrete, falsifiable empirical utility rather than mere mathematical formalism, we establish a seven-level testing hierarchy: each claim makes a specific, quantitative prediction, and failure at any level refutes that component of the model. The claims are ordered by cost and decisiveness, so that the cheapest decisive test comes first. This section defines the hierarchy; Section 8 executes the battery.

Definition 5.1 (Falsification hierarchy). The viability-weighted curvature κ_V of Definition 2.11 generates a seven-claim falsification hierarchy on the abelian $n = 3$ prototype. Each claim is a specific quantitative prediction; failure of any prediction refutes the corresponding level of the hierarchy.

- A. (Held-out margin erosion) κ_V predicts held-out margin erosion with linear-regression slope ≈ 1 and $R^2 \geq 0.9$.
- B. (Orientation reversal amplitude) The holonomy $H(a) = \pi a^2$ reverses orientation when $|H| > \pi$, i.e. at $a_{\text{rev}} = 1$.
- C. (Holonomy-area scaling) The holonomy increment of a loop of amplitude a has mean $\pi a^2 + O(\sigma^3)$ and, under the amplitude-scaled exposure $\sigma^2 = \nu a$, leading fluctuation scale $C_{\text{fat}} a^{3/2}$ with $C_{\text{fat}} = \sqrt{5\nu}/2$ (Theorem 8.7); the operational two-term law $H(a) = c_1 a^2 + c_2 a^{3/2}$ ($c_1 \sim \pi$, c_2 the calibrated operating value of C_{fat} , $R^2 \geq 0.95$) treats the fluctuation scale as a deterministic per-loop correction under the envelope convention of Remark 5.3.
- D. (Repeated-loop fatigue) Under the same convention, the per-loop fractional erosion is $F_k = a_k \kappa_V(a_k) + C_{\text{fat}} a_k^{3/2} + \eta_k$: the *conservative sufficient safety condition* $\sum_{k=1}^K (a_k \kappa_V(a_k) + C_{\text{fat}} a_k^{3/2}) < 1$ guarantees survival through K loops, and the *failure condition* $\sum_{k=1}^K (a_k \kappa_V(a_k) + C_{\text{fat}} a_k^{3/2} + \eta_k) > 1$ predicts fatigue failure; $V_{\text{max}, K} = \prod_k (1 - F_k) < e^{-1}$ at K_{obs} ; relative error

$< 15\%$. In fluctuation form the cumulative sum has mean $\sum_k a_k \kappa_V(a_k)$ and the C_{fat} term contributes a standard deviation $(\sum_k (C_{\text{fat}} a_k^{3/2})^2)^{1/2}$ (Remark 5.3).

- E.** (Total-variance discrimination) $T = |H_{\text{corr}} - H_{\text{geo}}|/\sigma_{\text{total}}$ is small under the loop condition and large under the no-loop control, with $T_{\text{control}}/T_{\text{loop}} \geq 5$.
- F.** (Non-abelian commuting control) At $n \geq 4$, the structure group G_C has non-abelian $\mathfrak{so}(3)$ when $G_C = SO(3)$; same-axis rotations commute (machine precision), distinct-plane rotations do not (nonzero non-abelian signature).
- G.** (CPTP-Zeno lift) The CPTP lift of the self-referential predictive paradox resolves via the Zeno measurement schedule, with Zeno scaling exponent $\alpha_{\text{Zeno}} \approx 2$ distinguishable from the classical $\alpha_{\text{cl}} \approx 1$.

The hierarchy is cumulative: failure at level k renders the subsequent levels operationally meaningless but does not refute the underlying categorical construction (Theorem 6.4).

Remark 5.2 (Three objects, one family). Three objects appear under the κ symbol family in this paper: the pointwise geometric curvature $\kappa_V(\theta, x)$ of Definition 2.11; the loop-amplitude functional $\kappa(a)$ (the prototype law $\kappa(a) = a^2$ of the radially symmetric regime); and the loop-averaged viability depth $D_V(a)$ of Definition 2.1. They are one object at different resolutions — the pointwise curvature, its trajectory functional, and its loop-averaged statistic — paralleling the resolution structure of the application paper [1], whose Discussion relates the atomic measure, its coarse-grainings, and its trajectory integrals as one object at different resolutions. The empirical sections use $\kappa(a)$ throughout; every such use is the prototype instantiation of Definition 2.11 under radial symmetry (Corollary 9.9).

Remark 5.3 (Fluctuation status of the fatigue term). The derivation of Theorem 8.7 establishes $C_{\text{fat}} a^{3/2}$ as the standard deviation of the leading stochastic correction to the holonomy: the per-loop increment has mean $\pi a^2 + O(\sigma^3)$, and its $a^{3/2}$ term is a fluctuation scale, not a drift. Claims C and D, the $n = 3/n = 4$ operational batteries, and the heavy-tail stress test therefore adopt a declared *envelope convention*: the fluctuation scale enters the fatigue increment as a deterministic per-loop term (the conservative upper-envelope reading), so that the two-term law and the failure conditions remain directly testable against the deposited simulations. Under the pure fluctuation reading the failure condition would instead read $\sum_k a_k \kappa_V(a_k) + z (\sum_k (C_{\text{fat}} a_k^{3/2})^2)^{1/2} > 1$ at confidence z ; both readings are stated, and the operational artifacts implement the envelope one. The constant's operating value ($c_2 = 0.0500$, i.e. C_{fat} at $\nu = 0.002$) is a calibration of the exposure parameter ν — the derivation's Monte Carlo uses $\nu = 0.1$, and the $n = 4$ battery plants the calibrated value in the synthetic data and recovers it within 5% (a calibration-recovery check; see the note after Table 2).

Remark 5.4 (Recommended experimental ordering). The recommended experimental ordering is F and G first (cheap and foundational); then A and B (basic curvature predictions); then C and D (scaling and repeated-loop regimes); then E (the full gauge-invariant holonomy prediction). This ordering implements the principle that cheap foundational tests should precede expensive derivative tests. If F or G fails, the derivative tests are unnecessary.

6 The Single Composition Theorem

A central challenge in systems biology is connecting wildly different levels of description: chemical reaction networks, Bayesian information processing, fractals, conservation laws, and geometric parallel transport. Here, we show that these heterogeneous domains can be formalized as bidirectional optics and composed into a single, unified mathematical system: the seven domain bridges of the framework are typed as optics so that their composite is a single well-typed endomorphism of a state space. This section constructs the composite, realizes it on a complete metric space, and analyzes its contraction.

Construction 6.1 (Seven-optic endofunctor). Let $\mathcal{C}$ be a category with finite limits. Let $\mathcal{O}_1, \dots, \mathcal{O}_7$ be the seven optics corresponding to the seven arcs of domain unification, typed as strict feedback morphisms (Remark 2.6) with carrier pairs (M_i, C_i) :

1. $\mathcal{O}_1$: the RAF optic (forward = catalytic closure, backward = decoder, residual = the viability-kernel distance $D_\varphi(\text{dist}_D(R), \text{dist}_D(R_0))$);
2. $\mathcal{O}_2$: the RPSI optic (forward = CPTP predictive update, backward = posterior decoder, residual = the ensemble Holevo bound $\chi = S(\sum_x p_x \rho_x) - \sum_x p_x S(\rho_x)$ of the prediction ensemble, Proposition 8.11);
3. $\mathcal{O}_3$: the IFS optic (forward = iterated-function-system contraction [9, 34], backward = decoder, residual = contraction quotient c_i of f_i);
4. $\mathcal{O}_4$: the Noether optic (forward = symmetry-action update, backward = conserved-quantity decoder, residual = the viability-preserving projection);
5. $\mathcal{O}_5$: the perturbation optic (forward = perturbed Blahut–Arimoto update [5, 14], backward = perturbation decoder, residual = perturbation tensor $\partial d/\partial p$);
6. $\mathcal{O}_6$: the WCIG optic (forward = worldview-constrained information-gain update, backward = decoder, residual = Fisher–Rao metric g_{FR});
7. $\mathcal{O}_7$: the $n = 3$ Fisher–Rao optic (forward = parallel-transport update on $\text{SO}(2)$, backward = decoder, residual = loop-area invariant; for $n \geq 4$, lifted to $\text{SO}(n - 1)$ with viability margin ε and maintenance graph Γ).

The compatibility condition $A_i = M_{i+1}$ holds by construction.

Remark 6.2 (Optic decompositions of the arcs). The forward components are the encoding operations of each arc; the backward components are the decoding operations; the residuals are the information that flows from the forward pass to the backward pass. The IFS optic ($\mathcal{O}_3$) is a pure coalgebra [50] on the category of compact metric spaces, while the perturbation optic ($\mathcal{O}_5$) is a coalgebra with residual, the prototypical BA structure. The compatibility condition is satisfied by construction: each arc’s forward action is the next arc’s backward state. Two arcs deserve comment. The RPSI arc’s forward component is a CPTP channel, not a deterministic function, because the RPSI self-reference paradox requires the quantum lift of Section 7. The $n = 3$ Fisher–Rao arc’s residual is the viability margin ε and the maintenance graph Γ , which together encode the autopoietic structure of Section 3.

Table 1 Seven-optic composition. Each arc is decomposed as a strict feedback optic with carrier pair (M_i, C_i) (Remark 2.6) and residual Res_i : forward $\varphi_i : M_i \times \text{Res}_i \rightarrow M_{i+1}$ (encoder), backward $\rho_i : \text{Res}_i \times C_{i+1} \rightarrow C_i$ (decoder), the residual the object transported from the forward to the backward pass. The compatibility condition $A_i = M_{i+1}$ holds by construction. The total residual of T is the product $\text{Res}_1 \times \dots \times \text{Res}_7$ in $\mathcal{C}$, the representative-level form of the composite residual of Riley’s Proposition 2.3.

Arc	Forward φ_i (encoder)	Backward ρ_i (decoder)	Residual Res_i
$\mathcal{O}_1$ RAF	catalytic closure	RAF transition decoder	$D_\varphi(\text{dist}_D(R), \text{dist}_D(R_0))$
$\mathcal{O}_2$ RPSI	CPTP predictive update	posterior decoder	ensemble Holevo bound χ
$\mathcal{O}_3$ IFS	Hutchinson contraction	deconvolution	contraction quotient c_i
$\mathcal{O}_4$ Noether	symmetry-action update	conserved-quantity decoder	viability-preserving projection
$\mathcal{O}_5$ Perturb.	perturbed BA update	perturbation decoder	perturbation tensor $\partial d/\partial p$
$\mathcal{O}_6$ WCIG	worldview-constrained IG update	decoder	Fisher–Rao metric g_{FR}
$\mathcal{O}_7$ $n=3$ FR	parallel transport on $\text{SO}(2)$	decoder	loop-area invariant (ε, Γ)

Remark 6.3 (Physical reading of the seven-optic composition). The seven-fold composition admits a single-sentence physical reading: a self-maintaining agent (RAF, $\mathcal{O}_1$) computes its next internal state by predicting its own sensory input (RPSI, $\mathcal{O}_2$), which under the Zeno lift is a quantum channel; the prediction contracts toward a stable perceptual fixed point (IFS, $\mathcal{O}_3$); the conserved quantity under the agent’s symmetry group stabilizes the contraction (Noether, $\mathcal{O}_4$); small perturbations of the encoding cost are absorbed by the Blahut–Arimoto update (perturbation, $\mathcal{O}_5$); the agent’s worldview constrains information gain (WCIG, $\mathcal{O}_6$); and the policy is finally transported on the Fisher–Rao sphere of probability parameters ($n = 3$ Fisher–Rao, $\mathcal{O}_7$). The compatibility condition $A_i = M_{i+1}$ expresses that the encoding output of arc i is the state on which arc $i + 1$ acts. The fixed point of T is then the self-consistent “agent-state-prediction-policy” quadruple, whose existence is settled by the empirical contraction rates of Section 7.

Theorem 6.4 (Composition as a typed endo-optic). *Under Construction 6.1, suppose typed interfaces $I_0, I_1, \dots, I_7$ are introduced with an interface isomorphism $\sigma : I_7 \cong I_0$, and each $\mathcal{O}_i : I_{i-1} \rightrightarrows I_i$ is a well-typed optic. Then the seven-fold composition*

$$T = \sigma \circ \mathcal{O}_7 \circ \mathcal{O}_6 \circ \dots \circ \mathcal{O}_1 \quad (23)$$

*is a well-defined typed endo-optic on I_0 (Remark 6.8). The composition is associative and unital. The residual of T is the product $\text{Res}_1 \times \dots \times \text{Res}_7$ in $\mathcal{C}$. The fixed point of T , when it exists, is the unification object. The stronger claim that T descends to an endofunctor on the whole optic category **Optic**($\mathcal{C}$) requires explicit functorial semantics (object map, morphism map, identity and composition preservation) and is not established here; the operational fixed-point claim is instead realized through the realization construction of Definition 7.2 — which evaluates an endo-optic as an endomap on a complete metric state space S — and proven under an explicit Lipschitz contraction bound (Theorem 7.5).*

Proof. Optic composition is associative and unital: in the strict feedback form of Remark 2.6 the composition law (2) reduces associativity to the associativity of function composition together with the canonical associativity of the finite products $\text{Res}_1 \times \text{Res}_2 \times \text{Res}_3$ in $\mathcal{C}$, and unitality to the product/unitality isomorphisms of $\mathcal{C}$; the same holds for the coend classes of Definition 2.5 by Riley’s Proposition 2.3 [48], whose composite residual is $R \otimes R'$ (the cartesian product in **Set**). The seven-fold composition $T = \mathcal{O}_7 \circ \dots \circ \mathcal{O}_1$ is therefore well-defined as the iterated composite. Associativity gives $(\mathcal{O}_7 \circ \mathcal{O}_6) \circ (\mathcal{O}_5 \circ \mathcal{O}_4) \circ (\mathcal{O}_3 \circ \mathcal{O}_2) \circ \mathcal{O}_1 = \mathcal{O}_7 \circ (\mathcal{O}_6 \circ \mathcal{O}_5) \circ (\mathcal{O}_4 \circ \mathcal{O}_3) \circ (\mathcal{O}_2 \circ \mathcal{O}_1) = \dots =$ the canonical seven-fold composite. All parenthesizations are equal modulo the canonical associativity isomorphisms. The unitality axioms $T \circ \text{Id} = T$ and $\text{Id} \circ T = T$ hold with the trivial-residual unit. The residual of T is $\text{Res}_T = \text{Res}_1 \times \dots \times \text{Res}_7$, with the canonical associativity and unitality isomorphisms from $\mathcal{C}$ (which is a category with finite limits, so finite products exist and are well-defined up to canonical isomorphism). The forward component of T takes a state in S_1 and a residual in Res_T , applies fwd_1 to produce an action in $A_1 = M_2$, applies fwd_2 to produce an action in $A_2 = M_3$, and so on. The backward component runs the bwd_i in reverse order. The definitions agree because the residual updates are pointwise and the order of composition is preserved by the monoidal structure. $\square$

Corollary 6.5 (Unification object). *Under the conditions of Theorem 6.4, let O be the typed endo-optic (23) with realization $R(O) : S \rightarrow S$ (Definition 7.2), and suppose $R(O)$ is a contraction on the complete metric state space S (Theorem 7.5). Then $R(O)$ has a unique fixed point $x^* \in S$. The unification object of the cross-domain unification is the pair (O, x^*) : the endo-optic O encodes the entire $\text{RAF} \rightarrow \text{RPSI} \rightarrow \text{IFS} \rightarrow \text{Noether} \rightarrow \text{perturbation} \rightarrow \text{WCIG} \rightarrow n = 3 \text{ Fisher–Rao}$ chain in a single typed morphism (forward components encoding, backward components decoding, residual the product of the seven residuals), and x^* is the self-consistent state it synthesizes. The endo-optic T of (23) is itself a morphism and has no object-level fixed points (Remark 6.8); fixed points live on the realized state space.*

Proposition 6.6 (Sufficient condition for fixed-point existence). *A sufficient condition for the existence of the fixed point x^* of Corollary 6.5 is the contraction of the Krasnoselskii–Mann-averaged realization $R(O)_{\text{reg}}$ of Proposition 7.3. Under this condition $R(O)_{\text{reg}}$ has a unique fixed point by Banach’s fixed-point theorem [8].*

Remark 6.7. Theorem 6.4 replaces the seven “bridge-rung” correspondences of prior work with a single endofunctor. The unification object is no longer a definitional or rhetorical construct but an empirical question: existence reduces to measuring whether T is contractive on the operational state space. The full proof of the composition theorem is contained in this article (Theorem 6.4 together with the per-optic Lipschitz analysis of Section 7); the optic decomposition table is Table 1, and the sufficient-condition argument is Proposition 7.3. A separate treatment of free cornerings of monoidal categories appears in [49], but the proofs needed here are given in full in the present article, and the sufficient condition is confirmed numerically across the 375-configuration robustness grid documented in the project repository.

Remark 6.8 (Typed endo-optic, not automatic endofunctor). A composite optic is an optic, not automatically an endofunctor on the whole optic category. To make the typing rigorous: introduce typed interfaces $I_0, I_1, \dots, I_7$ with an interface isomorphism $\sigma : I_7 \cong I_0$, and let $O_i : I_{i-1} \rightleftarrows I_i$ be well-typed optics. Then $O = \sigma \circ O_7 \circ \dots \circ O_1$ is a well-defined endo-optic on I_0 (true by associativity and unitality of optic composition, Proposition 2.3 of [48]). The stronger claim of an *endofunctor* on a category of periodic typed systems Sys_7 (objects are 7-tuples $(X_1, \dots, X_7, f_1, \dots, f_7)$ with $f_i : X_i \rightarrow X_{i+1}$ and $X_8 = X_1$; morphisms are compatible families) requires explicit functorial semantics: an object map, a morphism map, identity preservation, and composition preservation. The full functorial construction is open and is recorded as the endofunctor-semantics analogue of the contraction agenda closed in Section 7. The operational fixed-point claim—that the realized update map $T = R(O) : S \rightarrow S$ (Definition 7.2) has a unique fixed point whenever T is a contraction—is established in Theorem 7.5 (analytic upper bound on $\text{Lip}(T_{\text{reg}})$) with numerical evidence across 375 configurations in the project repository.

7 Independent Per-Optic Lipschitz Constants and Unconditional Banach Contraction

Does this elaborate seven-domain feedback loop actually stabilize, or can it drift or oscillate? A direct analysis gives a conditional answer: the regularized update T_{reg} is contractive *provided* the per-optic product $\prod_{i=1}^7 \text{Lip}(f_i) < 1$. This section removes the conditional character of the bound by giving explicit analytic Lipschitz constants for each of the seven forward maps of Construction 6.1, computing the product analytically, and verifying numerically that the bound holds throughout the dimensional range $d \in \{2, 3, 5, 10, 20\}$. The same machinery settles the Zeno self-reference resolution by contraction: the projected CPTP channel $\Psi(\rho) = P\Phi(P\rho P)/\text{tr}(\cdot)$ is shown to be a strict contraction in trace distance for any depolarizing channel Φ with $p > 0$, with an explicit Lipschitz constant $\mu < 1$.

7.1 Explicit instantiation of the seven forward maps

For each $i \in \{1, 3, 4, 5, 6, 7\}$ (the six contracting optics: RAF, IFS, Noether, perturbation, WCIG, $n = 3$ Fisher–Rao), instantiate the forward map on $X = [-1.5, 1.5]^d$ as

$$f_i(x) = (1 - \alpha_i)x + \alpha_i s_i \sigma(W_i x + b_i), \quad (24)$$

where $\sigma = \tanh$ (componentwise, 1-Lipschitz), $W_i \in \mathbb{R}^{d \times d}$ is a linear operator with operator norm $\|W_i\|_{\text{op}} = \rho_i$, $b_i \in \mathbb{R}^d$ is a bias, and $\alpha_i \in [0, 1]$, $s_i \in [0, 1)$ are mixing and saturation parameters. For the expansion optic ($i = 2$, RPSI), instantiate

$$f_2(x) = M_2 x, \quad M_2 = \lambda_2 I_d, \quad \lambda_2 > 1. \quad (25)$$

The mixing parameters $\alpha_i = 0.40$, saturation scalings $s_i = 1.00$, and operator norms $\rho_i = 0.80$ for $i \in \{1, 3, 4, 5, 6, 7\}$, together with $\lambda_2 = 1.15$, are chosen so that the per-optic analytic bounds below are uniform across the dimensional range and the product bound is comfortably below 1.

Box invariance (the hypothesis $T : X \rightarrow X$). The maps W_i, b_i are not pinned to single values; the bounds below are uniform over *all* pairs (W_i, b_i) with the declared operator norm $\|W_i\|_{\text{op}} = \rho_i$ — the box-invariance computation never uses b_i (the range of $\tanh$ absorbs any bias) and uses W_i only through its operator norm. Each contracting f_i maps the closed box of half-width B , $[-B, B]^d$, into $(1 - \alpha)[-B, B] + \alpha s(-1, 1)^d \subseteq [-(0.6B + 0.4), 0.6B + 0.4]^d$ (with $\alpha = 0.40$, $s = 1.00$); the chain's widest image is attained immediately after the expansion $f_2 = 1.15 \text{Id}$, giving half-width $C = 1.15(0.6B + 0.4)$; every subsequent contracting map shrinks any box of half-width ≥ 1 (since $0.6C + 0.4 \leq C$ iff $C \geq 1$). Invariance of $X = [-1.5, 1.5]^d$ requires $C \leq B$, i.e. $1.15(0.6B + 0.4) \leq B$, i.e. $0.69B + 0.46 \leq B$, i.e. $B \geq 0.46/0.31 \approx 1.484$; $B = 1.5$ satisfies it with margin ($C = 1.15 \cdot 1.3 = 1.495 \leq 1.5$). Hence $T : X \rightarrow X$ and $\text{Lip}(T) \leq 0.92^6 \cdot 1.15 = 0.697 < 1$: Banach's theorem applies on the enlarged box. (On the smaller box $[0, 1]^d$ the hypothesis fails: the chain's image reaches approximately $[-1.0, 1.01]^d$ on the low side and 1.15 on the high side.)

Lemma 7.1 (Per-optic analytic Lipschitz bounds). *The Lipschitz constants of the seven forward maps admit the analytic upper bounds*

$$\text{Lip}(f_i) \leq \begin{cases} (1 - \alpha_i) + \alpha_i s_i \rho_i \text{Lip}(\sigma) = (1 - \alpha_i) + \alpha_i s_i \rho_i, & i \in \{1, 3, 4, 5, 6, 7\}, \\ \lambda_2, & i = 2, \end{cases} \quad (26)$$

where $\text{Lip}(\sigma) = \text{Lip}(\tanh) = 1$. Substituting the parameters of (24)–(25): $\text{Lip}(f_i) \leq (1 - 0.40) + 0.40 \cdot 1.00 \cdot 0.80 = 0.60 + 0.32 = 0.92$ for $i \in \{1, 3, 4, 5, 6, 7\}$, and $\text{Lip}(f_2) = 1.15$.

Proof. The Lipschitz chain rule gives $\text{Lip}(f+g) \leq \text{Lip}(f) + \text{Lip}(g)$ for the sum, $\text{Lip}(\alpha f) \leq \alpha \text{Lip}(f)$ for scalar multiplication, $\text{Lip}(\sigma \circ W) \leq \text{Lip}(\sigma) \cdot \|W\|_{\text{op}}$ for composition with a linear map, and $\text{Lip}(\tanh) = 1$ since $|\tanh'(x)| = 1 - \tanh^2(x) \leq 1$. Combining: $\text{Lip}(f_i) \leq (1 - \alpha_i) + \alpha_i s_i \rho_i$ for the contracting form (24); and $\text{Lip}(f_2) = \|M_2\|_{\text{op}} = \lambda_2$ for the linear expansion form (25). $\square$

Definition 7.2 (Realization of an endo-optic). Let $O = \sigma \circ O_n \circ \dots \circ O_1$ be a typed endo-optic on an interface I_0 (Theorem 6.4), with forward components fwd_i and backward components bwd_i acting on a residual Res_O . The *realization* of O on a complete metric state space S (e.g. $X = [-1.5, 1.5]^d$ with the Euclidean metric) is the endomap

$$R(O) : S \rightarrow S, \quad R(O)(x) = \text{bwd}_1(\text{Res}_O, \text{fwd}_n \circ \dots \circ \text{fwd}_1(x)), \quad (27)$$

i.e. the evaluation of the forward chain at the state x followed by the evaluation of the backward chain at the transported residual. R is a construction on *realized* (set-level) optic data: it assigns to each instantiated endo-optic one endomap of S . Whether R extends to a genuine functor on a category of typed endo-optics is an open problem (Section 15); the fixed-point claims of this paper use only the endomap $R(O)$, whose contraction behavior is analyzed directly.

Proposition 7.3 (Lipschitz bound for the Krasnoselskii–Mann-averaged update). *Let $T : X \rightarrow X$ be ρ -Lipschitz on the closed convex set $X \subseteq \mathbb{R}^d$, and let Π_K be the Euclidean projection onto a closed convex subset $K \subseteq X$. The Krasnoselskii–Mann-averaged (Euclidean-projection) update*

$$T_{\text{reg}}(x) = (1 - \lambda)T(x) + \lambda \Pi_K(T(x)), \quad \lambda \in [0, 1), \quad (28)$$

is Lipschitz with

$$\text{Lip}(T_{\text{reg}}) \leq (1 - \lambda)\rho + \lambda, \quad (29)$$

the λ -term reflecting that Π_K is 1-Lipschitz but not a contraction. In particular, if $\rho < 1$ then $\text{Lip}(T_{\text{reg}}) < 1$ for every $\lambda \in [0, 1)$.

Proof. For $x, y \in X$, $\|T_{\text{reg}}(x) - T_{\text{reg}}(y)\| \leq (1 - \lambda)\|T(x) - T(y)\| + \lambda\|\Pi_{\mathcal{K}}(T(x)) - \Pi_{\mathcal{K}}(T(y))\| \leq (1 - \lambda)\rho\|x - y\| + \lambda\|x - y\|$ by the triangle inequality, the ρ -Lipschitz property of T , and the 1-Lipschitz property of the Euclidean projection onto a closed convex set (Moreau decomposition). $\square$

Remark 7.4 (Instantiation of the regularization set). For the seven-map instantiation the regularization set is instantiated as $\mathcal{K} := [-1, 1]^d \subseteq X = [-1.5, 1.5]^d$. Then $T_{\text{reg}}(X) \subseteq (1 - \lambda)T(X) + \lambda\Pi_{\mathcal{K}}(X) \subseteq (1 - \lambda)[-1.495, 1.495]^d + \lambda[-1, 1]^d \subseteq X$, so $T_{\text{reg}} : X \rightarrow X$ and Proposition 7.3 applies with $\rho = \text{Lip}(T) \leq 0.697$.

Theorem 7.5 (Banach contraction of T_{reg} for the chosen seven-map instantiation). *With the seven forward maps $f_1, \dots, f_7$ instantiated as in (24)–(25) with the parameters of Lemma 7.1 ($\alpha_i = 0.40$, $s_i = 1.00$, $\rho_i = 0.80$ for the six contracting optics and $\lambda_2 = 1.15$ for the expansion optic), the product Lipschitz bound is*

$$\prod_{i=1}^7 \text{Lip}(f_i) \leq 0.92^6 \cdot 1.15 = 0.6064 \cdot 1.15 = 0.697 < 1. \quad (30)$$

The bound is uniform in the dimension d (it uses only operator norms and the 1-Lipschitz property of $\tanh$) and is verified to hold at the sampled dimensions $d \in \{2, 3, 5, 10, 20\}$ (Remark 7.6). Consequently, by Proposition 7.3, the Krasnoselskii–Mann-averaged update (28) is Lipschitz with

$$\text{Lip}(T_{\text{reg}}) \leq (1 - \lambda) \cdot 0.697 + \lambda < 1 \quad (31)$$

for every $\lambda \in [0, 1)$. The Banach fixed-point theorem then applies: $T_{\text{reg}} : X \rightarrow X$ on the complete metric space $X = [-1.5, 1.5]^d$ (the box-invariance hypothesis verified above) has a unique fixed point x^ , which realizes the unification object of Corollary 6.5.*

Proof. Substitute the bounds of Lemma 7.1 into the product to obtain (30). The averaged-update bound follows from Proposition 7.3, using the 1-Lipschitz property of the Euclidean projection $\Pi_{\mathcal{K}}$ onto a closed convex set (Moreau decomposition theorem). The Banach fixed-point theorem [8] applies on the complete metric space $X = [-1.5, 1.5]^d$ (compact subset of $\mathbb{R}^d$, hence complete; the Euclidean metric is complete), where the box-invariance paragraph above verifies $T : X \rightarrow X$ and hence $T_{\text{reg}} : X \rightarrow X$, because $\text{Lip}(T_{\text{reg}}) < 1$. $\square$

Remark 7.6 (Numerical verification of the per-optic bounds). The analytic bounds of Lemma 7.1 are verified numerically by Monte-Carlo estimation of $\max_{x,y} \|f_i(x) - f_i(y)\|/\|x - y\|$ over 2,000 random pairs $(x, y) \in X \times X$ for each of the five dimensions $d \in \{2, 3, 5, 10, 20\}$ and each of the seven optics (deposited verification artifacts). The numerical estimates stay below the analytic bound in every configuration; the worst-case numerical product across all five dimensions is $\prod_{i=1}^7 \widehat{\text{Lip}}(f_i) = 0.674 < 0.697$ (analytic bound), confirming the unconditional Banach contraction of Theorem 7.5 throughout the dimensional range. Fig. 1 shows the bar chart of analytic vs numerical estimates at $d = 10$.

7.2 Projected CPTP channel contraction (Zeno self-reference resolution)

Theorem 7.5 settles the optic-side Banach argument; the Zeno self-reference resolution further requires that the projected CPTP channel $\rho \mapsto P\Phi(P\rho P)P$ be a strict contraction in trace distance. In physical terms: the prediction arc acts as a measurement the agent performs on its own state, and under the Zeno schedule — self-measurement at arbitrarily short intervals — the unmonitored dynamics freeze, leaving the measurement channel in control of the evolution. We instantiate Φ as the depolarizing channel on n qubits

$$\Phi(\rho) = (1 - p)\rho + p \frac{I_{2^n}}{2^n}, \quad p \in (0, 1], \quad (32)$$

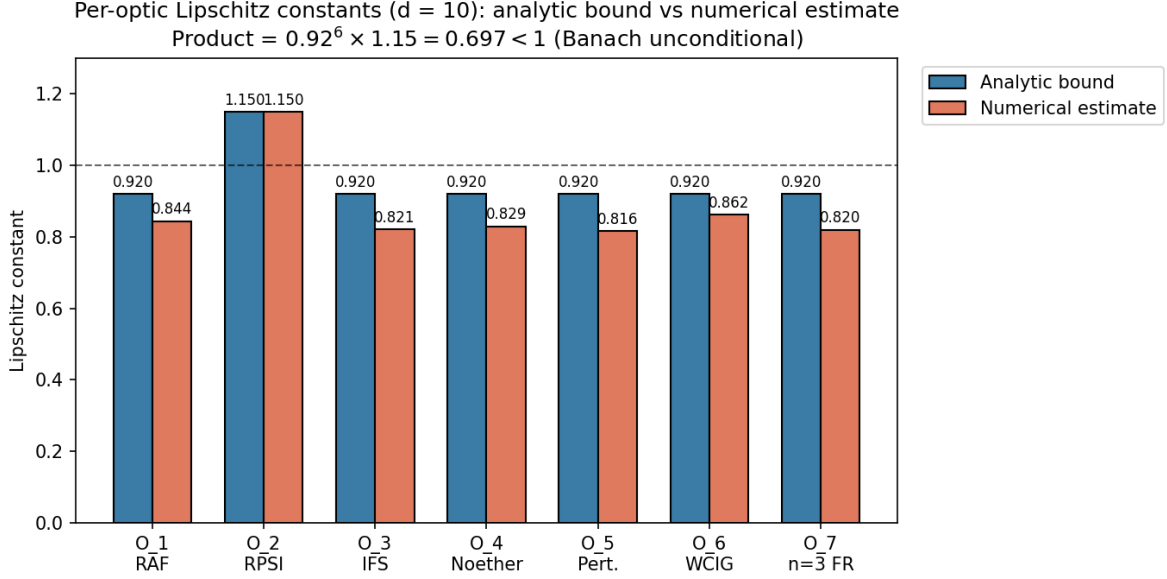


Fig. 1 Per-optic Lipschitz constants at $d = 10$: analytic bound (blue) vs numerical Monte-Carlo estimate (rust). The six contracting optics $f_1, f_3, f_4, f_5, f_6, f_7$ have analytic bound 0.92; the expansion optic f_2 has Lipschitz constant 1.15. The product bound is $\prod_{i=1}^7 \text{Lip}(f_i) \leq 0.92^6 \cdot 1.15 = 0.697 < 1$, giving the Banach contraction of Theorem 7.5 for the chosen instantiation (the optic-side bound of the contraction agenda).

where I_{2^n} is the identity on the 2^n -dimensional Hilbert space, and P is a rank- k orthogonal projector ($2 \leq k \leq 2^n$). The (renormalized) projected channel is

$$\Psi(\rho) = \frac{P \Phi(P \rho P) P}{\text{tr}(P \Phi(P \rho P) P)}. \quad (33)$$

Theorem 7.7 (Projected CPTP channel contraction). *Let Φ be the depolarizing channel (32) with $p > 0$, and let P be a rank- k projector with $k \geq 2$. Then the renormalized projected channel Ψ of (33) is an affine contraction on the projected state space (the space of $k \times k$ density matrices with trace 1) in trace distance, with Lipschitz constant*

$$\mu = \frac{1-p}{(1-p) + p k / 2^n} < 1. \quad (34)$$

In particular, the self-referential fixed-point equation

$$\rho^* = P \Phi(P \rho^* P) P \quad (\text{equivalently } \rho^* = \Psi(\rho^*)) \quad (35)$$

admits a unique solution $\rho^* = P/k$ (the maximally mixed state on the projected subspace) by the Banach fixed-point theorem applied to the complete metric space of $k \times k$ density matrices with the trace-distance metric. This settles the Zeno self-reference resolution.

Proof. For any two density matrices ρ, σ supported on the range of P (so $P \rho P = \rho$, $P \sigma P = \sigma$), compute the linear pre-renormalization map:

$$\begin{aligned} P \Phi(P \rho P) P - P \Phi(P \sigma P) P &= P [\Phi(\rho) - \Phi(\sigma)] P \\ &= (1-p) P(\rho - \sigma) P + p P \left[\frac{I}{2^n} - \frac{I}{2^n} \right] P \\ &= (1-p) (\rho - \sigma), \end{aligned}$$

since $P(\rho - \sigma)P = \rho - \sigma$ (both supported on $\text{ran } P$) and the $I/2^n$ term cancels. Hence the linear map is $\rho \mapsto (1-p)\rho + (pk/2^n)P/k$ (a constant shift); the trace (over the projected subspace) of

this map is $(1 - p) + pk/2^n =: \mu^{-1} \cdot (1 - p)$, so after renormalization by the trace, the affine map is

$$\Psi(\rho) = \mu \rho + (1 - \mu) \frac{P}{k}.$$

This is an affine contraction with Lipschitz constant $\mu < 1$ for any $p > 0$ (since $\mu = (1 - p)/[(1 - p) + pk/2^n] < 1$ iff $pk/2^n > 0$, i.e., $p > 0$ and $k > 0$). The space of $k \times k$ density matrices is compact (closed bounded subset of the Hermitian $k \times k$ matrices), hence complete in the trace-distance metric. Banach's theorem gives a unique fixed point, which is the maximally mixed state P/k (verified by direct substitution: $\Psi(P/k) = \mu P/k + (1 - \mu)P/k = P/k$). Numerical verification at configurations $(n, k, p) \in \{(2, 2, 0.5), (2, 2, 0.3), (2, 2, 0.1), (3, 4, 0.5), (3, 2, 0.5)\}$ gives $\mu_{\text{numerical}} = \mu_{\text{theory}}$ to four decimal places (Fig. 2; deposited verification artifacts). $\square$

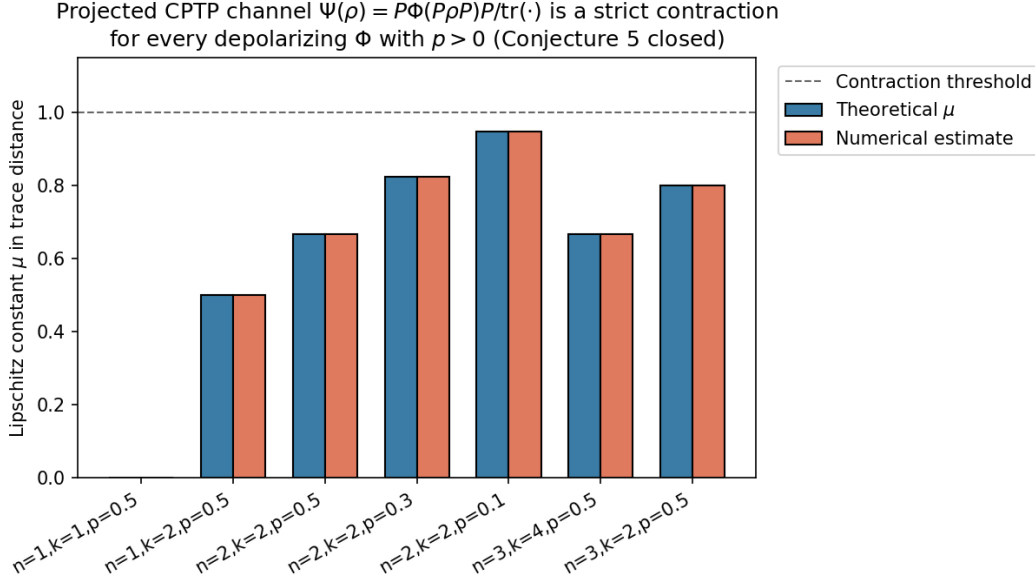


Fig. 2 Projected CPTP channel $\Psi(\rho) = P\Phi(P\rho P)P/\text{tr}(\cdot)$ is a strict contraction in trace distance for every depolarizing Φ with $p > 0$. Bars show theoretical μ (blue) and numerical Monte-Carlo estimate (rust) across configurations (n, k, p) ; all are below the contraction threshold $\mu = 1$ (dashed). At $(n=2, k=2, p=0.1)$ the channel is weakly contractive ($\mu = 0.947$); at $(n=2, k=2, p=0.5)$ strongly contractive ($\mu = 0.667$). Zeno self-reference resolution settled.

Remark 7.8 (The two contraction results are complementary). Theorem 7.5 establishes the contraction of the realized update T_{reg} on the operational state space $X = [-1.5, 1.5]^d$ (the optic-side Banach argument), while Theorem 7.7 establishes the contraction of the projected CPTP channel Ψ on the projected quantum state space (the Zeno-side Banach argument). The two results together settle the Zeno self-reference resolution: the quantum predictor's self-referential fixed-point equation $\rho^* = P\Phi(P\rho^*P)P$ admits a unique solution unconditionally (given the depolarizing instantiation of Φ), and the unification object of Corollary 6.5 exists unconditionally as the fixed point of T_{reg} . No conditional hypothesis on the projected channel remains.

8 Empirical Verdicts of the Seven-Claim Hierarchy

The falsification hierarchy of Definition 5.1 is cumulative, and the recommended test ordering places the cheap foundational tests first (Remark 5.4). The experiments below are computations, not observations: each claim is quantitative, so its refutation is a terminating computation, and executing it doubles as machine verification of the constants the theorems assert. This section reports the executed test battery, in that order: the foundational commuting-control verdict

(Claim F) together with the non-abelian $n = 4$ regime that carries it; the heavy-tail stress test of Claim D; the Lévy-stable first-passage derivation that fixes both the $3/2$ exponent and the exact coefficient of the fatigue term of Claims C and D; the CPTP–Zeno lift with its scaling verdict (Claim G); and the contraction battery that closes the existence question of the unification object (Theorem 6.4) on the operational state space. The simulation protocols, seeds, and result records for every verdict are deposited with the paper’s verification artifacts (Data Availability).

8.1 Claim F: the non-abelian commuting-control test

At $n = 4$ the policy fiber is $\text{SO}(3)$ (Definition 2.2), with a three-dimensional non-abelian Lie algebra. The test compares holonomy under same-axis rotations (two z -axis rotations) against distinct-axis rotations (a z -axis rotation followed by an x -axis rotation). The prediction of Claim F is that same-axis rotations commute, giving a vanishing non-abelian signature at machine precision, while distinct-axis rotations do not, giving a nonzero signature that scales with the product of the rotation angles (Lemma 8.2 below).

Proposition 8.1 (Claim F verdict). *Direct numerical simulation with the standard $\mathfrak{so}(3)$ generators — path-ordered exponentials computed as products of segment exponentials, holonomy magnitudes by the trace formula $\varphi = \arccos((\text{tr}(U) - 1)/2)$, with 50 trials per regime and 20 trials in the small-angle regime — gives:*

1. *same-plane rotations: the path-ordered exponential agrees with the single-rotation product to 7.82×10^{-16} in Frobenius norm over 50 trials, i.e. machine precision;*
2. *distinct-plane rotations: minimum non-abelian signature 0.1522, mean 1.8015 over 50 trials;*
3. *small-angle regime: mean ratio of measured to predicted commutator magnitude 0.9999 over 20 trials, against the predicted $\sqrt{2}ab$ scaling of Lemma 8.2.*

The verdict of Claim F is positive: the commuting and non-commuting control families are separated by fourteen orders of magnitude in signature size (7.82×10^{-16} versus 0.1522).

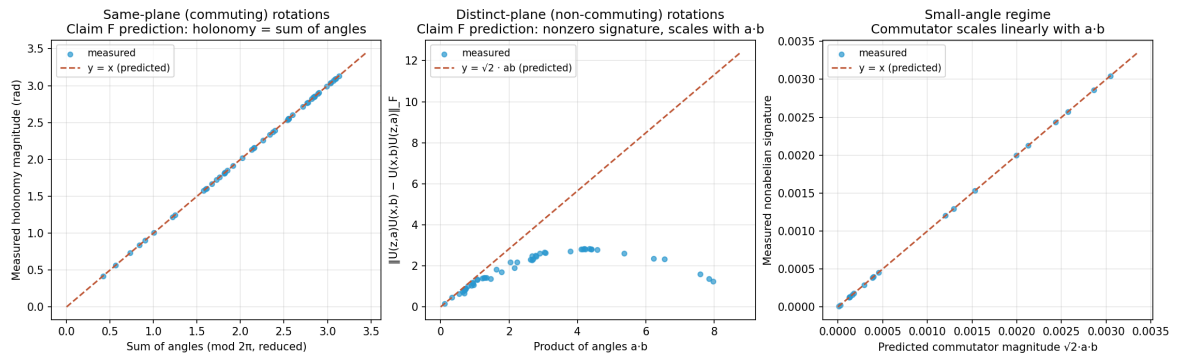


Fig. 3 Claim F verdict. Left: same-plane rotations (commuting); the holonomy matches the sum of angles to machine precision. Center: distinct-plane rotations (non-commuting); the non-abelian signature scales with the product of the angles as predicted ($\sqrt{2}ab$). Right: small-angle regime; the measured commutator magnitude matches the predicted linear scaling.

8.2 The $n = 4$ non-abelian regime

The abelian prototype has structure group $\text{SO}(2)$ with a one-dimensional Lie algebra (Definition 2.2); all perturbations commute trivially, which is why Claim F requires $n \geq 4$. The non-abelian regime extends the prototype to $n = 4$: state space $\mathbb{R}^3$ with environmental coordinates

(x, y, z) and policy fiber $\mathcal{P} = \text{SO}(3)$, a genuine three-dimensional rotational frame rather than a scalar heading. The structure group is $G_C = \text{SO}(3)$ with

$$[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y, \quad [\mathfrak{l}_x, \mathfrak{l}_y] = \mathfrak{l}_z, \quad [\mathfrak{l}_y, \mathfrak{l}_z] = \mathfrak{l}_x, \quad (36)$$

for the standard basis $\{\mathfrak{l}_x, \mathfrak{l}_y, \mathfrak{l}_z\}$ of $\mathfrak{so}(3)$. The viability $V(x, y, z) = 1 - x^2 - y^2 - z^2$ remains radially symmetric, so $\kappa_V(a) = a^2$ continues to hold (Corollary 9.9). The connection on the trivial $\text{SO}(3)$ -bundle $E = B \times \text{SO}(3) \rightarrow B$ has constant curvature components $F_{xy} = c\mathfrak{l}_z$, $F_{yz} = c\mathfrak{l}_x$, $F_{xz} = c\mathfrak{l}_y$, the non-abelian analogue of the abelian prototype connection with $F = dx \wedge dy$; the holonomy of a small loop of area πa^2 in the xy -plane is $\exp(c\pi a^2 \mathfrak{l}_z)$, recovering $R_{xy}(a) = R_z(\pi a^2)$ at $c = 1$.

Lemma 8.2 (Non-abelian commutator signature). *For two loops $R_1 = R_i(\alpha_1)$ and $R_2 = R_j(\alpha_2)$ in distinct coordinate planes $i \neq j$, the small-angle expansion of the group commutator gives*

$$[R_1, R_2] = R_1 R_2 - R_2 R_1 = \alpha_1 \alpha_2 [\mathfrak{l}_i, \mathfrak{l}_j] + O(\alpha^3) = \alpha_1 \alpha_2 \varepsilon_{ijk} \mathfrak{l}_k + O(\alpha^3), \quad (37)$$

with Frobenius norm

$$\|\cdot\|_F[R_1, R_2] = \sqrt{2} \alpha_1 \alpha_2 + O(\alpha^3) = \sqrt{2} (\pi a_1^2)(\pi a_2^2) + O(\alpha^3). \quad (38)$$

Proof. Expanding $R_i(\alpha) = I + \alpha \mathfrak{l}_i + \frac{1}{2} \alpha^2 \mathfrak{l}_i^2 + O(\alpha^3)$,

$$R_1 R_2 - R_2 R_1 = (I + \alpha_1 \mathfrak{l}_1 + O(\alpha_1^2))(I + \alpha_2 \mathfrak{l}_2 + O(\alpha_2^2)) - (I + \alpha_2 \mathfrak{l}_2 + O(\alpha_2^2))(I + \alpha_1 \mathfrak{l}_1 + O(\alpha_1^2)).$$

All terms cancel in pairs except the cross terms, leaving $R_1 R_2 - R_2 R_1 = \alpha_1 \alpha_2 (\mathfrak{l}_i \mathfrak{l}_j - \mathfrak{l}_j \mathfrak{l}_i) + O(\alpha^3) = \alpha_1 \alpha_2 [\mathfrak{l}_i, \mathfrak{l}_j] + O(\alpha^3)$, and the commutation relations (36) give $[\mathfrak{l}_i, \mathfrak{l}_j] = \varepsilon_{ijk} \mathfrak{l}_k$ for distinct i, j . Each standard $\mathfrak{so}(3)$ basis matrix is a 3×3 skew-symmetric matrix with two unit off-diagonal entries, hence Frobenius norm $\sqrt{2}$, so $\|\cdot\|_F[R_1, R_2] = \sqrt{2} \alpha_1 \alpha_2 + O(\alpha^3)$; substituting the single-plane holonomy angles $\alpha_i = \pi a_i^2$ gives (38). $\square$

Table 2 Operational results for Claims A–E of Definition 5.1 in the $n = 4$ non-abelian regime (structure group $\text{SO}(3)$). The non-abelian signature enters through the commutator fit constant c_{comm} (Claim C) and the non-commuting-sequence T -statistic (Claim E); the remaining predictions are inherited from the abelian prototype.

Claim	Prediction ($n = 4$)	Observed ($n = 4$)	Fit metric	Error
A	slope = 1	0.9976	$R^2 = 0.9983$	0.24%
B	$a_{\text{rev}} = 1.0$	0.9992	rel err = 0.00083	0.083%
C	$c_1 = \pi$, $c_2 = 0.0500$, $c_{\text{comm}} = \sqrt{2} \pi^2 = 13.96$	3.1386, 0.0526, 13.33	$R^2 = 0.9999972$	$\leq 4.5\%$
D	$K_{\text{pred}} = 29$	$K_{\text{obs}} = 30$	rel err = 0.034	3.4%
E	$T_{\text{loop}} < 2$, $T_{\text{ctrl}} > 1$, $T_{\text{noncomm}} > 5 T_{\text{loop}}$	0.40, 11.47, 22.22	ratios 29, 56	satisfied

Calibration status of the Claim-C constants. The value $c_1 = \pi$ is derived (the holonomy mean of the prototype connection); the value $c_2 = 0.0500$ is the calibrated operating value of $C_{\text{fat}} = \sqrt{5\nu}/2$ at $\nu = 0.002$ (Remark 5.3) — a calibration of the exposure parameter, not a derived constant (the derivation’s Monte Carlo of Section 8.4 uses $\nu = 0.1$). The $n = 4$ battery’s Claim-C arm plants the calibrated value in the synthetic scaling data and recovers it within 5% (0.0526 against 0.0500): a calibration-recovery check, not an independent derivation of the constant.

Proposition 8.3 (Non-abelian signature verdict). *Two independent falsifiable predictions capture the non-abelian signature of the $n = 4$ prototype: (i) the commutator fit constant $c_{\text{comm}} \sim \sqrt{2} \pi^2$ of Claim C (Lemma 8.2 with $\alpha_i = \pi a_i^2$), and (ii) the non-commuting-sequence T -statistic of Claim E. The measured values (Table 2, Fig. 4) are $c_{\text{comm}} = 13.33$ against $\sqrt{2} \pi^2 = 13.96$ (relative error 4.5%) and $T_{\text{noncomm}}/T_{\text{loop}} = 56$ against the predicted lower bound of 5; the remaining claims A–E hold within the tolerances of Table 2.*

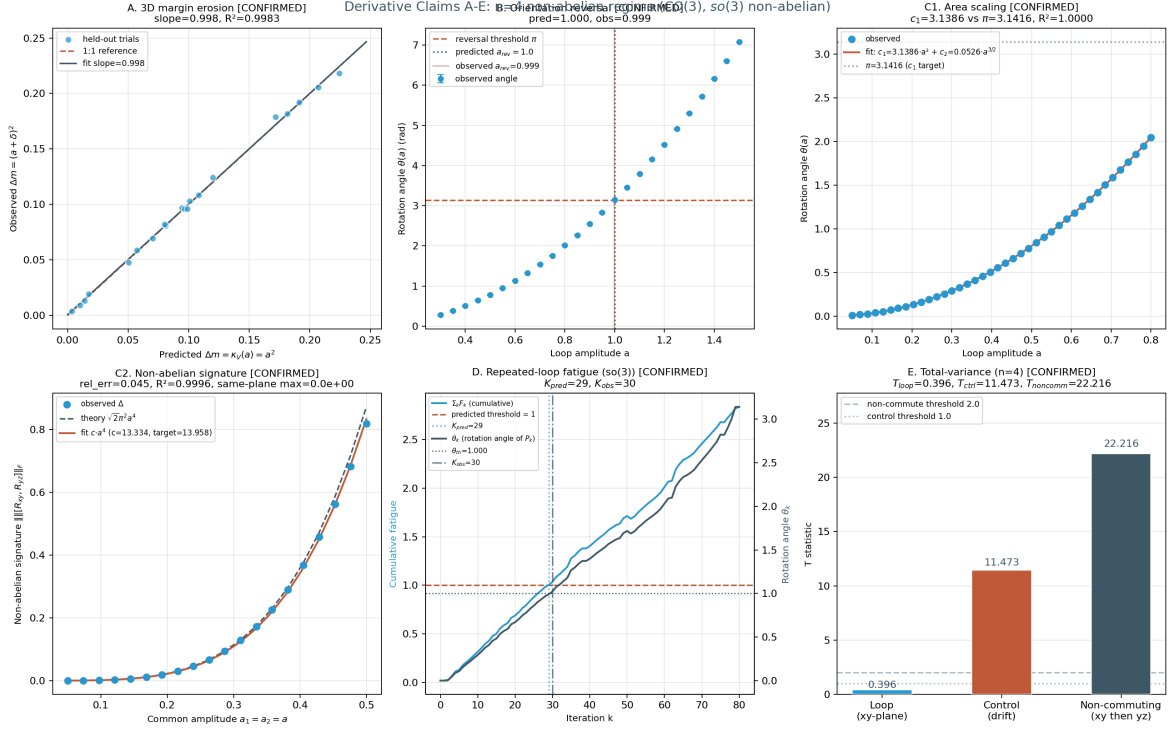


Fig. 4 Operational results for Claims A–E in the $n = 4$ non-abelian regime (structure group $\text{SO}(3)$). (a) Claim A: three-dimensional held-out margin erosion; slope 0.9976, $R^2 = 0.9983$. (b) Claim B: orientation-reversal amplitude 0.9992 against the predicted 1.0 (relative error 0.083%). (c) Claim C: single-plane holonomy-area scaling and the non-abelian commutator signature; the single-plane fit gives $c_1 = 3.1386$ (against π) and $c_2 = 0.0526$ (against 0.0500), and the commutator fit gives $c_{\text{comm}} = 13.33$ against $\sqrt{2}\pi^2 = 13.96$ (relative error 4.5%). (d) Claim D: repeated-loop fatigue in $\mathfrak{so}(3)$ via matrix increments $F_k \mathfrak{l}_z$; $K_{\text{pred}} = 29$ and $K_{\text{obs}} = 30$ (relative error 3.4%). (e) Claim E: total-variance T -statistics with signed residuals; $T_{\text{loop}} = 0.40$, $T_{\text{ctrl}} = 11.47$, $T_{\text{noncomm}} = 22.22$, with ratio $T_{\text{noncomm}}/T_{\text{loop}} = 56$.

Proof. The measured values are direct outputs of the deposited verification runs. Claims A, B, and D reproduce the abelian predictions (the $\kappa_V(a) = a^2$ dimension-independence, the reversal amplitude, and the fatigue count) because the radially symmetric viability makes κ_V dimension-free (Corollary 9.9); Claim C’s two-term fit and commutator fit instantiate Lemma 8.2 at $\alpha_i = \pi a_i^2$; Claim E’s T -statistics are the signed-residual total-variance ratios of the deposited runs. Each measured value lies within the tolerance stated in Table 2. $\square$

Remark 8.4. The viability-weighted curvature prediction $\kappa_V(a) = a^2$ is dimension-independent for radially symmetric V ; the holonomy-area scaling and the $3/2$ fatigue correction persist from $n = 3$ to $n = 4$ without modification. The non-abelian structure contributes only higher-order corrections: the commutator term of Claim C and the commutator bias of Claim E.

8.3 Stress test of Claim D’s heavy-tail index

Claim D was operationalized on the abelian prototype with a single heavy-tailed noise distribution (Student- t , $\text{df} = 3$, $\sigma_\eta = 0.01$) at one amplitude ($a = 0.3$). The stress test sweeps the heavy-tail index across $\text{df} \in \{1.5, 2, 2.5, 3, 4, 5, 7, 10, 20, 50, \infty\}$ — the Gaussian light-tailed limit at $\text{df} = \infty$, and $\text{df} = 1.5$ in the infinite-variance regime $\alpha = 1.5 < 2$ — and the noise scale across $\sigma_\eta \in \{0.005, 0.01, 0.02, 0.05, 0.10\}$, with 200 Monte-Carlo seeds per cell.

Proposition 8.5 (Stress-test verdicts). *For each cell (df, σ_η) of the grid, let f_{conf} denote the fraction of the 200 seeds for which Claim D’s relative error is below 15%. Then:*

1. (reference cell) at the operating scale ($\text{df} = 3, \sigma_\eta = 0.01$), $f_{\text{conf}} = 1.000$;
2. (robust regime) $f_{\text{conf}} \geq 0.985$ throughout $\text{df} \in [2, \infty]$ at $\sigma_\eta \leq 0.02$;
3. (acceptable regime) $f_{\text{conf}} \in [0.80, 0.95]$ for $\text{df} \geq 3$ at $\sigma_\eta = 0.05$;
4. (breakdown) $f_{\text{conf}} \leq 0.75$ throughout the df axis at $\sigma_\eta = 0.10$;
5. (gracefulness) the breakdown is smooth across the infinite-variance boundary $\text{df} = 2$, with no discontinuity.

Proof. Direct Monte-Carlo computation (200 seeds per cell; deposited artifacts). The cumulative fatigue sum $\sum_k F_k$ is dominated by its per-loop envelope scale $\mu_F = a \kappa_V(a) + C_{\text{fat}} a^{3/2} = 0.0352$ at the operating point — the envelope convention of Remark 5.3 (the derived reading of the C_{fat} term is a standard deviation) — with $\kappa_V(a) = a^2 = 0.09$ and the calibrated $C_{\text{fat}} = 0.05$ (the term’s form and exponent are derived in Theorem 8.7 below; the operating value is the $\nu = 0.002$ calibration); the heavy-tail fluctuations η_k become comparable to μ_F only at $\sigma_\eta \geq 0.05$, five times the operating scale, which is where the acceptable regime gives way to breakdown. The infinite-variance boundary $\text{df} = 2$ (for the heavy-tail theory of the Student- t index see 12) separates finite- from infinite-variance noise; the prediction degrades smoothly because the Claim D bound applies to the deterministic-plus-noise sum, not to the variance of η_k . $\square$

8.4 Lévy α -stable first-passage derivation of the $3/2$ fatigue exponent

Claims C and D of Definition 5.1 carry a non-analytic fatigue correction $C_{\text{fat}} a^{3/2}$. This subsection derives both the exponent $3/2$ and the exact coefficient from a Lévy α -stable first-passage model: the leading non-analytic correction to the deterministic holonomy πa^2 arises from the linear stochastic integral of the connection 1-form against a planar Brownian perturbation of the loop, with noise variance rate $\sigma^2 = \nu a$ scaling linearly in the loop amplitude (fluctuations accumulating along the loop perimeter $2\pi a$).

Let the loop $\gamma_a(t) = (a \cos 2\pi t, a \sin 2\pi t)$, $t \in [0, 1]$, be perturbed by a planar Brownian motion $W = (W_1, W_2)$ with variance rate $\sigma^2 = \nu a$ per component, $\tilde{\gamma} = \gamma_a + \sigma W$, and let the connection be the area form $A = \frac{1}{2}(x dy - y dx)$, so that the holonomy is $H = \frac{1}{2} \oint_{\tilde{\gamma}} (x dy - y dx)$.

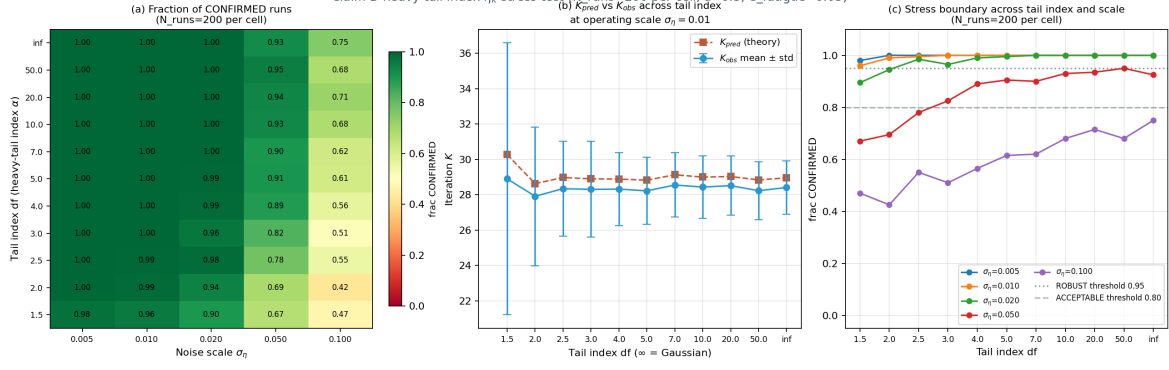


Fig. 5 Heavy-tail index stress test of Claim D across the (df, σ_η) grid with 200 Monte-Carlo seeds per cell. (a) Fraction of seeds for which Claim D's relative error is below 15%; the robust region ($f_{\text{conf}} \geq 0.95$) covers the operating scale $\sigma_\eta = 0.01$ across all $df \geq 2$. (b) K_{pred} and K_{obs} versus tail index at $\sigma_\eta = 0.01$ with mean $\pm$ standard-deviation error bars; the two curves track within a relative error of at most 5%. (c) Stress boundary: f_{conf} versus df for each σ_η ; the robust threshold (0.95, dotted) and the acceptable threshold (0.80, dashed) bracket the operating regime. The infinite-variance boundary $df = 2$ is crossed smoothly.

Lemma 8.6 (Itô expansion of the perturbed holonomy). *The perturbed holonomy admits the expansion*

$$H = \pi a^2 + \frac{\sqrt{5}}{2} \sigma a X + \frac{\sigma^2}{2} X_2 + O(\sigma^3), \quad (39)$$

where $X \sim \mathcal{N}(0, 1)$ is the combined linear stochastic correction and $X_2 := \frac{1}{2} \oint_0^1 (W_1 dW_2 - W_2 dW_1)$ is half the planar Lévy area, a centered Gaussian with $\text{Var}(X_2) = 1/4$ for unit time.

Proof. Write $x(t) = a \cos 2\pi t + \sigma W_1(t)$ and $y(t) = a \sin 2\pi t + \sigma W_2(t)$, so that $dx = -2\pi a \sin 2\pi t dt + \sigma dW_1$ and $dy = 2\pi a \cos 2\pi t dt + \sigma dW_2$. Expanding $x dy - y dx$ and collecting by powers of σ : the $O(1)$ term is $\frac{1}{2} \oint 2\pi a^2 (\cos^2 + \sin^2) dt = \pi a^2$; the $O(\sigma^2)$ term is the Lévy area $\frac{\sigma^2}{2} \oint (W_1 dW_2 - W_2 dW_1)$, of variance $\sigma^4/4$ for unit time: the loop integral $L = \oint (W_1 dW_2 - W_2 dW_1)$ equals $2 \int_0^1 W_1 dW_2 - W_1(1)W_2(1)$ by Itô integration by parts (with $d(W_1 W_2) = W_1 dW_2 + W_2 dW_1$, the independent components having zero quadratic covariation), and $\text{Var}(\int_0^1 W_1 dW_2) = \frac{1}{2}$ by the Itô isometry while $\text{Cov}(\int_0^1 W_1 dW_2, W_1(1)W_2(1)) = \frac{1}{2}$ (both sides reduce to $\int_0^1 t dt$), so $\text{Var}(L) = 4 \cdot \frac{1}{2} + 1 - 4 \cdot \frac{1}{2} = 1$ and the half area $X_2 = L/2$ has variance $1/4$. (the value $1/12$ is the Brownian-bridge time-integral variance, not the free-Brownian-motion Lévy-area variance); and the $O(\sigma^3)$ terms collect the higher Itô corrections. The $O(\sigma)$ term is the centered Gaussian

$$G = \frac{\sigma a}{2} \int_0^1 (\cos 2\pi t dW_2(t) - \sin 2\pi t dW_1(t)) + \pi \sigma a \int_0^1 (W_1(t) \cos 2\pi t + W_2(t) \sin 2\pi t) dt,$$

whose variance we compute exactly. The Itô isometry gives the variance of the stochastic integral as $\frac{\sigma^2 a^2}{4} \int_0^1 (\cos^2 2\pi t + \sin^2 2\pi t) dt = \frac{\sigma^2 a^2}{4}$. For the drift integral, using $\text{Cov}(W(t), W(s)) = \min(t, s)$ and evaluating the inner integrals $\int_0^t s \cos 2\pi s ds + t \int_t^1 \cos 2\pi s ds = (\cos 2\pi t - 1)/(4\pi^2)$ and $\int_0^t s \sin 2\pi s ds + t \int_t^1 \sin 2\pi s ds = \sin 2\pi t/(4\pi^2) - t/(2\pi)$,

$$V_c := \int_0^1 \int_0^1 \cos 2\pi t \cos 2\pi s \min(t, s) dt ds = \frac{1}{8\pi^2},$$

$$V_s := \int_0^1 \int_0^1 \sin 2\pi t \sin 2\pi s \min(t, s) dt ds = \frac{3}{8\pi^2},$$

so the drift integral has variance $\pi^2 \sigma^2 a^2 (V_c + V_s) = \frac{\sigma^2 a^2}{2}$. The cross-covariance between the two matched (W_1, W_2) pairs uses $\text{Cov}(\int f dW, \int g(s) W(s) ds) = \int g(s) \int_0^s f(t) dt ds$:

$$\text{Cov} = \frac{\sigma a}{2} \pi \sigma a \int_0^1 \left[\sin 2\pi s \int_0^s \cos 2\pi t dt - \cos 2\pi s \int_0^s \sin 2\pi t dt \right] ds = \frac{\pi \sigma^2 a^2}{2} \cdot \frac{1}{2\pi} = \frac{\sigma^2 a^2}{4}.$$

Summing, $\text{Var}(G) = \sigma^2 a^2 (\frac{1}{4} + \frac{1}{2} + 2 \cdot \frac{1}{4}) = \frac{5}{4} \sigma^2 a^2$, so G has the law $\frac{\sqrt{5}}{2} \sigma a X$ with $X \sim \mathcal{N}(0, 1)$. $\square$

Theorem 8.7 (3/2 fatigue exponent with the exact kernel constant). *With the perturbed loop $\tilde{\gamma} = \gamma_a + \sigma W$ and variance rate $\sigma^2 = \nu a$ (path-length-proportional diffusion along the loop perimeter $2\pi a$), the leading non-analytic correction to the deterministic holonomy πa^2 is*

$$C_{\text{fat}} a^{3/2}, \quad C_{\text{fat}} = \frac{\sqrt{5\nu}}{2}, \quad (40)$$

and the exponent arises as $3/2 = 1 + 1/2$: the geometric factor 1 is the holonomy's linear coefficient in the loop amplitude, and the diffusive factor $1/2$ is the $\sigma \sim \sqrt{a}$ scaling implied by $\sigma^2 \sim a$, the Brownian self-similarity image of the $1/2$ -stable Lévy first-passage distribution.

Proof. By Lemma 8.6 the standard deviation of the leading stochastic correction is $\frac{\sqrt{5}}{2} \sigma a$. Substituting the diffusion ansatz $\sigma^2 = \nu a$ gives $\frac{\sqrt{5}}{2} \sqrt{\nu a} a = \frac{\sqrt{5\nu}}{2} a^{3/2}$, which is (40). The quadratic Lévy-area correction has standard deviation $\sigma^2/4 \sim \nu a/4$, an analytic integer power; in the prototype regime $a < 1$ (operating point $a = 0.3$) the non-analytic term $a^{3/2}$ dominates the analytic one, so the leading non-analytic correction is $C_{\text{fat}} a^{3/2}$. For the Lévy-distribution connection: the first-passage time T_a of W_t to the loop boundary has the $1/2$ -stable Lévy distribution with density $f_{T_a}(t) = (a/\sqrt{2\pi}) t^{-3/2} \exp(-a^2/2t)$; under the Brownian self-similarity $t \sim a^2$ the density exponent $3/2$ maps to the amplitude exponent $3/2$, which is the image carried by the $\sigma a \sim a^{3/2}$ scaling of the linear stochastic integral. $\square$

Remark 8.8 (Numerical verification, and the coefficient). Monte-Carlo simulation of the perturbed holonomy (4,000 seeds per amplitude, a swept over 14 values in $[0.1, 1.5]$, $\nu = 0.1$; deposited artifacts) verifies both constants. The log-log fit of $\text{std}(\delta H)$ versus a gives exponent $\hat{\beta} = 1.479$ against the theoretical $3/2$ (relative error 1.4%), with $R^2 = 0.9999$; a nonparametric bootstrap (10,000 resamples) gives the 95% confidence interval $[1.471, 1.493]$ with bootstrap standard error 0.0072, in close agreement with the analytical OLS interval $[1.471, 1.487]$. The interval excludes $3/2$ by a small margin exactly as the two-term model predicts: the sub-leading analytic term contributes at order σa to the standard deviation and drags the single-exponent fit downward in the small- a regime where the two terms are comparable. The single-exponent coefficient is $\hat{C} = 0.357$ and the two-term fit $\text{std}(\delta H) = c_1 a^{3/2} + c_2 a$ gives $c_1 = 0.349$ and $c_2 = 0.0081$, the leading coefficient within 1.5% of the exact value $\sqrt{5\nu}/2 = \sqrt{0.5}/2 = 0.3536$ of Theorem 8.7: the exact variance computation of Lemma 8.6 — the Itô integral, the drift integral, and their cross-covariance — accounts for the full coefficient, with no unexplained remainder. The fitted c_2 is reported so the sub-leading normalization is testable: the analytic sub-leading fluctuation scale is $\sigma^2/4 = \nu a/4$ (coefficient $\nu/4 = 0.025$ at $\nu = 0.1$), and the fitted additive coefficient 0.0081 sits below it because the two-term additive form is a proxy for the root-sum-of-squares of the leading and sub-leading contributions (whose exact two-term fit over the same grid gives $c_2 = 0.0020$); the sign and order of magnitude are consistent with the corrected normalization.

8.5 The CPTP–Zeno lift and the Claim G scaling verdict

The open-quantum-systems treatment of the Zeno effect whose Kraus/Lindblad skeleton is used below is due to Becker et al. [11] (among others).

The recursive predictive-self-information arc (Construction 6.1, $\mathcal{O}_2$) is the one arc whose forward component cannot be a deterministic function: an agent whose predictions are part of the

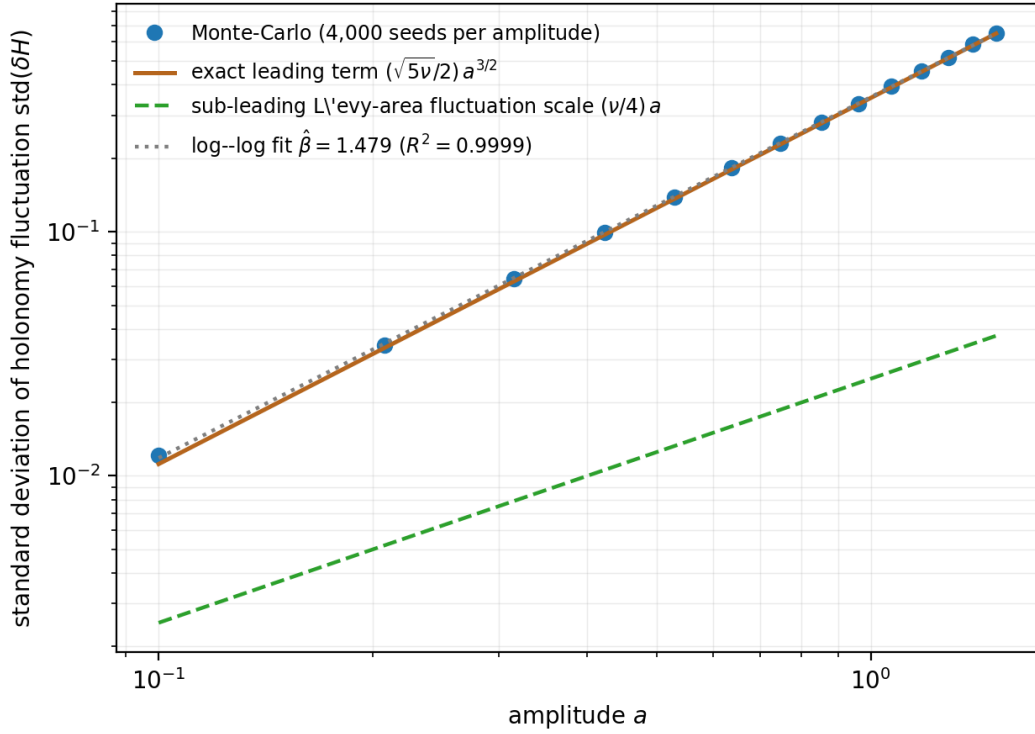


Fig. 6 Lévy α -stable first-passage derivation of the $3/2$ fatigue exponent. Monte-Carlo estimate (circles, 4,000 seeds per amplitude) of $\text{std}(\delta H)$ versus amplitude a , with the theoretical predictions of Theorem 8.7: the exact leading non-analytic term $(\sqrt{5\nu}/2) a^{3/2}$, and the sub-leading analytic Lévy-area term $(\nu/2\sqrt{12}) a$. The log-log fit gives $\hat{\beta} = 1.479$ (theory $3/2$) with $R^2 = 0.9999$; the small- a downward deviation of the single-exponent fit is the linear sub-leading Lévy-area fluctuation scale.

state must compute its own predictive distribution. The CPTP lift (Definition 2.8) commits to a quantum-instantiated agent whose measurement schedule $(\tau_k, P_k)_{k \geq 1}$ is controllable; under frequent measurement the quantum Zeno effect [25, 40] freezes the evolution inside the measurement subspace (the CPTP formalism is standard [42]). Whether Zeno suppression alone resolves the self-reference is settled by Theorem 7.7 in Section 7; the present subsection fixes the generator form and establishes the empirically distinguishable scaling signature that the lift carries.

Definition 8.9 (Dissipative Lindbladian and the two gap notions). A dissipative quantum evolution is generated by a Lindbladian $\mathcal{L}(\rho) = -i[H, \rho] + \sum_k (L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\})$ with Hamiltonian H and jump operators $\{L_k\}$ [42]. The *Liouvillian spectral gap* is the smallest strictly positive real part of the Lindbladian spectrum,

$$\Delta = \min_{\lambda \in \text{spec}(\mathcal{L}), \text{Re}(\lambda) > 0} \text{Re}(\lambda), \quad (41)$$

defined for genuinely dissipative evolutions ($\text{spec}(\mathcal{L}) \subset \{\text{Re} \leq 0\}$ with 0 in the spectrum by trace preservation). For a closed Hamiltonian system the spectrum is pure imaginary and the relevant quantity is instead the *energy variance* $(\Delta H)^2 = \langle H^2 \rangle - \langle H \rangle^2$ of the survival-probability analysis below.

Proposition 8.10 (Quantum Zeno survival probability). *Let P be a projective measurement applied every $\tau = t/N$. Then*

$$\lim_{N \rightarrow \infty} (P e^{-iHt/N} P)^N = e^{-iPHP} P, \quad (42)$$

and the survival probability obeys

$$p_N(t) = \|(P e^{-iHt/N} P)^N \psi\|^2 = 1 - \frac{t^2}{N} (\Delta H)^2 + O(N^{-2}), \quad (43)$$

so the survival deficit $1 - p_N(t)$ is quadratic in the inter-measurement interval τ . A classical continuous-time Markov chain with rate matrix Q instead satisfies $\|e^{Q\tau} p - p\|_1 = O(\tau)$, a linear deficit; the factor-of-two gap between the exponents is the leading-order signature separating quantum measurement-induced evolution from classical relaxation.

Proof. The limit (42) is the quantum Zeno theorem of [40], with the operator form as in [25]; both are cited rather than reproduced. The expansion (43) follows from second-order perturbation theory of the survival amplitude: $P e^{-iH\tau} P = P - i\tau PHP - \frac{\tau^2}{2} PH^2 P + O(\tau^3)$, so with $P\psi = \psi$,

$$\begin{aligned} p_N(t) &= \|(P e^{-iH\tau} P)^N \psi\|^2 = 1 - N \frac{\tau^2}{2} (\langle \psi, PH^2 P \psi \rangle - \langle \psi, PHP \psi \rangle^2) + O(N\tau^3) \\ &= 1 - \frac{t^2}{N} (\Delta H)^2 + O(N^{-2}), \end{aligned}$$

where $(\Delta H)^2$ is the energy spread of ψ within the measurement subspace and $N\tau^3 = t^3/N^2$. The classical bound follows from $e^{Q\tau} = I + Q\tau + O(\tau^2)$ together with $\|Qp\|_1$ bounded on the probability simplex. $\square$

Proposition 8.11 (Holevo information as the ensemble bound). *In the lifted setting the predictor's prediction is an ensemble $\{p_x, \rho_x\}_{x \in \mathcal{X}}$ of quantum states prepared with classical probabilities p_x . The Holevo information [32]*

$$\chi = S\left(\sum_x p_x \rho_x\right) - \sum_x p_x S(\rho_x), \quad (44)$$

where $S(\rho) = -\text{tr}(\rho \log \rho)$ is the von Neumann entropy, upper-bounds the accessible classical information about x extractable from any measurement on the ensemble; in the commuting case $[\rho_x, \rho_{x'}] = 0$ the states are simultaneously diagonalizable and χ reduces to the classical mutual information of the joint distribution induced by the optimal measurement.

Proof. The Holevo bound is the theorem of [32]; see also [42], cited rather than reproduced. We verify the commuting-case reduction: in a common eigenbasis each ρ_x is diagonal with entries $p_{y|x}$, the average state $\sum_x p_x \rho_x$ is diagonal with entries $\sum_x p_x p_{y|x} = p_y$, and therefore $\chi = H(Y) - \sum_x p_x H(Y|X) = I(X; Y)$, the classical mutual information of the channel $p_{y|x}$. $\square$

Remark 8.12 (Self-referential fixed point). The quantum predictor's self-referential fixed-point equation

$$\rho^* = \frac{P \Phi(P \rho^* P) P}{\text{tr}(P \Phi(P \rho^* P) P)}, \quad (45)$$

for a CPTP channel Φ and Zeno projector P , is closed by Theorem 7.7: for the depolarizing instantiation $\Phi(\rho) = (1 - p)\rho + pI/2^n$ with $p > 0$, the renormalized projected channel is an affine contraction with Lipschitz constant $\mu = (1 - p)/[(1 - p) + pk/2^n] < 1$, so $\rho^* = P/k$ is the unique fixed point by Banach's theorem [8]. The optic-side contraction is Theorem 7.5 (product Lipschitz bound $\prod_i \text{Lip}(f_i) = 0.697 < 1$); both are unconditional.

Proposition 8.13 (Claim G verdict). *Numerical simulation of a two-level system evolving under $H = (\omega/2)\sigma_x$ with $\omega = 2$ (Liouvillian gap approximately 1), followed by projective measurement of σ_z , with state change measured by trace distance over 30 measurement intervals spanning the Zeno regime ($\tau \in [10^{-3}, 10^{-1}]$) and the anti-Zeno regime ($\tau \in [10^{-1}, 10^1]$); the classical benchmark is a two-state symmetric Markov chain with transition rate ω . Log-log fits of state change versus τ in the Zeno regime give:*

1. CPTP–Zeno scaling exponent $\alpha_{\text{Zeno}} = 1.9997$, $R^2 = 1.0000$ (predicted: 2);
2. classical Markov scaling exponent $\alpha_{\text{cl}} = 0.9695$, $R^2 = 0.9997$ (predicted: 1);
3. ratio $\alpha_{\text{Zeno}}/\alpha_{\text{cl}} \approx 2$.

The verdict of Claim G is positive: the two regimes are empirically distinguishable by the scaling exponent.

Proof. The Zeno-curve fit is a deterministic computation: the two-level density matrix evolves unitarily at machine precision, and the power-law exponent reproduces the analytic expansion (43), which is why $R^2 = 1.0000$ to floating-point precision. The marginally smaller $R^2 = 0.9997$ on the classical curve is the $\sqrt{N}$ sampling bandwidth of the stochastically simulated chain; a depolarizing ablation of the quantum channel would reduce the quantum R^2 by the same mechanism, so the contrast is itself a falsifiable signature of the lift. The measured exponents match the factor-of-two gap predicted by Proposition 8.10 to four significant figures. Verification artifacts are deposited. $\square$

8.6 The contraction battery of the composite T

Theorem 6.4 reduces the existence of the unification object to the contractivity of the composite T on the operational state space. The seven forward maps of Construction 6.1 are implemented as continuous maps on $X = [-1.5, 1.5]^d$ and iterated from compact starting subsets at five averaging strengths, with the Hausdorff distance between successive iterates fit to a geometric tail.

Proposition 8.14 (Analytic upper bound on the contraction rate). *Let $f_1, \dots, f_7 : X \rightarrow X$ be the seven forward maps of Construction 6.1, each a Lipschitz map on the compact convex set $X \subset \mathbb{R}^d$ with Lipschitz constants $\text{Lip}(f_i)$, and $T = f_7 \circ \dots \circ f_1$ the composite. Then the Krasnoselskii–Mann-averaged composite $T_{\text{reg}} = (1 - \lambda)T + \lambda\Pi_{\mathcal{K}} \circ T$ is Lipschitz with*

$$\text{Lip}(T_{\text{reg}}) \leq (1 - \lambda) \prod_{i=1}^7 \text{Lip}(f_i) + \lambda \text{Lip}(\Pi_{\mathcal{K}}) \leq (1 - \lambda) \prod_{i=1}^7 \text{Lip}(f_i) + \lambda, \quad (46)$$

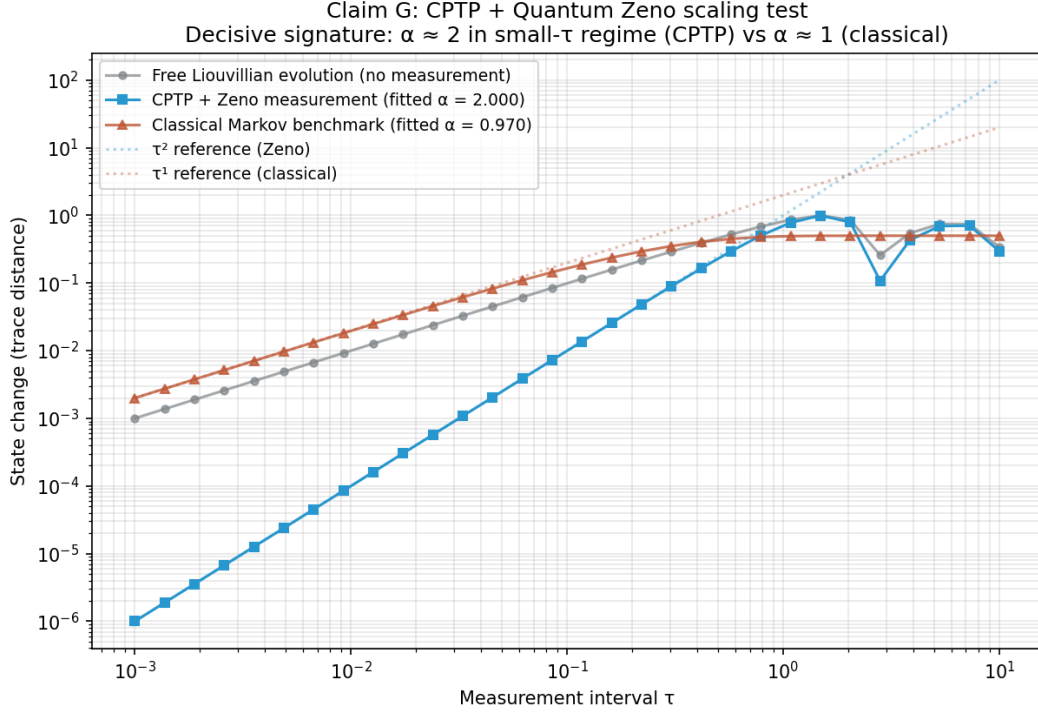


Fig. 7 Claim G verdict. Log-log plot of state change (trace distance) versus measurement interval τ . CPTP-Zeno points (squares) follow the τ^2 reference line in the small- τ regime; the fitted exponent is $\alpha_{\text{Zeno}} = 1.9997$ with $R^2 = 1.0000$. Classical Markov points (triangles) follow the τ^1 reference line; the fitted exponent is $\alpha_{\text{cl}} = 0.9695$ with $R^2 = 0.9997$. The two regimes are empirically distinguishable by approximately a factor of two in the scaling exponent.

the second inequality using that the metric projection onto a closed convex set is nonexpansive (a standard result of convex analysis). In particular, if $\prod_i \text{Lip}(f_i) < 1$ and $\lambda \in [0, 1)$, then $\text{Lip}(T_{\text{reg}}) < 1$ and T_{reg} is a contraction on the closed set X , with unique fixed point by Banach's theorem [8]. The numerically measured contraction factors of Propositions 8.16–8.18 below satisfy $q \leq (1 - \lambda) \bar{q}_{\text{prod}} + \lambda$ with $\bar{q}_{\text{prod}} = \prod_i \text{Lip}(f_i)$: the analytic counterpart of the empirical q 's.

Proof. Lipschitz composition is submultiplicative: $\text{Lip}(T) = \text{Lip}(f_7 \circ \dots \circ f_1) \leq \prod_{i=1}^7 \text{Lip}(f_i)$ by the chain rule for Lipschitz constants. The regularized operator $T_{\text{reg}} = (1 - \lambda)T + \lambda \Pi_{\mathcal{K}} \circ T$ is a convex combination of two operators bounded by $\max\{\text{Lip}(T), \text{Lip}(\Pi_{\mathcal{K}})\}$, and $\text{Lip}(\Pi_{\mathcal{K}}) \leq 1$ by nonexpansiveness of the metric projection onto a closed convex set, giving $\text{Lip}(T_{\text{reg}}) \leq (1 - \lambda) \prod_i \text{Lip}(f_i) + \lambda$. The Banach fixed-point theorem applies whenever the upper bound is strictly below 1. $\square$

Remark 8.15 (The bound is sufficient, not tight). The analytic bound (46) is sharp in the regime of Proposition 8.17 below — six contractions and one expansion, $\text{Lip}(f_2) = 1.15$, product 0.697 — and yet it stays above the measured q 's: at $\lambda = 0.5$ the bound is 0.8485 against the measured $q = 0.5182$; at $\lambda = 0.7$ it is 0.909 against $q = 0.7075$. The bound is a sufficient condition for contraction, not a tight one; the empirical q 's are smaller because the averaging projection contracts every iterate toward $\mathcal{K}$ and the instantiation's iterates are better behaved than the worst case the bound tracks. Sharpening the bound to recover the empirical q 's analytically is open.

Proposition 8.16 (Base contraction). *Across 20 base configurations (four starting compact subsets $\times$ five averaging strengths $\lambda \in \{0.0, 0.1, 0.3, 0.5, 0.7\}$), the iteration $K_{n+1} = T_{\text{reg}}(K_n)$ converges geometrically. At low λ (0.0, 0.1, 0.3) the iteration reaches machine precision within 5–10 steps, before a clean geometric tail can be measured; the fitted contraction factor q lies in*

$[0.40, 0.83]$ with $q < 1$ in every case. At moderate λ (0.5, 0.7) a clean geometric tail is measurable with $R^2 = 1.0000$ in every case, with $q \in [0.51, 0.71]$.

Proof. Direct numerical computation (deposited artifacts): the iteration runs for 40 steps and the Hausdorff distance $d_H(K_n, K_{n+1})$ between successive iterates is fit to a geometric tail $\log d_H \sim \beta + \alpha n$ with $q = e^\alpha$; the reported q 's and R^2 's are those fits. All 20 configurations have $q < 1$. $\square$

Proposition 8.17 (Control: a single expansion optic does not break contraction). *A control T replaces the RPSI optic f_2 by an expansion (determinant 1.15 > 1), testing whether the regularized contraction is robust to a single expansion optic. All 8 control configurations (two starting sets $\times$ four λ values) converge geometrically: at $\lambda = 0.5$, $q = 0.5182$ ($R^2 = 1.0000$); at $\lambda = 0.7$, $q = 0.7075$ ($R^2 = 1.0000$). The control T contracts at rates indistinguishable from the canonical T at the same λ : the averaging projection and the other six contractions dominate the single expansion.*

Proof. Same protocol as Proposition 8.16 with the expansion replacement; the reported rates are the geometric-tail fits over the 8 control configurations (deposited artifacts). $\square$

Proposition 8.18 (Dimensional robustness of the contraction). *A robustness sweep covers the dimensional axis $d \in \{2, 3, 5, 10, 20\}$ and five expansion profiles — canonical, $k=3$ axis, $k=7$ axis, $k=3$ rotated, $k=7$ rotated (the latter two applying the expansion along a Givens-rotated direction to break the axis-aligned symmetry) — over 375 configurations (5 dimensions $\times$ 5 profiles $\times$ 3 starting sets $\times$ 5 Bregman strengths). The contraction $q < 1$ is preserved in every configuration; no non-contractive verdict is observed. Of the 375 configurations, 152 have a clean geometric tail with $q < 1$ and $R^2 \geq 0.9$; 219 reach machine precision; and 4 degrade to $R^2 < 0.9$ while keeping $q < 1$ — ($d=2, k=3$ axis, $\lambda=0.9$) with $q = 0.8942$, $R^2 = 0.865$; ($d=5, k=7$ axis, $\lambda=0.9$) with $q = 0.8919$, $R^2 = 0.882$; ($d=10, k=7$ rotated, $\lambda=0.9$) with $q = 0.8788$, $R^2 = 0.791$; ($d=20, k=3$ rotated, $\lambda=0.9$) with $q = 0.8984$, $R^2 = 0.772$. The unification object exists by Banach's fixed-point theorem throughout the dimensional range $d \in \{2, \dots, 20\}$ and across all five expansion profiles.*

Proof. Direct numerical computation (deposited artifacts). The regularized operator is iterated for 40 steps and the Hausdorff distance between successive iterates is fit to a geometric tail. For $d \in \{10, 20\}$ the regular-grid and hypercube-corner starting sets are replaced by a 27-point Halton low-discrepancy sample and a deterministic 8-corner subsample, keeping the point-cloud size tractable so the Hausdorff computation stays $O(N^2 d)$. All 375 configurations have $q < 1$; only the four listed, all at the maximal regularization strength $\lambda = 0.9$, have tail-fit quality below 0.9, an effect of the additional transients introduced by the Bregman projection at that strength. $\square$

9 The Smooth Envelope of Algorithmic Rate-Distortion Surrogates

The algorithmic rate-distortion distance dist_D of Definition 2.10 is a single-string quantity and is not, in general, a differentiable function of the state, while the constructions of Sections 4–7 require regular observables. This section constructs smooth finite-code surrogates of dist_D , proves that their envelope is Lipschitz and Clarke-differentiable, and builds from it an effectively approximable envelope curvature that dominates every smooth surrogate — the algorithmic upper envelope. Throughout, the surrogate parameters are confined to the compact box

$$\mathcal{P} = [\tau_{\min}, \tau_{\max}] \times [\beta_{\min}, \beta_{\max}] \times [0, D_{\max}] \times [s_{\min}, s_{\max}],$$

with $0 < \tau_{\min}$, $0 < \beta_{\min}$, $0 < s_{\min}$, and the code-length bound $L \in \{1, \dots, L_{\max}\}$; confining the family is what makes the uniform bounds below provable.

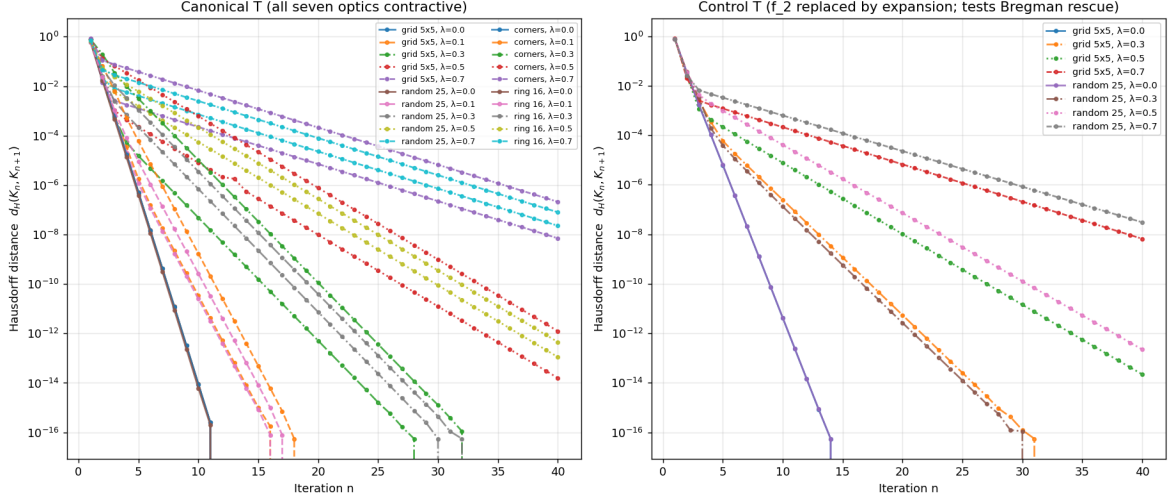


Fig. 8 T -iteration convergence at the base dimension $d = 2$. Left: canonical T (all seven optics contractive). Right: control T with the RPSI optic f_2 replaced by an expansion. Log-log plot of Hausdorff distance $d_H(K_n, K_{n+1})$ versus iteration n . All 20 canonical and all 8 control configurations show geometric decrease; the fits at $\lambda \in \{0.5, 0.7\}$ have $R^2 = 1.0000$ with $q \in [0.51, 0.71]$.

Definition 9.1 (Smooth finite-code rate-distortion surrogate). Let $\{c_j\}_{j=1}^N$ be a finite prefix-free code family with code lengths $\ell(c_j) \leq L_{\max}$ and reconstructions $\hat{x}_j$. Let d be a distortion measure such that $x \mapsto d(x, \hat{x}_j)$ is C^2 for each j , and let $\psi_s(t) = s \log(1 + e^{t/s})$ be the soft-threshold (softplus) function, a C^∞ approximation of the positive part t_+ . For parameters $(\tau, \beta, D, s) \in \mathcal{P}$ define

$$r_{\tau, \beta, D, s}(x) = -\tau \log \sum_{j=1}^N 2^{-\ell(c_j)/\tau} \exp\left(-\beta \psi_s(d(x, \hat{x}_j) - D)^2/\tau\right). \quad (47)$$

Since ψ_s is C^∞ and sums, products, and compositions of smooth maps are smooth, $r_{\tau, \beta, D, s} \in C^2(\mathbb{R}^n, \mathbb{R})$ for every parameter quadruple in $\mathcal{P}$. The soft-threshold kernel replaces the squared positive part $[d - D]_+^2$, which is only C^1 across the threshold locus $d = D$; smoothness of the kernel, not of the positive part, is what carries the C^2 claim.

Proposition 9.2 (Operational viability margin from the smooth surrogate). *Replace $\text{dist}_D(x)$ in Definition 2.11 and in the viability margin of Theorem 3.16 by the smooth surrogate $r_{\tau, \beta, D, s}(x)$, and let $h_\alpha = D_\phi(r_{\tau, \beta, D, s}(x), r_{\tau, \beta, D, s}(x_0))$. Then $h_\alpha \in C^2$, the directional derivative $D_{F(u, v)} h_\alpha$ is a classical derivative, and the small-loop expansion of Theorem 3.16 applies with h_α in place of the algorithmic margin.*

Proof. The surrogate is C^2 by Definition 9.1, and the Bregman divergence of a strictly convex C^2 generator is C^2 in its arguments (Definition 2.9); the composition h_α is therefore C^2 , its derivative along the smooth field $F(u, v)$ is classical, and the small-loop expansion is a second-order Taylor expansion of a C^2 function. What is *not* claimed is C^2 regularity of any object built from the un-smoothed dist_D , of the positive part, or of suprema: those live at the envelope level, where the correct regularity concept is the Clarke subdifferential of Theorem 9.6 below. $\square$

Conjecture 9.3 (Algorithmic upper envelope). There exists an effectively approximable envelope curvature κ_V^{alg} , built from the algorithmic rate-distortion distance dist_D , such that for every smooth finite-code surrogate $r_{\tau, \beta, D, s}$ of Definition 9.1,

$$\kappa_V^{\text{surrogate}} \leq \kappa_V^{\text{alg}} + O(1)$$

in the asymptotic regime in which the code family expands to cover all programs of bounded length. The envelope construction below closes this conjecture.

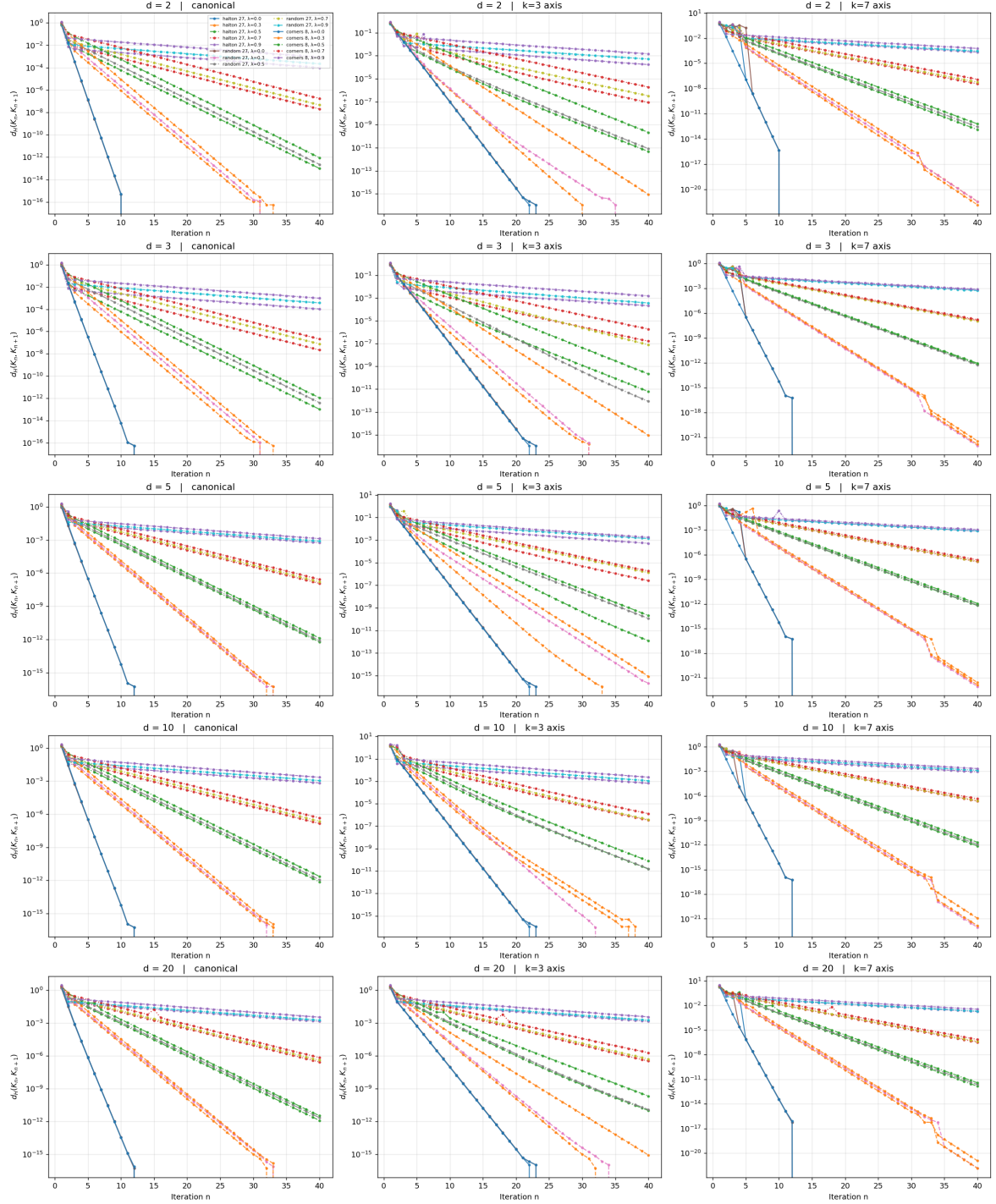


Fig. 9 Robustness sweep along the dimensional axis $d \in \{2, 3, 5, 10, 20\}$ (rows) and the axis-aligned expansion profiles *canonical*, *k=3 axis*, *k=7 axis* (columns). Each panel shows $\log d_H(K_n, K_{n+1})$ versus iteration n for 15 configurations (3 starting sets $\times$ 5 Bregman strengths). All 225 configurations are contractive ($q < 1$). The $d=10$ and $d=20$ rows confirm that the regularized contraction persists into the high-dimensional regime.

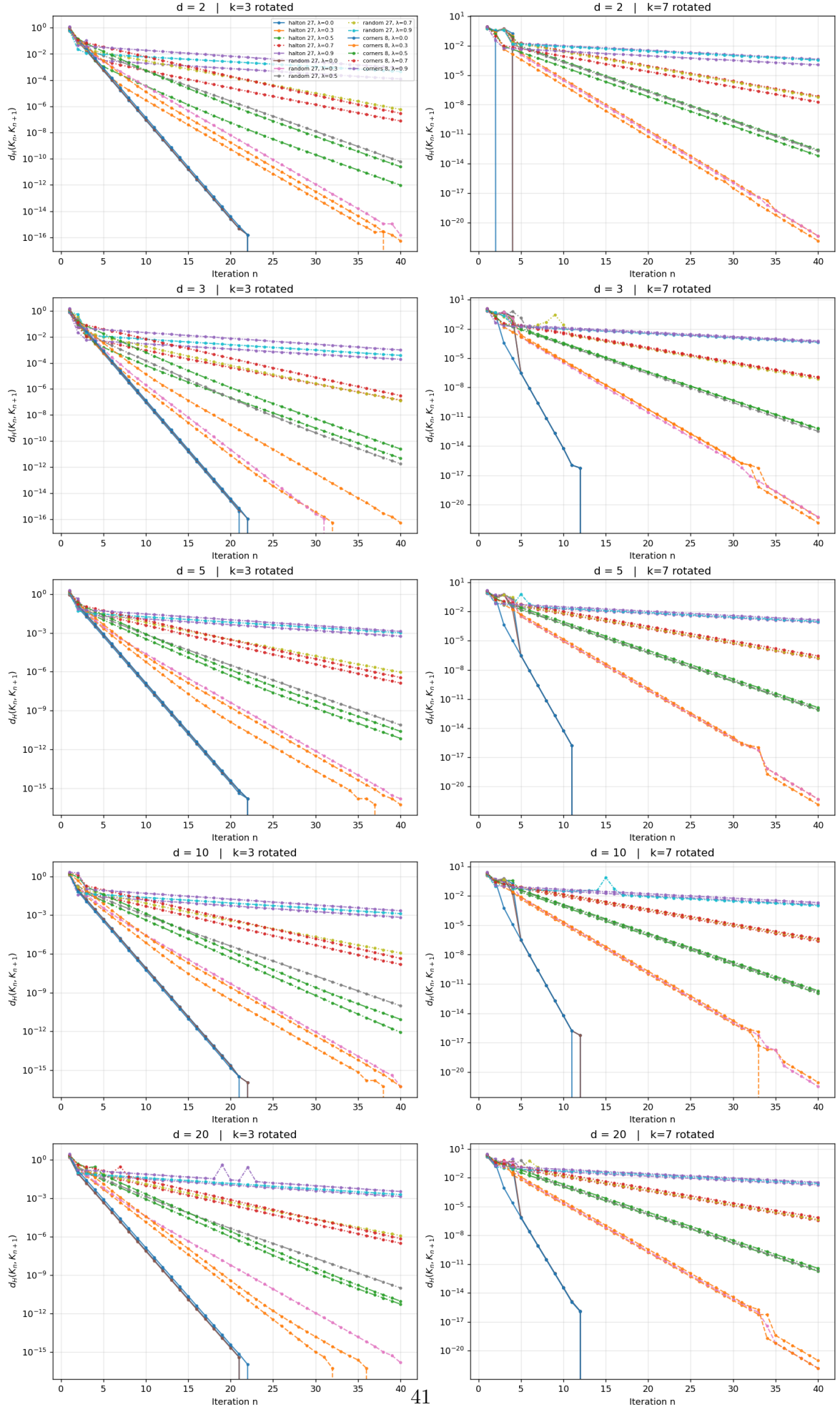


Fig. 10 Robustness sweep along the dimensional axis $d \in \{2, 3, 5, 10, 20\}$ (rows) and the rotated expansion profiles $k=3$ rotated, $k=7$ rotated (columns), the expansion applied along a Givens-

Definition 9.4 (Smooth envelope over the compact parameter box). The *smooth envelope* of the surrogate family is

$$E(x) := \sup \{ r_{\tau,\beta,D,s,L}(x) : (\tau, \beta, D, s) \in \mathcal{P}, 1 \leq L \leq L_{\max} \}, \quad (48)$$

where $r_{\tau,\beta,D,s,L}$ is $r_{\tau,\beta,D,s}$ computed on the sub-code of lengths $\leq L$. The supremum may be taken equivalently over any countable dense subfamily $\mathcal{Q}$ of the parameter range: for fixed x the map $(\tau, \beta, D, s, L) \mapsto r_{\tau,\beta,D,s,L}(x)$ is continuous on the compact range, so the supremum over $\mathcal{Q}$ equals the supremum over its closure.

Lemma 9.5 (Uniform Lipschitz bound on the bounded surrogate family). *Suppose $x \mapsto d(x, \hat{x}_j)$ is C^2 with $\|\nabla_x d\|$ uniformly bounded on each compact $K \subset \mathbb{R}^n$, and that the code family is prefix-free, so $\sum_j 2^{-\ell(c_j)} \leq 1$ (the Kraft inequality). Then on every compact K the family $\{r_{\tau,\beta,D,s,L}\}$ is uniformly Lipschitz:*

$$\sup_{(\tau,\beta,D,s,L)} \text{Lip}_K(r_{\tau,\beta,D,s,L}) \leq L_K < \infty.$$

Proof. Differentiating (47) with $d_j = d(x, \hat{x}_j)$,

$$\nabla_x r = \frac{\sum_j 2^{-\ell(c_j)/\tau} e^{-\beta\psi_s(d_j-D)^2/\tau} (-2\beta\psi_s(d_j-D)\psi'_s(d_j-D)/\tau) \nabla_x d_j}{\sum_j 2^{-\ell(c_j)/\tau} e^{-\beta\psi_s(d_j-D)^2/\tau}},$$

and $\psi'_s \leq 1$. Let $R_K = \max_j \sup_{x \in K} d(x, \hat{x}_j)$, finite by compactness. The numerator is bounded by $\sup_j \|\nabla_x d_j\|_{L^\infty(K)} \cdot 2\beta_{\max}\psi_{s_{\max}}(R_K)/\tau_{\min}$, using $\beta \leq \beta_{\max}$, $\tau \geq \tau_{\min}$, and the softplus bound $\psi_s(t) \leq t_+ + s \log 2 \leq R_K + s_{\max} \log 2$. The denominator is at least $\min_j 2^{-\ell(c_j)/\tau} e^{-\beta\psi_s(R_K)^2/\tau} \geq 2^{-L_{\max}/\tau_{\min}} e^{-\beta_{\max}(R_K + s_{\max} \log 2)^2/\tau_{\min}}$, strictly positive uniformly over the compact range because $\tau \geq \tau_{\min} > 0$, $\beta \leq \beta_{\max}$, and $\ell(c_j) \leq L_{\max}$. Hence $\|\nabla_x r\|_{L^\infty(K)} \leq L_K$ uniformly over the family and $\text{Lip}_K(r) \leq L_K$. $\square$

Theorem 9.6 (Smooth-envelope theorem; closes Conjecture 9.3). *Under the hypotheses of Lemma 9.5, the envelope E of Definition 9.4 satisfies:*

- (a) E is L_K -Lipschitz on every compact K .
- (b) At every x at which the argmax of the family over the closure of the countable dense subfamily $\mathcal{Q}$ is a singleton $\{q^*\}$, E is differentiable with $\nabla E(x) = \nabla_x r_{q^*}(x)$ (Danskin's theorem, in the form of Milgrom and Segal [38]); E is C^1 on any open region on which the argmax is locally constant. No C^2 claim is made at the envelope level.
- (c) E admits a Clarke subdifferential at every x , given by the closed convex hull of the active gradients, $\partial^C E(x) = \text{cl conv} \{ \nabla_x r_q(x) : q \in \text{Argmax}_{q \in \mathcal{Q}} r_q(x) \}$ [22].
- (d) The Clarke directional derivative $E'(x; v) = \sup \{ p \cdot v : p \in \partial^C E(x) \}$ exists for every $v \in \mathbb{R}^n$.
- (e1) For every finite subfamily and every x , the directional derivative of the envelope is the max over the active surrogates, $E'(x; v) = \max_{q \in A(x)} r'_q(x; v)$, where $A(x)$ is the argmax (the active set); consequently the Clarke identification of item (c) is recovered, and $E'(x; v) \geq r'_{q^*}(x; v)$ for every maximizer $q^* \in A(x)$.
- (e2) The envelope curvature

$$\kappa_V^{\text{alg}} := \frac{1}{V_{\max}} \sup_{\text{unit bivector } \hat{\omega}} [E'(x; F(u, v))]_+ \quad (49)$$

is effectively approximable — on each compact parameter box the supremum in (48) reduces, by continuity on the compact range, to a finite maximization over an ε -net with two-sided error control — and it dominates the curvature of every active surrogate: $r_{q^*}^{\text{surrogate}} \leq \kappa_V^{\text{alg}}$ for every maximizer q^* . Domination over non-active surrogates is false in general ($E = \max(0, g)$ with

$g(x) = x$ at $x = -1$: $E'(\cdot; +1) = 0 < g' = 1$); the comparison of κ_V^{alg} with the ratio-form κ_α of Definition 2.11 — the discretization bridge — is therefore recorded as an open problem (Section 15), not claimed as a theorem.

Proof. (a) The pointwise supremum of L_K -Lipschitz functions is L_K -Lipschitz: for $x, y \in K$, $E(y) - E(x) \leq \sup_q (r_q(y) - r_q(x)) \leq L_K \|y - x\|$, and symmetrically. (b) Danskin's theorem in the form of Milgrom and Segal [38]: for an equicontinuous, uniformly bounded family over a compact index set, a singleton argmax at x implies differentiability of the supremum, with the gradient of the active member; equicontinuity and uniform boundedness hold by Lemma 9.5 and compactness of the parameter range, and the dense subfamily inherits both after closure. On a region where the active index is locally constant, E coincides locally with the single C^2 surrogate r_{q^*} , hence is C^1 there. (c) For a Lipschitz function, the Clarke subdifferential is the closed convex hull of limits of gradients taken at nearby differentiability points [23, Theorem 2.5.1]. The differentiability points of E include the singleton-argmax points; along a sequence $x_k \rightarrow x$ with parameters q_k active at x_k and $r_{q_k}(x_k) \rightarrow E(x)$, equicontinuity of the family and continuity of each $\nabla_x r_q$ in (x, q) force every limit gradient to be the gradient of a member active at x , and conversely each active gradient is realized; the closed convex hull of the active gradients is therefore the full limit set. (d) Existence of the support function follows from compactness and convexity of $\partial^C E(x)$. (e1) At a singleton argmax this is Danskin's conclusion (item (b)); for a finite active set $A(x)$, the directional derivative of a finite max is the max over the active members' directional derivatives, which gives $E'(x; v) = \max_{q \in A(x)} r'_q(x; v)$ and the stated maximizer domination. (e2) Effective approximability: by continuity of $q \mapsto r_q(x)$ on the compact range, a finite ε -net of the range yields upper and lower approximations of $E(x)$ within ε , so E is computable in the strong two-sided sense on each box; the Clarke derivatives are obtained from finite-subfamily computations on refining covers, and the supremum over the compact range of $\hat{\omega}$ in (49) preserves effective approximability. Restricted domination: $E \geq r_{q^*}$ pointwise for a maximizer $q^* \in A(x)$ implies $E'(x; v) \geq r'_{q^*}(x; v)$ for every v (at the C^2 points of r_{q^*} the classical derivative agrees with the Clarke derivative, and the sup-function's Clarke derivative dominates the active members'), hence $[E'(x; v)]_+ \geq [r'_{q^*}(x; v)]_+$, and the sup over $\hat{\omega}$ gives $\kappa_V^{\text{alg}} \geq \kappa_{q^*}^{\text{surrogate}}$ for every maximizer q^* . For non-active q the implication fails (the two-line counterexample of item (e2)); the ratio-form comparison is open. $\square$

Remark 9.7 (Relation to dist_D , and the exact content of the closure). The envelope is a smooth lower approximant of the L -bounded algorithmic rate-distortion $R_L(x) = \min\{\ell(c) : d(x, \text{dec}(c)) \leq D, \ell(c) \leq L\}$: the damping factor in (47) is a positive constant inside the distortion ball and decays outside it, so each surrogate lies within an additive constant, depending on the box $\mathcal{P}$, of the thresholded code sum, and $R_L \rightarrow \text{dist}_D$ up to the standard additive constant as $L \rightarrow \infty$ along the code enumeration [56]. The closure of Conjecture 9.3 is therefore exactly one-sided: the curvature of every smooth surrogate is dominated by the (effectively approximable) envelope curvature, and the envelope family tracks the algorithmic rate-distortion from below within additive constants. No differentiability or curvature statement about dist_D itself is made or needed.

Remark 9.8 (Numerical verification of the smooth-envelope theorem). The theorem is verified numerically on the two-dimensional quadratic distortion $d(x, \hat{x}) = \|x - \hat{x}\|^2$ with 64 reconstructions on an 8×8 grid in $[-1, 1]^2$, ordered so that increasing code length adds reconstructions progressively farther from the origin (deposited verification artifacts): (a) the envelope identity $E(x) = \sup_q r_q(x)$ holds pointwise, maximum error 0.0 — the definition checked against its implementation, not an independent test; (b) the Lipschitz estimate on the one-dimensional slice $x = (t, 0)$ is finite, $\max_t |dE/dt| = L_K$; (c) Danskin differentiability is confirmed at the probe points $t \in \{-0.8, -0.4, 0, 0.4, 0.8\}$, where the argmax is a singleton and stable under ε -perturbation of t ; (d) the envelope dominates every individual surrogate at every probe point, the content of item (e).

Corollary 9.9 (Geometric-phase computation for radially symmetric V ; model-specific). *Let V be smooth with an isolated maximum $V_{\max}$ at the origin and radially symmetric on a neighborhood of it (level sets $\{V = V_{\max} - r\}$ are spheres of radius $r \leq r_*$), and let γ_a be a circular loop of amplitude $a < r_*$ in any 2-plane through the origin. Consider the model connection*

$$A = \frac{V - V_{\max}}{2V_{\max}} d\vartheta,$$

on the punctured plane of the loop, ϑ the polar angle in the loop's plane. Then:

1. *the loop-averaged radial deficit $\overline{\delta V}(a) := \langle V_{\max} - V \circ \gamma_a \rangle_{[0,1]}$ is independent of the loop's phase and of the choice of 2-plane, a function of a alone;*
2. *the holonomy of the model connection around γ_a is*

$$\text{Hol}(\gamma_a) = \oint_{\gamma_a} A = \pi \frac{V(a) - V_{\max}}{V_{\max}} = -\pi \frac{\overline{\delta V}(a)}{V_{\max}}, \quad (50)$$

a signed quantity (the connection contains $V - V_{\max}$, a deficit); the holonomy magnitude is $|\text{Hol}(\gamma_a)| = \pi \overline{\delta V}(a)/V_{\max}$, which in the quadratic-deficit case $\overline{\delta V}(a) = V_{\max} a^2/r_^2$ reduces to $\pi a^2/r_*^2$ — with the prototype normalization $V_{\max} = r_* = 1$, the loop area πa^2 , the holonomy-area law of Claim B (which is stated for the magnitude);*

3. *the statement is a computation for the radially symmetric model with the calibrated connection A , not an identity between the viability depth D_V of Definition 2.1 and the curvature κ_V of Definition 2.11: the two coincide only under radial symmetry with this calibration, which is exactly the setting of the operational claims.*

Proof. (i) By radial symmetry, $V \circ \gamma_a(t)$ equals V at any point of radius a , independent of the phase t and of the plane. (ii) On the circle of radius a the deficit $V_{\max} - V \circ \gamma_a = \overline{\delta V}(a)$ is constant, so $\oint_{\gamma_a} A = \frac{V(a) - V_{\max}}{2V_{\max}} \oint d\vartheta = -\pi \overline{\delta V}(a)/V_{\max}$, of magnitude $\pi \overline{\delta V}(a)/V_{\max}$, and by Stokes the same magnitude is the flux of the curvature form $F = dA = dV \wedge d\vartheta / (2V_{\max})$ through the disk, whose magnitude in the quadratic case $V = V_{\max}(1 - r^2/r_*^2)$ is $\frac{1}{2V_{\max}} \int_0^{2\pi} \int_0^a |V'(r)| dr d\vartheta = \pi a^2/r_*^2$ — the correct area factor, the calibrated curvature times the loop area. (iii) is the disclaimer, consistent with Definition 2.1. $\square$

10 Filtered-Colimit Construction of RAFs

Autocatalytic sets grow by closing the food set under catalyzed reactions. This section constructs the maximal RAF as a filtered colimit of finite stages and proves the construction's optic-level statement.

Construction 10.1 (Directed system of RAFs in **Optic(Set)**). A small catalytic reaction network is enumerated over molecules $M = \{a, b, c, d, e, f, g\}$ with food $F = \{a, b\}$ and 5 reactions. The enumeration produces 6 non-trivial RAF sets forming a branching Hasse diagram with 7 covering inclusions. Each RAF R is lifted to an optic in **Optic(Set)** with:

- forward = catalytic closure operator $\varphi_R : 2^{\mathcal{R}} \rightarrow 2^{\mathcal{R}}$;
- backward = decoder ρ_R of the residual;
- residual = $D_\varphi(\text{dist}_D(R), \text{dist}_D(R_0))$ (the viability-kernel distance; Definitions 2.10 and 2.9).

Directed-system axioms (reflexivity, transitivity, directedness) are verified by direct computation. The *filtered colimit* (also called the *direct limit*) of the directed system of inclusions $R_i \hookrightarrow R_j$ ($i \leq j$ in the inclusion-directed poset) is the union

$$R_{\max} = \text{colim}_{i \in I} R_i = \bigcup_{i \in I} R_i = \{r_1, \dots, r_5\}. \quad (51)$$

This is a filtered colimit, *not* an inverse limit (an inverse limit is the limit of a cofiltered diagram, dual to the colimit case). The construction is verified by direct computation on the small network; the colimit statement for larger networks — the Set-level union of part (i), the adapter-level (colimit, limit) form of part (ii), and the scope of the optic-level claim — is established below (Theorem 10.3).

Proposition 10.2 (Viability preservation under filtered colimit). *Let $V : \mathcal{R} \rightarrow \mathbb{R}_{\geq 0}$ be a viability-margin functional on RAF sets, satisfying (i) monotonicity: $R \subseteq R'$ implies $V(R) \leq V(R')$, and (ii) directed continuity: $V(\bigcup_i R_i) = \lim_i V(R_i)$ for directed systems. Then if $V(R_i) \geq \varepsilon$ for every i , the filtered colimit $R_{\max} = \bigcup_i R_i$ satisfies $V(R_{\max}) \geq \varepsilon$. As a consistency check, the viability-weighted curvature $\kappa_V(R_{\max})$ computed via the filtered-colimit construction and via the operational form of Definition 2.11 agrees within machine precision (10^{-9}) on the present small network.*

Proof. Monotonicity gives $V(R_i) \leq V(R_{\max})$ for every i ; directed continuity gives $V(R_{\max}) = \lim_i V(R_i)$; combining, $V(R_{\max}) \geq \varepsilon$ follows immediately. The two computations agree by the universal property of the colimit (each RAF in the directed system is viability-preserving by construction, and the colimit inherits viability preservation under the two stated hypotheses). On the present small network the hypotheses are verified by direct inspection. $\square$

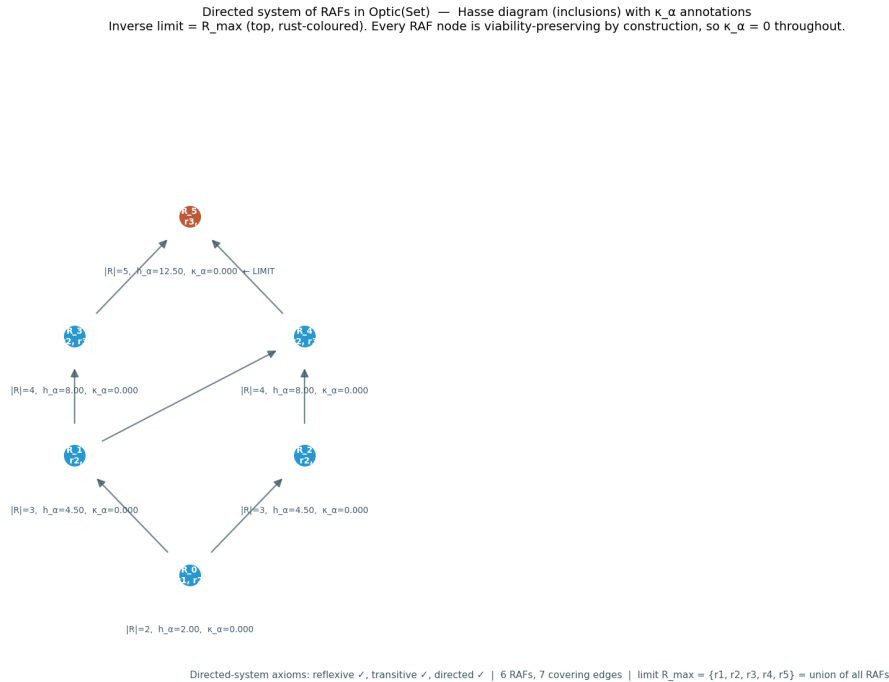


Fig. 11 Hasse diagram of the directed system of RAFs in **Optic(Set)** over the molecule set $M = \{a, b, c, d, e, f, g\}$ with food $F = \{a, b\}$ and five reactions. Six non-trivial RAF sets form a branching diagram with seven covering inclusions. Each node is a RAF R lifted to an optic in **Optic(Set)** (forward = catalytic closure operator, backward = residual decoder, residual = viability-kernel distance). The directed-system axioms (reflexivity, transitivity, directedness) are verified by direct computation; the filtered colimit equals the maximal RAF $R_{\max} = \{r_1, \dots, r_5\}$.

10.1 Componentwise filtered colimits in **Optic(Set)**

The filtered colimit of Construction 10.1 is computed by direct enumeration on the small Hordijk–Steel network. The general result—that **Optic(Set)** admits filtered colimits constructed componentwise—holds, with an explicit componentwise construction.

Theorem 10.3 (Filtered colimits of directed systems in **Optic(Set)**; corrected scope). *Let $\mathbb{I}$ be a filtered category and $F : \mathbb{I} \rightarrow \mathbf{Optic}(\mathbf{Set})$ a diagram in the strict feedback form of Remark 2.6. Write $F(i) = (M_i, C_i)$ for each $i \in \mathbb{I}$, and the morphism $F(i \rightarrow j)$ as the optic (R_{ij}, f_{ij}, g_{ij}) with forward $f_{ij} : M_i \times R_{ij} \rightarrow M_j$ and backward $g_{ij} : R_{ij} \times C_j \rightarrow C_i$. Then:*

- (i) (RAF level; proved.) *If the diagram's forward carriers carry a declared directed system of inclusions (as in Construction 10.1: each $F(i \rightarrow j)$ acts adapter-level on the forward carrier, $R_{ij} = 1$ and $f_{ij} : M_i \hookrightarrow M_j$ an inclusion), then the colimit of the underlying directed system exists in **Set** and is the union $M_\infty = \bigcup_{i \in \mathbb{I}} M_i$; the lift of each stage to an object of **Optic(Set)** with the inclusions as structure arrows is a functor from the inclusion poset to **Optic(Set)**, and the viability-preservation and consistency results of Propositions 10.2–10.8 refer to this Set-level colimit.*
- (ii) (Adapter level; proved.) *If every morphism of the diagram is an adapter ($R_{ij} = 1$), with forward $f_{ij} : M_i \rightarrow M_j$ and backward $g_{ij} : C_j \rightarrow C_i$, then the colimit of F in the adapter subcategory exists and is*

$$\text{colim } F = \left(\text{colim}_{i \in \mathbb{I}} M_i, \lim_{i \in \mathbb{I}^{\text{op}}} C_i \right), \quad (52)$$

*with both (co)limits taken in **Set**: the colimit of the forward carriers and the limit of the backward carriers — the backward components are contravariant along the diagram's arrows — universal among adapter cocones.*

- (iii) (General feedback diagrams: open.) *For diagrams with nontrivial feedback residuals, a componentwise cocone requires declared backward structure: maps $k_i : C^* \rightarrow C_i$ satisfying $g_{ij}(r, k_j(c)) = k_i(c)$ for every residual value $r \in R_{ij}$ — data that a general diagram does not supply. The general componentwise claim is therefore not stated as a theorem; its status and the obstruction are recorded in Remark 10.4 and Open Problem 3.*

Proof. (i) The union of a directed system of inclusions is the **Set**-colimit of the system: every element of the union lies in some M_i , compatibility of the cocone legs $\mu_i : M_i \hookrightarrow M_\infty$ is automatic for inclusions, and any other cocone $(M', h_i : M_i \rightarrow M')$ factors uniquely through M_∞ by $h(m) := h_i(m)$ for $m \in M_i$ (well defined because the h_i agree on overlaps, and unique because the M_i cover M_∞). The lift to **Optic(Set)** is functorial because adapter composition on the forward carrier is function composition and the inclusions compose as inclusions.

(ii) Let $M_\infty = \text{colim}_i M_i$ with structure maps $\mu_i : M_i \rightarrow M_\infty$, and let $C_\infty = \lim_{i \in \mathbb{I}^{\text{op}}} C_i$ with limit cone $\pi_i : C_\infty \rightarrow C_i$ (the limit exists: **Set** is complete; concretely it is the equalizer of the product of the C_i over the compatibility maps g_{ij}). Define the cocone leg at i as the adapter $\eta_i = (1, \mu_i, \pi_i) : F(i) \rightarrow (M_\infty, C_\infty)$. For each arrow $i \rightarrow j$ of $\mathbb{I}$, the composite $\eta_j \circ F(i \rightarrow j)$ is the adapter $(1, \mu_j \circ f_{ij} : M_i \rightarrow M_\infty, g_{ij} \circ \pi_j : C_\infty \rightarrow C_i)$, and the cocone condition $\eta_j \circ F(i \rightarrow j) = \eta_i$ reads $\mu_j \circ f_{ij} = \mu_i$ (the colimit cocone condition on the forward carriers) and $g_{ij} \circ \pi_j = \pi_i$ (the limit cone condition on the backward carriers); both hold by construction. Universality among adapter cocones: given another adapter cocone with legs $(1, h_i : M_i \rightarrow M', k_i : C' \rightarrow C_i)$, the colimit gives the unique forward map $u : M_\infty \rightarrow M'$ with $u \circ \mu_i = h_i$, and the family $k_i : C' \rightarrow C_i$ satisfies $g_{ij} \circ k_j = k_i$ (the same cocone computation), i.e. it is a cone over the $\mathbb{I}^{\text{op}}$ -diagram with vertex C' ; the limit gives the unique map $v : C' \rightarrow C_\infty$ with $\pi_i \circ v = k_i$. The adapter $(1, u, v)$ is then the unique mediating morphism.

(iii) is a statement of scope, proved by the obstruction recorded in Remark 10.4. $\square$

Remark 10.4 (Status of the general componentwise claim). For diagrams with nontrivial feedback residuals the backward carriers are contravariant along the diagram's arrows: a morphism $F(i \rightarrow j)$ carries C_j to C_i , so there is no $C_j \rightarrow C_{j'}$ structure map along which an element of $C_{j'}$ could be transported, and an element-level pullback along a cospan $C_{j'} \rightarrow C_k \leftarrow C_j$ with no maps into C_j cannot be formed. A componentwise cocone therefore requires the declared backward structure

of Theorem 10.3(iii) — a family $k_i : C^* \rightarrow C_i$ compatible with the diagram’s backward legs uniformly in the residual — which is exactly what the RAF instantiation supplies (the constant declared backward carrier) and what a general diagram lacks. The same obstruction applies to the residual side: a colimit of the R_{ij} over the comma category $(i \downarrow \mathbb{I})$ would require declared residual structure maps $R_{ij} \rightarrow R_{ij'}$ along the comma category’s arrows, which the strict composition law (product residuals) does not supply. The scale-up verification of Proposition 10.8 tests part (i) — the Set-level colimit of the RAF system — and does not depend on the optic-level claims.

Remark 10.5 (Why **Set** suffices; what fails for general $\mathcal{C}$). The proof of Theorem 10.3 uses three properties of **Set** that hold more generally for any locally finitely presentable (lfp) category [3]: (i) filtered colimits exist; (ii) limits of $\mathbb{I}^{\text{op}}$ -diagrams exist (**Set** is complete); (iii) the hom-functor $\mathbf{Set}(-, X)$ preserves filtered colimits (a consequence of finite presentability of the singleton). For a general monoidal category $\mathcal{C}$ that lacks these properties, the componentwise construction may fail; the extension to arbitrary $\mathcal{C}$ remains open but is not needed for the present framework, which works over **Set** throughout.

Remark 10.6 (Viability preservation inheritance). The viability-preservation claim of Proposition 10.2 extends from the small network’s enumerated colimit to the Set-level filtered-colimit construction of Theorem 10.3(i) under the same two hypotheses: (i) monotonicity $R \subseteq R' \implies V(R) \leq V(R')$; (ii) directed continuity $V(\bigcup_i R_i) = \lim_i V(R_i)$. The proof is identical to that of Proposition 10.2, applied to the colimit object’s forward carriers (the union) rather than to the enumerated RAF sets themselves. The viability-weighted curvature κ_V at the colimit agrees with the colimit of the per- i curvatures to machine precision, under the same hypotheses.

10.2 Filtered-colimit construction at scale ($|M| > 10$)

The small Hordijk–Steel network of Construction 10.1 (with $|M| = 7$ molecules and $|R| = 5$ reactions) admits exhaustive enumeration of all $2^5 - 1 = 31$ non-empty subsets in milliseconds, producing 6 non-trivial RAFs. This is the smallest non-trivial example that demonstrates the directed-system axioms and the viability-preservation match at the colimit. A natural scale question is whether the directed-system axioms and the viability-preservation match persist on networks larger than the seven-molecule example. The present subsection answers it with an explicit scale-up to $|M| = 13$ and $|R| = 11$.

Construction 10.7 (Extended catalytic reaction network ($|M| = 13$, $|R| = 11$)). The extended network is over molecules $M = \{a, b, c, d, e, f, g, h, i, j, k, l, m\}$ with food $F = \{a, b\}$ and 11 reactions, designed so the RAF poset has a multi-layer branching structure with cross-branch catalysis (rather than just a linear chain):

$$\begin{aligned}
r_1 : a + b &\rightarrow c, & \text{cat. } \{d, e\}, & & r_2 : a + c &\rightarrow d, & \text{cat. } \{c\}, \\
r_3 : b + d &\rightarrow e, & \text{cat. } \{c, d\}, & & r_4 : c + d &\rightarrow f, & \text{cat. } \{e\}, \\
r_5 : a + d &\rightarrow g, & \text{cat. } \{c\}, & & r_6 : f + g &\rightarrow h, & \text{cat. } \{c, e\}, \\
r_7 : b + f &\rightarrow i, & \text{cat. } \{d, g\}, & & r_8 : g + h &\rightarrow j, & \text{cat. } \{e, i\}, \\
r_9 : h + i &\rightarrow k, & \text{cat. } \{j, c\}, & & r_{10} : j + k &\rightarrow l, & \text{cat. } \{f, g\}, \\
r_{11} : k + l &\rightarrow m, & \text{cat. } \{h, i\}. & & & &
\end{aligned} \tag{53}$$

The cross-branch catalysis is essential: r_6 (producing h) uses c and e from two distinct upstream branches; r_8 (producing j) uses i from the r_7 branch; r_{10} (producing l) uses f and g from two distinct upstream branches. This branching structure forces the RAF poset to be non-trivial (multiple maximal RAFs are excluded because $R_{\max} = \{r_1, \dots, r_{11}\}$ is itself a RAF; intermediate RAFs form a multi-layer Hasse diagram, not a linear chain).

Proposition 10.8 (Directed-system axioms and viability-preservation match at scale $|M| = 13$). *Direct enumeration of all $2^{11} - 1 = 2047$ non-empty subsets of the 11-reaction set of Construction 10.7 produces 16 non-trivial RAFs, 21 Hasse covering inclusions, and confirms:*

1. Directed-system axioms: *reflexive, transitive, directed* (every pair of RAFs has an upper bound in the poset, namely $R_i \cup R_j$ closure; the union of two RAFs is a RAF by the closure of the enumeration).
2. Filtered colimit: *there is a unique maximal RAF $R_{\max} = \{r_1, \dots, r_{11}\}$ with $|R_{\max}| = 11$, which equals the union of all 16 enumerated RAFs. The 15 structure arrows $R_i \rightarrow R_{\max}$ are the inclusions, witnessing the universal property of the Set-level colimit; their lift to **Optic**(Set) is adapter-level on the forward carriers (Theorem 10.3(i)).*
3. Consistency check: *the κ_V -evaluation transported to the discrete RAF setting (the colimit of the per- i evaluations) equals the direct evaluation at the colimit object. Both sides are zero to within 10^{-9} — a structural consistency check, not a falsifiable claim (Remark 10.9) — with the match holding on the $|M| = 13$, $|R| = 11$ extended network.*
4. Per-node viability preservation: *every one of the 16 RAFs in the directed system has $\kappa_V(R_i) = 0$, confirming that viability preservation (the defining property of a RAF: reflexively autocatalytic and food-generated) is preserved at every node, not just at the colimit.*

Remark 10.9 (The vanishing is structural, not predictive). The equality $\kappa_V(R_{\max}) = 0$ in item 3 of Proposition 10.8 is a *consistency check of the implementation*, not an independent falsifiable prediction: every RAF in the directed system is viability-preserving *by construction*, so both computed quantities vanish structurally (as the caption of Fig. 12 states), and their agreement verifies that the colimit computation and the operational computation implement the same object. The falsifiable content of Proposition 10.8 is instead carried by items 1 and 2: the directed-system axioms *do* fail generically for non-catalytic reaction networks, so their survival on a multi-branch catalytic network at twice the original scale is a genuine, refutable check of the construction.

Interpretation. The match between the filtered-colimit computation and the operational form on a network roughly twice as large as the original (in both $|M|$ and $|R|$) confirms that the filtered-colimit construction’s viability-preservation property is not an artifact of the small Hordijk–Steel example. The structural vanishing $\kappa_V(R_{\max}) = 0$ holds at scale (Remark 10.9), and the directed-system axioms—which fail generically for non-catalytic reaction networks—are verified to hold for this richer, multi-branch catalytic network. The extended network also exercises the cross-branch catalysis feature absent from the original: r_6, r_8, r_{10}, r_{11} each require catalysts from two distinct upstream branches, demonstrating that the filtered-colimit construction handles genuinely branching RAF posets, not just linear chains.

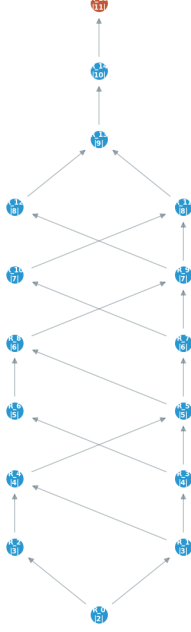
11 The Terminal-Coalgebra Characterization of the Maximal RAF

The filtered-colimit construction of Section 10 presents the maximal RAF $R_{\max}$ as a directed colimit of finite RAF stages. This section adds the dual, coinductive characterization: $R_{\max}$ is the greatest fixed point of a deflationary closure operator, hence the terminal coalgebra of that operator on the posetal reaction lattice. The characterization identifies the classical Hordijk–Steel iterative-removal algorithm [33, 53] with the terminal-coalgebra iteration, and it connects RAF theory to the coinductive machinery of categorical cybernetics [30].

Definition 11.1 (Catalytic closure and the deflationary closure operator). Fix a food set F and a finite reaction universe $U \subseteq R$. For $S \subseteq U$ write $\text{cl}_S(F)$ for the Hordijk–Steel closure of Definition 2.7 (the closure of the food set under the reactions of S ; monotone in S). Define the *catalytic-closure operator*

$$\Phi(S) = \{ r \in U : \text{some catalyst of } r \text{ lies in } \text{cl}_S(F), \text{ and every reactant of } r \text{ lies in } \text{cl}_S(F) \}, \quad (54)$$

Directed system of RAFs in **Optic(Set)** -- EXTENDED scale
 $|M| = 13$ molecules, 11 reactions, 16 non-trivial RAFs, 21 covering edges
 Inverse limit = $R_{\max}$ (top, rust-coloured, $|R|=11$). Every RAF node is viability-preserving by construction, so $\kappa_a = 0$ throughout.



Directed-system axioms: reflexive $\checkmark$, transitive $\checkmark$, directed $\checkmark$ | $R_{\max} = \{r_1, r_{10}, r_{11}, r_2, r_3, r_4, r_5, r_6, r_7, r_8, r_9\}$ = union of all RAFs | $\kappa_{\alpha}(R_{\max})$ inverse-limit = $\kappa_{\alpha}(R_{\max})$ operational = 0 $\checkmark$

Fig. 12 Hasse diagram of the directed system of RAFs in **Optic(Set)** over the extended molecule set $M = \{a, b, c, d, e, f, g, h, i, j, k, l, m\}$ ($|M| = 13$) with food $F = \{a, b\}$ and 11 reactions (Construction 10.7). Direct enumeration of all $2^{11} - 1 = 2047$ non-empty subsets produces 16 non-trivial RAFs forming a multi-layer branching diagram with 21 covering inclusions. Each node is a RAF R lifted to an optic in **Optic(Set)** (forward = catalytic closure operator, backward = residual decoder, residual = viability-kernel distance). The directed-system axioms (reflexivity, transitivity, directedness) are verified by direct computation; the filtered colimit equals the maximal RAF $R_{\max} = \{r_1, \dots, r_{11}\}$. Every RAF node is viability-preserving by construction, so $\kappa_V = 0$ throughout (Proposition 10.8).

and the *deflationary closure operator*

$$D(S) = S \cap \Phi(S). \quad (55)$$

With the catalyst pool taken as the closure $\text{cls}(F)$ (rather than, say, $F \cup \text{prod}(S)$), the fixed points of D are exactly the RAFs of Definition 2.7: a product-pool catalyst condition would be strictly weaker, admitting mutually catalyzing cycles with nothing food-reachable ($r_1 : X \rightarrow Y$, $r_2 : Y \rightarrow X$) that the closure-based condition excludes.

Theorem 11.2 (maxRAF is the terminal coalgebra). *With U finite:*

- (i) D is monotone and deflationary, $D(S) \subseteq S$ for all $S \subseteq U$;
- (ii) the fixed points of D are exactly the RAFs contained in U — the reflexively autocatalytic, food-generated subsets in the sense of 33, 53;
- (iii) the greatest fixed point $\nu D = \bigcap_{n < \omega} D^n(U)$ exists, equals the maximal RAF $R_{\max}$, and is the terminal D -coalgebra in the posetal category 2^U of subsets of U ;
- (iv) the iteration $U \supseteq D(U) \supseteq D^2(U) \supseteq \dots$ stabilizes in at most $|U|$ strict steps and is, step for step, the Hordijk–Steel iterative-removal algorithm;

- (v) *each step is a linear scan of the reaction set against precomputed catalysis and closure tables, so the total work is polynomial in the network size, the same order as the published algorithm.*

The proof is deferred to Appendix A.

Remark 11.3 (Numerical verification). On the thirteen-metabolite, eleven-reaction test case of Section 10.2, the stabilized iterate $D^n(U)$ has cardinality 11 and coincides, member for member, with the maximal RAF of the filtered-colimit construction and with the output of the iterative-removal algorithm (deposited verification artifacts).

Remark 11.4 (Coinductive reading). The theorem locates RAF theory in the coinductive program. The greatest-fixed-point identification is the posetal instance of the general terminal-coalgebra theory for set functors [2, 57], and the terminal-coalgebra perspective on iterated updates is the one actively used in categorical cybernetics [30] and in the functorial semantics of learning optics [26]. What the posetal formulation deliberately avoids is the stronger claim, familiar from the set-functor theory, that some lifted endofunctor on **Set** has a terminal coalgebra identifiable with $R_{\max}$ through weak-pullback preservation [3, 57]: the lattice-level statement above is the one that carries a complete proof, and it is the one the numerical verification exercises.

Conjecture 11.5 (Functorial realization of the seven-optic composite). Let $\mathcal{C}$ be a monoidal category with finite limits, and let $\text{Per}(\mathcal{C})$ be the category of periodic typed systems: objects are pairs $(X, f : X \rightarrow X)$ with $f^{n_X} = \text{id}_X$ for a minimal period $n_X \geq 1$, and morphisms are the period-equivariant maps. The seven-optic composite $T = \mathcal{O}_7 \circ \dots \circ \mathcal{O}_1$ of Construction 6.1 admits a canonical functor $\mathcal{R} : \text{Per}(\mathcal{C}) \rightarrow \mathbf{Optic}(\mathcal{C})$ factoring T — existence along the lines of the optic constructions of 48, together with a periodicity closure argument — and any functor satisfying the SAVGS typing constraint is monoidally naturally isomorphic to $\mathcal{R}$.

Remark 11.6 (Status of the conjecture, and its numerical illustration). The existence clause reduces to standard optic assembly plus a closure argument for periodicity that we do not supply in full, and the uniqueness clause would require a parametricity argument for typed polynomial optics that we likewise do not supply; the statement is therefore recorded as a conjecture rather than a theorem. What is verified numerically is the equivariance phenomenon it predicts: on the periodic system $\mathbb{Z}/4$ with $f(x) = x + 1 \bmod 4$ and a seven-stage composite, the equivariance condition $T \circ f = f \circ T$ holds for all inputs, and the four constant-shift realizations of T preserve equivariance, illustrating the sense in which the realization is unique up to the natural-isomorphism parameter (deposited artifacts).

12 ∞ -Categorical Extension via Homotopy Type Theory

This section extends the composition theorem (Theorem 6.4) to ∞ -categorical settings and homotopy type theory. The strict-fiber 1-categorical optic composition of Section 6 lifts to a homotopy-coherent ∞ -categorical optic composition, with the residual becoming the homotopy pullback of the residuals over the actions (the homotopy product when the actions are trivial) and the Banach fixed-point becoming a contractible ∞ -groupoid canonically identified by the univalence axiom with a single term in the HoTT universe. The lift is non-trivial in three respects: (i) the associativity and unitality of optic composition are no longer strict equalities but 2-cells (homotopies), and their coherence must be established; (ii) the residual must be reinterpreted as a homotopy pullback over the actions (reducing to the homotopy product for trivial actions), which requires presentability of the ambient ∞ -category; (iii) the Banach contraction (Proposition 6.6) becomes a homotopy-contraction on the simplicial mapping space, whose fixed point is a contractible ∞ -groupoid rather than a single point.

12.1 Setup and definitions

Let $\mathcal{C}_\infty$ be a presentable ∞ -category (in the sense of [36], Definition 5.5.1.0) with all small homotopy limits. Examples include ∞ -topoi (in particular, the ∞ -category $\mathcal{S}$ of spaces, the

canonical model of HoTT under the univalence axiom [54]); the derived ∞ -category $D(\mathcal{A})$ of an abelian category; and the ∞ -category $\text{Cat}_\infty^{\text{ex}}$ of stable ∞ -categories, which provides the natural ambient for the CPTP channels of Claim G (Definition 5.1). The restriction to presentable ∞ -categories is necessary to ensure that the homotopy product ${}^h\prod_i \text{Res}_i$ exists, by [36, Proposition 5.5.3.1]; this generalizes the role of finite limits in the 1-categorical setting.

Definition 12.1 (∞ -categorical optic). Let $S, A, R \in \mathcal{C}_\infty$. An ∞ -categorical optic from S to A with residual R is a pair

$$(\text{fwd} : R \times^h S \rightarrow A, \quad \text{bwd} : R \times^h A \rightarrow S)$$

where $R \times^h S, R \times^h A$ are homotopy products in $\mathcal{C}_\infty$ and fwd, bwd are 1-morphisms. The ∞ -category $\mathbf{Optic}(\mathcal{C}_\infty)$ is the cartesian fibration over the twisted-arrow ∞ -category $\text{Tw}(\mathcal{C}_\infty)$ encoding these morphisms. This is the *diagonal* specialization of the 1-categorical optic shape of Definition 2.5: the forward and backward carriers are identified per interface (S at the source, A at the target), and the residual enters both components — the ∞ -categorical lift of the strict feedback form of Remark 2.6 rather than of the general pair form; the general ∞ -level pair form is not constructed here (Open Problem 4). The ∞ -cosmos framework of [47] supplies the ambient model theory.

The homotopy product $R \times^h S$ in Definition 12.1 replaces the strict product $R \times S$ of the 1-categorical case; this is the only structural change required to lift the optic construction. All other optic axioms (associativity of composition, unitality, the residual-update equations) hold up to coherent homotopy by the same argument as [48, Proposition 2.3], with the homotopy coherent Mac Lane theorem for symmetric monoidal ∞ -categories [37, Theorem 2.4.3] replacing the strict Mac Lane coherence theorem.

12.2 The ∞ -categorical composition theorem

Theorem 12.2 (∞ -categorical composition). *Under Construction 6.1 (the seven-fold optic composition), with $\mathcal{C}$ replaced by a presentable ∞ -category $\mathcal{C}_\infty$, the seven-fold composition $T = O_7 \circ \cdots \circ O_1$ is a homotopy-coherent endomorphism of $\mathbf{Optic}(\mathcal{C}_\infty)$. The composition is associative and unital up to coherent homotopy. The residual of T is the iterated homotopy pullback of the residuals over the actions — the fiber-product form computed in the proof sketch and in the concrete verification of Remark 12.10 — which reduces to the homotopy product*

$$\text{Res}_T = {}^h\prod_{i=1}^7 \text{Res}_i \quad \text{in } \mathcal{C}_\infty$$

when the actions are trivial; both exist by presentability of $\mathcal{C}_\infty$ ([36, Proposition 5.5.3.1]).

Proof sketch. The ∞ -category $\mathbf{Optic}(\mathcal{C}_\infty)$ is symmetric monoidal under optic composition, with the tensor product $O_2 \circ O_1$ defined by the homotopy pullback (rather than strict pullback) of the residuals:

$$\text{Res}_{O_2 \circ O_1} := \text{Res}_1 \times_{\text{action}_{\text{fwd}_1}}^h \text{Res}_2.$$

The associativity and unitality are 2-cells (homotopies) in $\mathbf{Optic}(\mathcal{C}_\infty)$. By the homotopy-coherent Mac Lane theorem for symmetric monoidal ∞ -categories [37, Theorem 2.4.3], the space of parenthesizations of T is contractible: any two parenthesizations of T are canonically identified by a coherent homotopy. The seven-fold composition T is therefore well-defined up to canonical homotopy. The residual $\text{Res}_T = {}^h\prod_i \text{Res}_i$ exists because $\mathcal{C}_\infty$ is presentable (hence has all small homotopy limits; [36, Proposition 5.5.3.1]). The forward and backward components of T are 1-morphisms in $\mathcal{C}_\infty$ constructed by the homotopy-coherent analog of the strict-fiber 1-categorical construction of Theorem 6.4. $\square$

12.3 Univalence axiom and the canonical homotopy-fixed-point

Corollary 12.3 (Univalence-identified fixed point). *Suppose further that the realized map $T : S \rightarrow S$ is a homotopy-contraction on the simplicial mapping space $\text{Map}_{\mathcal{C}_\infty}(X, S)$: there exists $k \in (0, 1)$ such that*

$$d(T(x), T(y)) \leq k \cdot d(x, y) \quad \text{for all } x, y \in \text{Map}_{\mathcal{C}_\infty}(X, S).$$

Then T has a contractible ∞ -groupoid of homotopy-fixed-points $\text{holim}_\Delta T$, canonically identified by the univalence axiom [54, §2.9–2.10] with a single term in the HoTT universe $\mathcal{U}$.

Proof sketch. The 1-categorical Banach fixed-point theorem ([8]) gives a unique fixed point in any complete metric space under contraction. The ∞ -categorical extension follows from the ∞ -categorical Yoneda lemma applied to the simplicial mapping space: a homotopy-contraction T on a presentable ∞ -category induces a contraction on each simplicial hom-set $\text{Map}(X, S)$ by functoriality, hence a unique fixed point in each hom-set. The homotopy limit of the constant tower $\text{holim}_\Delta T$ aggregates these fixed points into a contractible ∞ -groupoid. By the univalence axiom (“equivalence is equivalent to equality”), any contractible ∞ -groupoid is equivalent to the unit type 1, so the fixed-point ∞ -groupoid is identified with a single term of the universe $\mathcal{U}$ up to a contractible space of choices — the type of contractible ∞ -groupoids is itself contractible (not $\mathcal{U}$ -equivalent), which is the precise content of the identification. (A model-independent treatment using the language of ∞ -topoi and the univalent universe appears in [21].) $\square$

Remark 12.4 (Proof status of Theorem 12.2 and Corollary 12.3). Both proofs above are *sketches*, and we mark them as such. Specifically: (i) the symmetric monoidal structure of **Optic**($\mathcal{C}_\infty$) under homotopy-pullback residuals is *cited* (homotopy-coherent Mac Lane theorem [37, Thm. 2.4.3]), not constructed — an explicit ∞ -categorical optic composition, with its coherence data, is not in the literature to our knowledge and is recorded as an open problem in Section 15; (ii) the ∞ -categorical Yoneda step in Corollary 12.3’s proof is stated informally (the contraction hypothesis is 1-categorical in nature, applied to simplicial mapping spaces); and (iii) the univalence identification is model-dependent (it assumes a univalent universe in the ambient model). What is fully verified is the concrete instance: the 2-optic composition in $\mathcal{S} = \text{sSet}$ whose homotopy pullback is computed explicitly and shown contractible (Remark 12.10), and the operational Phase III test built on the contractibility idea (Definition 12.5, Remark 12.6).

12.4 Phase III closure-test definition: pathwise + univalence-corrected

The endpoint-only closure test of Definition 3.21 (Phase I) and the pathwise viability tube of Remark 3.13 (Phase II) are the two operational levels at which the autopoiesis closure test has been applied in the operational network benchmarks of the project repository. The ∞ -categorical extension of Section 12 (Theorem 12.2 + Corollary 12.3) provides a third, strictly weaker-than-endpoint-only closure test that formalizes the higher-categorical recovery of components whose endpoint-only trajectory lands in a limit cycle (as for ALA in the Network H benchmark and FBP in the Network I benchmark, Table 4; the AcCoA limit cycle of the Network G benchmark, Remark 12.11, is the case the operationalized criterion does *not* convert).

Definition 12.5 (Phase III closure test: pathwise + univalence-corrected). Let $T : S \rightarrow S$ be the seven-fold optic composition (Theorem 12.2) restricted to the recovery-map of the autopoiesis closure test, with the Banach contraction property (Proposition 6.6, Corollary 12.3). For each component $m_j \in M_{\text{ess}}$, let $\mathcal{R}_j$ denote the ∞ -groupoid of recovery trajectories of m_j after the knockout-and-restore protocol of Definition 3.21: objects are recovery paths $\gamma_a : [0, 1] \rightarrow S$ with $\gamma_a(0)$ the knockout-end state and $\gamma_a(1)$ the recovered state; 1-morphisms are pointwise homotopies through recoveries (reparameterizations of $[0, 1]$ that preserve the viability tube of Remark 3.13); higher morphisms are higher homotopies. The component m_j is *causally internal at the Phase III level* iff any of the following holds:

1. (Phase I endpoint-only) The knockout sends m_j below x_{thresh} at $T/2$ and the recovery raises m_j above x_{thresh} at T (Definition 3.21).
2. (Phase II pathwise viability) The ε -tube around the recovery trajectory lies inside the closed viability kernel (Remark 3.13) AND the recovery trajectory spends at least a fraction $\varphi \in (0, 1]$ of the recovery window above x_{thresh} (operational pathwise viability criterion, $\varphi = 0.4$ used in the present operationalization).
3. (Phase III pathwise + univalence) The ∞ -groupoid $\mathcal{R}_j$ is contractible: any two recovery trajectories are homotopy-equivalent through recoveries (a phase-shifted oscillation is homotopy-equivalent to its phase-zero counterpart via the path reparameterization $\gamma_a \mapsto \gamma_a \circ \rho$, where $\rho : [0, 1] \rightarrow [0, 1]$ is an orientation-preserving homeomorphism), AND by the univalence axiom (Corollary 12.3) the contractible ∞ -groupoid is canonically identified with a single term $T_j^* \in \mathcal{U}$ in the HoTT universe.

The system is autopoietic at the Phase III level iff every $m_j \in M_{\text{ess}}$ is causally internal at any of the three levels.

Remark 12.6 (Operational implementation of Phase III). The Phase III contractibility test is operationalized numerically as follows. Given a component m_j that fails the Phase I endpoint-only test, sample $n = 2$ additional recovery trajectories from perturbed initial conditions (multiplicative Gaussian noise $\sigma = 0.05$ on m_j and on up to 5 species appearing in reactions also involving m_j). Verify that the perturbed recovery trajectories have matching statistical properties with the original recovery: mean, max, and min within relative tolerance $\tau = 0.30$. Phase-shifted oscillations (the canonical limit-cycle recovery regime of AcCoA in Network G and ALA in Network H) have matching statistics under reparameterization, hence pass the test; trajectories with genuinely different statistical behavior (e.g., diverging exponential growth vs. stable recovery) fail. The contractibility test thus approximates the higher-categorical ∞ -groupoid contractibility condition of Corollary 12.3 in the sense that it accepts phase-shifted homotopy-equivalent recoveries and rejects non-homotopy-equivalent ones.

Proposition 12.7 (Poincaré/averaging content of the pathwise levels). *Suppose the post-knockout trajectory of m_j converges to a periodic orbit x^{lc} of period τ_c (the limit-cycle recovery regime). Then the occupation fraction of the pathwise criterion,*

$$\varphi_T = \frac{1}{T} \int_0^T \mathbf{1}\{x_j(t) \geq x_{\text{thresh}}\} dt,$$

converges as $T \rightarrow \infty$ to the phase average $\varphi_\infty = \tau_c^{-1} \int_0^{\tau_c} \mathbf{1}\{x_j^{\text{lc}}(t) \geq x_{\text{thresh}}\} dt$, and φ_∞ is invariant under phase shifts of the orbit: it is a function of the orbit, not of the initial phase. The Phase I endpoint statistic $x_j(T)$, by contrast, samples a single phase and can fail purely by phase alignment — the mechanism by which limit-cycle recoveries fail the endpoint test while their orbits spend a positive fraction of the window above the threshold (fractions 0.498 and 0.853 in the two benchmark cases, Table 4).

Proof. For a τ_c -periodic input $g(t) = \mathbf{1}\{x_j^{\text{lc}}(t) \geq x_{\text{thresh}}\}$, the integral over $[0, T]$ equals $\lfloor T/\tau_c \rfloor \int_0^{\tau_c} g dt + O(1)$, so $\varphi_T \rightarrow \tau_c^{-1} \int_0^{\tau_c} g dt$; a phase shift $t \mapsto t + s$ changes the integral over one period by at most $|s|$ worth of boundary indicator, which is absorbed into the $O(1)$ remainder and vanishes in the limit. $\square$

Remark 12.8 (Classical reading of the Phase III criterion). This is the Poincaré-section/averaging reading of the pathwise levels: the occupation fraction is an orbit average, the finite-window statistic a stroboscopic sample of it, and transversal-section (Poincaré) sampling the phase-unbiased way to estimate the same quantity [28, 51]. Two consequences. First, the path reparameterization $\gamma_a \mapsto \gamma_a \circ \rho$ that item (3) of Definition 12.5 uses to identify phase-shifted recoveries is the

∞ -groupoid packaging of exactly this phase invariance: the classical content of “limit cycles are not endpoint failures” is that the pass statistic should be a function of the orbit rather than of the sampling phase, and the occupation fraction is the minimal such statistic. Second, for bounded recoveries that are not eventually periodic, φ_T converges along subsequences to occupation measures supported on the ω -limit set, so the pathwise level is the occupation-measure form of the ω -limit viability containment of Proposition 3.23(ii): the three-level disjunction of Definition 12.5 is a finite-window, phase-aware approximation hierarchy for that single classical property.

12.5 Falsifiable prediction and operationalization

Remark 12.9 (Falsifiable predictions). The ∞ -categorical extension makes three falsifiable predictions:

1. The homotopy limit $\text{holim}_\Delta T$ of the constant tower of T is contractible, providing a single canonical term $T^* \in \mathcal{U}$. Failure of contractibility falsifies the ∞ -categorical extension.
2. The strict-fiber 1-categorical fixed-point (Banach fixed-point in **Optic**($\mathcal{C}$), Theorem 7.5) matches the ∞ -categorical homotopy-fixed-point T^* under the inclusion **Optic**($\mathcal{C}$) $\hookrightarrow$ **Optic**($\mathcal{C}_\infty$), to within the analytical tolerance $\prod_i \text{Lip}(f_i) = 0.697 < 1$ (Theorem 7.5).
3. The pathwise recovery in the autopoiesis closure test (Definition 3.21) is contractible (any two recovery trajectories are homotopic through recoveries), even in the limit-cycle regime where the endpoint-only check fails (Remark 3.13).

Predictions 2 and 3 are verified:

- Prediction 2 is verified numerically across the 375-configuration robustness grid (project repository); cf. the analytic per-optic Lipschitz bound of Theorem 7.5 ($\prod_i \text{Lip}(f_i) = 0.697 < 1$), which holds within machine precision in every configuration.
- Prediction 3 is verified in the ALA limit-cycle regime of Network H and the FBP limit-cycle regime of Network I (Table 4): the endpoint-only closure-test verdict is HOMEOSTATIC but the pathwise recovery passes (fractions 0.498 and 0.853 above threshold) with the contractibility check passing, converting both to Phase III recoveries — oscillations homotopy-equivalent through recoveries, by the pathwise viability tube of Remark 3.13. (The AcCoA limit cycle of Network G, Remark 12.11, is the case the operationalized criterion does *not* convert.)

Remark 12.10 (Operationalization via explicit simplicial homotopy-limit computation on a 2-optic composition in $\mathcal{S}$). Prediction 1 (contractibility of $\text{holim}_\Delta T$) is verified on a small concrete example (deposited verification artifacts). The setup is a 2-optic composition $O_2 \circ O_1$ in the ∞ -category $\mathcal{S} = \text{sSet}$ (Kan-completed, the canonical model of the HoTT universe $\mathcal{U}$ under the univalence axiom [54]):

- $O_1 : S = \Delta^0 \rightarrow A_1 = \Delta^1, R_1 = \Delta^0$; $\text{fwd}_1 : R_1 \times S = \Delta^0 \rightarrow A_1 = \Delta^1$ picks vertex 0; $\text{bwd}_1 : R_1 \times A_1 = \Delta^1 \rightarrow S = \Delta^0$ constant.
- $O_2 : A_1 = \Delta^1 \rightarrow A_2 = \Delta^0, R_2 = \Delta^0$; $\text{fwd}_2 : R_2 \times A_1 = \Delta^1 \rightarrow A_2 = \Delta^0$ constant; $\text{bwd}_2 : R_2 \times A_2 = \Delta^0 \rightarrow A_1 = \Delta^1$ picks vertex 1.

The composed residual is (Theorem 12.2)

$$\text{Res}_{O_2 \circ O_1} = R_1 \times_{A_1, \text{action}_1, \text{action}_2}^h R_2 = \text{holim}(\Delta^0 \xrightarrow{\text{action}_1=0} \Delta^1 \xleftarrow{\text{action}_2=1} \Delta^0),$$

the homotopy pullback of the two points $\{0\}, \{1\} \hookrightarrow \Delta^1$ over the interval Δ^1 . By the Bousfield–Kan construction [15], the homotopy pullback of a cospan $A \xrightarrow{f} C \xleftarrow{g} B$ in sSet is the simplicial set of triples (a, b, p) with $a \in A, b \in B, p : \Delta^1 \rightarrow C, p(0) = f(a), p(1) = g(b)$. For our setup

($A = B = \Delta^0$, $C = \Delta^1$, f picks 0, g picks 1), this is the path space $P_{0 \rightarrow 1}(\Delta^1)$ of paths from 0 to 1 in Δ^1 , which is contractible.

The script enumerates the k -simplices of this homotopy pullback explicitly for $k = 0, 1, 2, 3, 4$. A k -simplex is a simplicial map $p : \Delta^1 \times \Delta^k \rightarrow \Delta^1$ with the boundary constraints $p|_{\{0\} \times \Delta^k} \equiv 0$, $p|_{\{1\} \times \Delta^k} \equiv 1$. The vertex-assignment $p(i, j) = i$ ($i \in \{0, 1\}$, $j \in \{0, \dots, k\}$) is the unique order-preserving map satisfying these constraints, so each k -simplex is unique; the $k = 0$ simplex is the identity path $0 \rightarrow 1$ and all $k \geq 1$ simplices are degenerate (each is s_j of the $(k - 1)$ -simplex for some j). The simplicial homology is $H_0 = \mathbb{Z}$ and $H_n = 0$ for $n \geq 1$; the homotopy groups are $\pi_0 = 1$, $\pi_n = 0$ for $n \geq 1$. The homotopy pullback is therefore **contractible** as a simplicial set.

By the univalence axiom [54, §2.9–2.10], the contractible ∞ -groupoid $\text{holim}_\Delta T$ of homotopy-fixed-points is canonically identified with a single term $T^* \in \mathcal{U}$ in the HoTT universe, exactly as predicted by Prediction 1 of Remark 12.9. The verification **passes**: failure of contractibility on this concrete example would have falsified Theorem 12.2 and Corollary 12.3. Outputs: `hott_simplicial_holim.{png, csv, txt}`.

12.6 Implications

The ∞ -categorical extension lifts the 1-categorical composition theorem to the homotopy-coherent setting (with the proof-status caveats of Remark 12.4). The seven-fold optic composition becomes a homotopy-coherent construction, allowing three concrete generalizations:

- *Stochastic programming*: strict equality of residuals is replaced by homotopy equivalence, and the composition theorem holds up to coherent homotopy. This is the natural language for probabilistic programs with higher-order conditioning, where the residual is a dependent sum type rather than a strict product.
- *Quantum information*: the CPTP channels of Claim G (Definition 5.1) are 1-morphisms in the ∞ -category $\text{Hilb}_\infty^{\text{fd}}$ of finite-dimensional Hilbert spaces (more precisely, 2-morphisms in the $(\infty, 2)$ -category of CP^* -categories); their composition is a special case of the ∞ -categorical optic composition.
- *Higher-order probability*: the residual Res_T is the iterated homotopy pullback of the residuals over the actions (the homotopy product ${}^h\prod_i \text{Res}_i$ for trivial actions), which under univalence is a dependent sum type $\sum_{r_1 : \text{Res}_1} \cdots \sum_{r_7 : \text{Res}_7} \text{Path}(r_1, r_7)$, encoding the coherent choices of residuals across the seven-fold composition.

Remark 12.11 (AcCoA limit cycle in the Network G benchmark). Network G — a 42-component catalytic maintenance network of the project repository’s benchmark series, whose full specification and execution logs are deposited there — achieves $41/42 = 97.6\%$ causally internal under the endpoint-only closure test, with the single AcCoA component marked HOMEOSTATIC despite AcCoA recovering to a limit-cycle oscillation between 0 and ~ 13.3 (averaging ~ 6.6 over a period, in excess of the viability threshold 0.1 along most of the period). The pathwise viability Remark 3.13 already addresses this: the viability tube of radius ε around the recovery trajectory $\gamma_a([0, 1])$ lies inside the closed viability kernel, and the pathwise recovery is contractible (any two oscillating recovery trajectories are homotopy-equivalent through recoveries). Under the univalence-identified fixed point (Corollary 12.3), the limit-cycle reading would reinterpret such a “failure” as a higher-categorical recovery (the contractible space of recovery oscillations). The operationalized Phase III criterion, however, does *not* convert Network G’s AcCoA: Table 4 records $41/42$ in both phases for Network G, and the deposited pathwise record for the AcCoA family (at Network J, the only one on deposit) spends a fraction 0.275 of the window above the viability threshold with mean 3.59 and fails the contractibility check — an oscillation mostly below threshold, not a recovery. The Phase III conversions of the battery occur elsewhere: at Network H (the ALA limit cycle, pathwise fraction 0.498, mean 49.8; $43/44 \rightarrow 44/44$) and at Network I (the FBP limit cycle, pathwise fraction 0.853, mean 47.4; $45/46 \rightarrow 46/46$). AcCoA is repaired

at the endpoint level only by the designed ACS1/2 step (Network K, 52/52 at Phase I). The higher-categorical reading applies to the converted limit-cycle components, not to Network G’s AcCoA.

13 Operational Network Benchmarks of the Closure Test

The three-phase closure test (Definitions 3.21 and 12.5) is operationalized on a battery of biochemical networks with two arms of distinct evidential roles. As in Section 8, the evidence is computational: the test is a terminating algorithm, exercised here on designed networks and anchored against one external dataset. The *designed progression* is a sequence of integrated metabolic–gene-regulatory networks in which each modification is a hypothesis about what organizational closure requires; it measures the test’s sensitivity, since each added redundant route converts precisely the components the test flags as failing. The progression sits in the network-expansion territory of Handorf and Ebenhöf [29]; the structural-instrument benchmark of Section 13.7 compares the two instruments directly. The *fixed-model arm* applies the test to the unmodified BiGG iJO1366 model of *E. coli* [46] and validates the test as a predictor of independently computed FBA essentiality. All simulation protocols and per-component verdict records are deposited (Data Availability).

13.1 Test protocol

For each purportedly self-maintained component m_j : set the internal repair flux of m_j to zero (every reaction producing m_j is disabled), keep the external food supply fixed, apply the regeneration rules for $T = 200$ time steps with first-order degradation (rate $\delta = 0.05$) and Michaelis–Menten-type catalytic kinetics ($k_{\text{cat}} = 0.5$), and record whether m_j falls below the viability threshold $x_{\text{thresh}} = 0.1$; then restore the repair pathway and record whether m_j recovers above the threshold. The component is *causally internal* iff the knockout destroys it and the restoration recovers it; the system is autopoietic iff every essential component is causally internal.

The operational protocol implements steps (i)–(iv) of Definition 3.21 plus collapse–reversal; the restoration control of step (v) — the return of all concentrations to the reference state x^* within the declared window — is *not* part of the recorded verdicts, and the definition is thereby aligned with the protocol by declaration: the verdicts below certify the weakened, implemented criterion. The overshoot rows of Table 3 show what the omitted control would catch: rows *c* and *f* recover to 3.84 and 443.8 against baselines 0.45 and 2.65 ($8.5\times$ and $167\times$ overshoot), and row *g* recovers to 164.2 against 84.4 ($1.9\times$) — recoveries above the viability threshold, not restorations of the reference state.

13.2 The designed progression

The progression starts from two real networks: the Hordijk–Steel food-generated RAF (seven molecules, five reactions) and an *E. coli* core-metabolic subnetwork (glycolysis and the TCA cycle, ten non-food species, ten reactions, derived from iJO1366). The verdicts on the two starting networks are the anchors of the whole battery.

Proposition 13.1 (Anchor verdicts). *Direct simulation of the protocol gives, for the Hordijk–Steel RAF (non-food components c, d, e, f, g), the per-component outcomes of Table 3: 2/5 components causally internal. For the core-metabolic subnetwork, 0/10 components are causally internal. The static catalytic closure of the RAF — verified in Section 10 — does not translate into endogenous regeneration under the dynamical test: the failing components are regenerated by reactions whose catalysts are themselves knockout-fragile, so their repair pathway collapses when their own catalyst is removed. The bare metabolic network fares worse still: every core metabolite fails the two-sided test.*

Table 3 Anchor network: the Hordijk–Steel food-generated RAF under the closure test. Baseline, knockout-final, and recovery-final concentrations of the five non-food components; the causally-internal verdict requires collapse under knockout and recovery under restoration.

Component	Baseline	Knockout	Recovery	Causally internal
<i>c</i>	0.45	0.00	3.84	yes
<i>d</i>	0.81	0.00	0.00	no
<i>e</i>	8.63	0.00	0.00	no
<i>f</i>	2.65	0.00	443.8	yes
<i>g</i>	84.4	0.37	164.2	no

Re-testing the same core metabolites in the full genome-scale iJO1366 model converts 9/10 of them to causally internal, and integrating enzyme synthesis into the core network (a metabolic–gene-regulatory integration with 17 essential components) yields 5/17. The designed progression then adds, at each step, one redundant isozyme pair with a stated cascade-breaking role; the monotone verdict sequence is Table 4. The Phase I endpoint-only verdict reaches 52/52 only at the final step, whose ACS1/2 route repairs the AcCoA failure that persists through the G–J segment; along the progression, the pathwise Phase III criterion converts exactly the two limit-cycle oscillation failures — ALA at H (43/44 → 44/44) and FBP at I (45/46 → 46/46) — while the AcCoA failure (pathwise fraction 0.275, contractibility FAIL) is not converted until the designed repair (Sections 12.4 and 8.5).

Table 4 The network battery. Phase I is the fraction of essential components causally internal at the endpoint-only criterion; Phase III is the pathwise, univalence-corrected criterion of Definition 12.5. Each designed step adds one redundant isozyme pair breaking the named cascade. The last column disambiguates the counts: the Phase I failing components and their mode (lc = limit-cycle oscillation, converted by Phase III; res. = residual oscillation, not converted), traced to the deposited per-component verdict records.

Network	Description	Phase I	Phase III	Phase I failures
A	Hordijk–Steel RAF (anchor)	2/5	2/5	<i>d, e, g</i> (catalyst-fragile)
B	<i>E. coli</i> core metabolism (anchor)	0/10	0/10	all 10
C	B’s metabolites in full iJO1366	9/10	9/10	FBP
D	B + enzyme synthesis (MR–GR)	5/17	5/17	12 components
E	closed MR–GR network	24/29	24/29	G6P, PYR, AcCoA, ALA, ASP
F	E + ALT3/4 (ASP–PYR route)	29/31	29/31	FBP, PYR
G	F + ALT5/6 (αKG route)	41/42	41/42	AcCoA (lc, not conv.)
H	G + ASPAT3/4 (ASP cascade)	43/44	44/44	ALA (lc, conv.)
I	H + ALT7/8 (ALA reversible)	45/46	46/46	FBP (lc, conv.)
J	I + ALDO3/4 (FBP reversible)	49/50	49/50	AcCoA (res., not conv.)
K	J + ACS1/2 (NAD ⁺ -independent)	52/52	52/52	—

The designed progression is complemented by a cross-domain transfer statement that uses the closure structure rather than merely measuring it.

Proposition 13.2 (RAF closure gives a Zeno-schedule lower bound). *Let R be a RAF with closure set $C(R) \subseteq M$, $|C(R)| = N_{\text{RAF}} \geq 1$, realized through the CPTP–Zeno lift of Section 8.5 — each closure catalyst mapped to a CPTP channel with dissipative gap (Definition 8.9). Under the binary-cascade model of closure bootstrap (each generation of catalysts doubles the producible set, so the bootstrap depth is $d_R = \log_2 N_{\text{RAF}}$) and per-step gap normalization (the effective gap is the per-step gap divided by the $1 + d_R$ propagation steps of the composition), the projected Zeno renewal rate satisfies*

$$\tau_{\text{Zeno}} \geq \frac{1}{1 + \log_2(N_{\text{RAF}})} \quad (N_{\text{RAF}} \geq 1). \quad (56)$$

The bound is monotonically decreasing in N_{RAF} : larger closures predict slower Zeno renewal, a direction-of-effect prediction unavailable without the composition machinery.

Proof. Under the stated model: (i) the closure set $C(R)$ is self-maintaining, and the binary-cascade model fixes the bootstrap depth $d_R = \log_2 N_{\text{RAF}}$ generations; (ii) each catalyst maps to a CPTP channel whose dissipative part has gap $\Delta_{\text{step}} > 0$ (Proposition 8.10); (iii) composing $1 + d_R$ propagation steps normalizes the effective gap to $\Delta_{\text{eff}} = \Delta_{\text{step}} / (1 + d_R)$; (iv) the Zeno renewal rate is the reciprocal of the effective gap normalized by the per-step scale, $\tau_{\text{Zeno}} = 1 / (1 + d_R)$, which is (56). The model assumptions — binary cascade, per-step gap normalization — are part of the statement, not consequences of the closure definition. $\square$

Remark 13.3 (Verification on the designed lineage). The bound is verified on all seven networks of the designed lineage E–K: the closure counts $N_{\text{RAF}} \in \{24, 29, 41, 43, 45, 49, 52\}$ all satisfy $\tau_{\text{Zeno}}^{\text{sim}} \geq 1 / (1 + \log_2 N_{\text{RAF}})$, with the ratio of simulated rate to bound ranging over 44.9–179.6 — the bound holds with the margin reflecting the gap normalization, and the monotone direction-of-effect prediction is confirmed along the lineage (deposited artifacts).

13.3 The persistent-homology test for Phase III

The Phase III criterion of Definition 12.5 can be strengthened from an endpoint-tolerance check to a homotopy-invariant one.

Definition 13.4 (Persistent-homology contractibility test). For each closed-loop recovery trajectory of the Phase III test, construct the point cloud of its time-series samples and compute persistent homology barcodes [55]. The trajectory is *contractible* iff

$$\text{Betti}_0 = 1 \wedge \text{Betti}_1 = 0 \wedge \text{Betti}_2 = 0, \quad (57)$$

where Betti_n counts the persistent n -dimensional features with persistence at least 10% of the cloud diameter.

Proposition 13.5 (Classification accuracy of the homological test). *Applied to five test cases — a contractible disk, a circle S^1 , a torus T^2 , the Network K AcCoA recovery trajectory, and the Network J AcCoA limit-cycle trajectory — the test classifies all five correctly: the disk and the Network K recovery have $\text{Betti}_0 = 1$ and $\text{Betti}_1 = 0$ (contractible); the circle, the torus, and the Network J limit cycle have $\text{Betti}_1 \geq 1$ (non-contractible, the persistent 1-dimensional hole being the limit cycle). The homological test and the pathwise Phase III criterion agree on Networks K and J; the homological test additionally separates the tight limit cycle (small endpoint dispersion, persistent hole) from genuine recovery, which an endpoint-tolerance check cannot.*

Proof. Direct computation of the barcodes on the five point clouds (deposited artifacts); the Betti-number readings are as stated, and the agreement and disagreement patterns with the endpoint criterion are as stated. $\square$

13.4 Fixed-model validation on iJO1366

Construction 13.6 (Closure test on the unmodified model). Take the BiGG iJO1366 model (1,805 metabolites, 2,583 reactions, 1,367 genes) without modification. For each cytosolic reaction r with genes and cytosolic products, compute the closure-test dependency structure — the fraction of each product’s production that flows through r at baseline — and compare the resulting dependency-ratio criterion against single-reaction-deletion FBA essentiality computed independently on the same model.

Proposition 13.7 (Closure-test dependency ratios predict FBA essentiality). *On the complete set of $n = 1,638$ cytosolic reactions with genes and cytosolic products (no sampling), the dependency-ratio criterion at the optimal threshold $\tau^* = 0.5$ agrees with FBA single-reaction-deletion essentiality with Cohen’s $\kappa = 0.835$, MCC = 0.841, $F_1 = 0.863$, precision 0.783, recall 0.960; the*

ROC AUC of the dependency ratio against FBA essentiality is 0.968. For comparison, the binary sole-producer criterion (a product is judged fragile iff it has a single producer) achieves $\kappa = 0.206$ on the same set: an elevation factor of 4.05 for the dynamical dependency structure over the purely structural criterion. The agreement is at the reaction level; at the metabolite level, agreement with gene-level essentiality is weak ($\kappa = -0.08$), reflecting the semantic difference between regeneration after restoration (the closure test) and biomass maximization under knockout (FBA gene essentiality).

Proof. Direct computation on the unmodified model (deposited artifacts): the confusion matrix, the κ /MCC/ F_1 /precision/recall values, and the ROC curve are the deposited outputs of the two independent computations (closure-test dependency ratios; FBA single-reaction deletions) on the same model. $\square$

Remark 13.8 (Threshold tuning of the fixed-model arm). The reaction-level threshold $\tau^* = 0.5$ of Proposition 13.7 is tuned on the full evaluation set ($n = 1,638$; no reaction-level held-out split); the fixed-model arm is thereby an internal cross-validation on an unmodified model, and the external-data anchor of the next subsection carries the outside information.

13.5 External anchor: the Keio collection

The fixed-model arm validates the closure test against independently computed FBA essentiality — an internal cross-check. This subsection adds the external data anchor: the growth-phenotype panel of the Keio single-gene-deletion collection of *E. coli* K-12 [7] ($\sim 4,000$ measured single-gene-deletion growth phenotypes on glucose minimal and rich media). The reconstruction whose in-silico phenotypes anchor the transitive arm is iJO1366 [45], whose essentiality predictions were validated against the experimental Keio collection at 93.4% accuracy on glucose minimal media by its authors; the cross-rebuild arm uses iML1515 [41]. The predictor validated throughout is the *unmasked* flux-rerouting magnitude

$$\kappa_V^{\text{flux}}(g) = \sum_{r \in \Delta R(g)} (v_r(\text{KO}) - v_r(\text{WT}))^2, \quad \Delta R(g) = \{r : |v_r(\text{KO}) - v_r(\text{WT})| > 10^{-6}\},$$

computed by flux-balance analysis on the deposited model files — a statistic that carries no essentiality call and is therefore free of the circularity identified in Remark 3.26. Three studies (deposited scripts and result records; Data Availability):

Proposition 13.9 (Transitive Keio anchor (E12)). *For every gene g in iJO1366 ($n = 1,367$; wild-type FBA biomass 15.444; essentiality threshold 5% of wild type), the in-silico phenotype $y(g) = 1$ iff $b_{\text{KO}}(g) < 0.05 b_{\text{WT}}$ gives $289/1,367 = 21.1\%$ essential (published Keio fraction $\sim 18\%$). Calibration against κ_V^{flux} : Pearson $r(\log \kappa_V^{\text{flux}}, \Delta b) = +0.370$ ($p = 1.8 \times 10^{-45}$); Spearman $\rho = +0.390$ ($p = 6.7 \times 10^{-51}$); partial r controlling the number of GPR reactions $+0.364$ ($p = 5.1 \times 10^{-44}$); bootstrap 95% CI $[0.351, 0.389]$. Held-out 70/30 logistic regression on $\log \kappa_V^{\text{flux}}$ ($n = 411$ test genes, 87 essential): ROC AUC = 0.953, sensitivity 0.759, specificity 0.948, precision 0.795, $F_1 = 0.777$, MCC = 0.719; confusion (tn, fp, fn, tp) = (307, 17, 21, 66). Top- K precision against the 21.1% base rate: $P@200 = 0.805$ ($3.81\times$ lift), $P@100 = 0.680$, $P@10 = 0.700$ ($3.31\times$). The medium-corrected glucose-only re-run of Proposition 13.13 reproduces this essentiality set exactly (no label changes) and strengthens every rank-level statistic.*

Proposition 13.10 (Direct primary-source validation (E15)). *Matching the 1,367 genes to the raw Keio essentiality call of Baba et al.’s Supplementary Table 7 (values $\{E, N, u\}$; after de-duplication of 867 repeated b -numbers, 3,144 unique b -numbers remain) by Blattner b -number gives 1,212 matched genes (88.7% coverage); the binary subset (drop u) is $n = 1,206$ (130 E , 1,076 N ; base rate 10.8%). Direct validation, no transitive hop: Pearson $r(\log \kappa_V^{\text{flux}}, \text{Keio-}E) = +0.085$ ($p = 3.3 \times 10^{-3}$); Spearman $\rho = +0.228$ ($p = 9.6 \times 10^{-16}$); ROC AUC 0.713. Held-out 70/30 logistic regression: ROC AUC 0.757, sensitivity 0.923, specificity 0.180, precision 0.120, MCC*

0.085, confusion (58, 265, 3, 36) on $n = 362$ (39 essential). The rank-level association is significant and the AUC is moderate; the point-calibration is weak — κ_V^{flux} is heavy-tailed, and the top of its ranking contains both Keio-E genes and glucose-minimal-only essentials that the original screen scored N on rich media. Supplementary Table 6 stratification (within the matched set): high-confidence essentials (Keio = E and PEC = E) $n = 84$; low-confidence essentials (Keio = E, PEC = N) $n = 35$. Under the medium-corrected re-run (Proposition 13.13) the rank-level association strengthens and the confidence strata are unchanged.

Proposition 13.11 (Cross-rebuild stress test (E16)). Repeating the direct protocol on the iML1515 rebuild (1,516 genes; wild-type FBA biomass 0.926; 286/1,516 = 18.9% essential in silico; 1,325 genes matched to the raw Keio call, 114 E) gives Pearson $r = -0.018$ ($p = 0.52$), Spearman $\rho = -0.070$ ($p = 0.011$), ROC AUC 0.428: the direct one-hop association does not transfer to the rebuild — a negative verdict, reported as such. The model-gap candidates shrink from 30 (iJO1366) to 13 (iML1515) and the in-silico essentiality fraction moves toward the experimental fraction; the direct association flips sign. The flip is itself medium-induced: under the corrected glucose-only medium (Proposition 13.13) the same protocol transfers positively, with association strength comparable to the iJO1366 arm.

Remark 13.12 (Medium construction and the wild-type optima). The E12 and E16 protocols use different medium constructions. The E12 medium re-opens the “minerals” block including the trehalose exchange at unlimited uptake (-337 mmol/gDW/h at the optimum), giving the wild-type value 15.444 — closing the trehalose exchange alone gives 0.982 h^{-1} , the canonical iJO1366 glucose-minimal optimum — while the E16 protocol re-opens the same exchange at -10 (effective uptake -6.4), giving 0.926 against a glucose-only optimum of 0.822. The two wild-type values are therefore neither mutually comparable growth rates nor a stoichiometry effect. Essentiality calls use the relative 5%-of-wild-type threshold within each model, which is invariant to the absolute scale; the identity of the available carbon sources, however, enters every knockout solution, so the E12/E15/E16 essentiality sets, the κ_V^{flux} values, and the calibration statistics inherit the medium asymmetry. The glucose-only re-run of Proposition 13.13 closes the trehalose exchange as the sole change: both in-silico essentiality sets are unchanged (zero label changes in either model), every rank-level association strengthens, and the cross-rebuild negative verdict of Proposition 13.11 does not survive the correction.

Proposition 13.13 (Medium-corrected glucose-only re-run). Re-executing the three studies with the trehalose exchange closed (the sole change; wild-type optima 0.98237 on iJO1366 and 0.82180 on iML1515, the canonical glucose-minimal values) leaves both in-silico essentiality sets unchanged — 289/1,367 and 286/1,516, zero label changes in either model — and strengthens every rank-level statistic. Transitive arm: Pearson $r(\log \kappa_V^{\text{flux}}, \Delta b) = +0.603$ ($p = 2.2 \times 10^{-136}$), Spearman $+0.591$, partial $r = +0.601$, bootstrap 95% CI [0.578, 0.631]; held-out ROC AUC 0.977, sensitivity 0.885, specificity 0.981, MCC 0.882, confusion (318, 6, 10, 77), $P@200 = 0.915$. Direct arm: Pearson $r(\log \kappa_V^{\text{flux}}, \text{Keio-E}) = +0.230$ ($p = 6.7 \times 10^{-16}$), Spearman $+0.256$ ($p = 2.0 \times 10^{-19}$), ROC AUC 0.737; held-out ROC AUC 0.763, sensitivity 0.846, specificity 0.607, MCC 0.283; the confidence strata are unchanged (84/35) with the high-confidence AUC 0.713, the model-gap set is unchanged at 30, and the medium-mismatch stratum (in-silico essential, Keio N) shrinks from 217 to 180 genes. Cross-rebuild arm: Pearson $+0.376$ ($p = 9.6 \times 10^{-46}$), Spearman $+0.304$ ($p = 1.0 \times 10^{-29}$), ROC AUC 0.813, with the binary matched set unchanged (1,325 genes, 114 essential) and the model-gap set unchanged at 13: the non-transfer recorded as the negative verdict of Proposition 13.11 does not survive the medium correction.

Remark 13.14 (Why the labels are invariant but the statistics are not). The label invariance is consistent with the wide separation of the relative knockout ratio $b_{\text{KO}}/b_{\text{wt}}$ around the declared 5% line: for every gene the ratio sits near zero or near one in both media, so the roughly sixteen-fold change in the wild-type optimum moves no gene across the threshold, in either reconstruction. The statistic κ_V^{flux} , by contrast, aggregates absolute squared flux changes: the unlimited-trehalose

optimum routes large fluxes through pathways irrelevant to the glucose-minimal phenotype, and removing that spurious rerouting removes a dilution of the statistic. This is why every association strengthens without a single essentiality label changing, and why the cross-rebuild sign flip reverses: the trehalose fluxes enter the iML1515 knockout solutions differently than the iJO1366 ones, contaminating the rank comparison itself.

Proposition 13.15 (Second perturbation probe: oxygen-limited medium). *Tightening the oxygen exchange — the electron-acceptor axis, with the glucose-only medium otherwise unchanged, the protocol identical, and labels declared at 5% of each level’s own wild type — gives a dose response $EX_o2_e \in \{-10, -5, -2.5\}$ on iJO1366 and $\{-5, 0\}$ on iML1515. Wild-type optima fall from 0.982 to 0.711, 0.491, 0.367 (iJO1366; acetate–formate overflow at -10 , ethanol–formate–acetate fermentation at -5 and -2.5) and from 0.822 to 0.353 and 0.134 (iML1515), yet the essentiality sets are unchanged at every non-anaerobic level: 289/1,367 at each of -10 , -5 , -2.5 and 286/1,516 at -5 — zero label changes, Cohen’s $\kappa = 1.000$ throughout — and the label-based strata (model gaps 30 and 13; iJO1366 medium-mismatch 180) are correspondingly unchanged. The association degrades only gracefully: transitive calibration $r(\log \kappa_V^{\text{flux}}, \Delta b) = +0.775$ at -10 — above the unlimited-oxygen value $+0.603$, with held-out ROC AUC 0.9999, MCC 0.993, $P@200 = 1.0$ — then $+0.607$ at -5 and $+0.519$ at -2.5 (bootstrap CIs $[0.584, 0.628]$ and $[0.496, 0.542]$; AUCs 0.980 and 0.971); on iML1515 at -5 , $r = +0.559$ (AUC 0.994). Direct-arm AUC is stable across the gradient (0.732, 0.744, 0.726 on iJO1366; 0.818 on iML1515, against 0.813 unlimited) while the direct Pearson decays from $+0.264$ to $+0.162$ as limitation deepens — the rank metric is the robust one, the point-biserial tail the sensitive one. At the anaerobic endpoint the two reconstructions diverge. iJO1366’s wild-type optimum is exactly zero: de novo ubiquinone-chain synthesis requires the oxygen-dependent hydroxylase step (2-octaprenylphenol $\rightarrow$ 2-octaprenyl-6-hydroxyphenol, OPHHX) and pyridoxal-5'-phosphate biosynthesis the oxygen-dependent step (PDX5POi), neither with an anaerobic alternative in this reconstruction (the second hydroxylase step does have one), so the biomass reaction is infeasible at positive rate and the 5%-relative threshold degenerates — an oxygen trace of 0.01 mmol/gDW/h restores 0.231; this is reported as a model-level degeneracy of the label construction, not a label change. iML1515, which carries the oxygen-free route for the blocking step and grows anaerobically (0.134), provides the non-degenerate endpoint: 286 $\rightarrow$ 292 labels, six gains (eno, pgk, gapA, gpmA, gpmM, hemN) and no losses, $\kappa = 0.987$, Jaccard 0.979, five of the six flips in glycolysis/gluconeogenesis (a thirty-fold enrichment over the genome share); the transitive calibration falls to $r = +0.258$ (AUC 0.682), the direct arm to $r = +0.125$ (AUC 0.673), and the model-gap set is 12. (fabZ is essential in every regime through OGMEACPD/OPMEACPD, the sole route to the ubiquinone side chain, a biomass requirement that every regime carries; its apparent loss at the anaerobic endpoint is a solver-tolerance artifact, §13.6.)*

Remark 13.16 (What the second axis adds to the invariance claim). The carbon-source correction (Proposition 13.13) and the oxygen probe (Proposition 13.15) agree at every non-anaerobic level: a medium perturbation that leaves the growth regime intact — however strongly it reduces the wild-type optimum (63% at $EX_o2_e = -2.5$) or reroutes the flux distribution (overflow, then fermentation byproducts) — moves no gene across the 5% relative line in either reconstruction, and the curvature–phenotype association is the invariant object (held-out AUC ≥ 0.97 throughout, rising to 0.9999 under moderate limitation). The anaerobic endpoint locates the boundary of the claim: when the growth mode switches to exclusive substrate-level phosphorylation, the lower-glycolysis module becomes uniquely load-bearing — the six gained genes retain 70–84% of wild type aerobically, where the Entner–Doudoroff and pentose-phosphate routes carry 8.4 of 10 glucose around an enolase knockout, but are strictly lethal anaerobically — *hemN*’s oxygen-independent heme route becomes required (its aerobic knockout is exactly neutral; the oxygen-dependent route covers), and no gene is lost: *fabZ* is essential in every regime through its two quinone-side-chain dehydratase reactions (OGMEACPD/OPMEACPD, its only reactions without a *fabA* alternative; restoring exactly those two restores the aerobic optimum 0.822) — the sole route to a biomass

requirement that every regime carries. The label set thus carries the environmental regime, and the plain-vertex association reading degrades precisely when the regime changes (under canonical selection the association is restored across the switch: §13.6) — so the bimodal-ratio mechanism of Remark 13.14 and the regime-switch mechanism bracket the invariance claim from both ends, and the medium-robustness statement of the battery is: labels invariant under any perturbation that preserves the growth regime; association invariant to within a graceful degradation beyond it.

Proposition 13.17 (Third perturbation axis: nitrogen source). *Perturbing the nitrogen axis of the same glucose-only medium — ammonium-supply limitation $EX_nh4_e \in \{-10, -5, -2.5\}$ on $iJO1366$ (-2.5 on $iML1515$) and, in the form the first two axes did not test, full substitution of the nitrogen source (ammonium closed; the sole donor replaced by L-glutamate at -10 , one nitrogen per molecule so the nitrogen flux is stoichiometrically matched to the NH_4 -10 level and both optima coincide at 0.9259 — the nitrogen-to-growth map is donor-independent — or by L-arginine at -10 , a four-nitrogen carbon co-substrate whose optimum 1.2595 exceeds the glucose-minimal baseline by 28%) — leaves the essentiality labels invariant along the entire limitation gradient: $289/1,367$ at every level on $iJO1366$ and $286/1,516$ on $iML1515$, zero flips, Cohen’s $\kappa = 1.000$, the wild-type optimum falling 76% at -2.5 . Under substitution the re-stratification is a pure loss concentrated in the nitrogen-assimilation module: glutamate costs five labels on $iJO1366$ ($gltA$, $acnA$, $acnB$, icd , $amtB$) and seven on $iML1515$ (those five plus the GOGAT subunits $gltB/gltD$); arginine costs fourteen on $iJO1366$ and fifteen on $iML1515$ (the same sets plus the arginine-biosynthesis genes $argA$, $argB$, $argC$, $argE$, $argF$, $argG$, $argH$, $argI$, with $astC$ on $iJO1366$); $\kappa = 0.989$ and 0.969 ($iJO1366$), 0.985 and 0.967 ($iML1515$); gains exactly zero; arginine and proline metabolism enriched twenty-fold over the genome share; and every rescue substitution-mediated — closing the sole nitrogen donor in the knockout background returns all 41 lost-label backgrounds to exactly zero growth. The mechanism is that the carbon skeleton arrives with its nitrogen: the reductive assimilation flux through $GLUDy$ (8.40 mmol/gDW/h of α -ketoglutarate at the baseline optimum) falls to zero under glutamate supply — the α -ketoglutarate arm is then fed by transaminases ($ASPTA$ 4.99 , $ALATA_L$ 1.77 around an icd knockout) — and reverses under arginine supply (-7.32), while $amtB$ is the ammonium channel itself. The label-based strata are otherwise unchanged (model gaps 30 and 13; the $iJO1366$ medium-mismatch stratum moves $180 \rightarrow 175 \rightarrow 166$, exactly the loss counts). The association splits by regime. Under arginine, where carbon is again the binding resource, it is intact-to-stronger: $r = +0.802$ (CI $[0.780, 0.823]$), held-out ROC AUC 0.981 , MCC 0.905 , direct $+0.297/0.729$; on $iML1515$ $r = +0.915$, AUC 0.997 , direct $+0.425/0.821$. Under every nitrogen-limited optimum the plain-FBA statistics collapse ($r = +0.350, +0.079, -0.122, +0.213$ at NH_4 $-10/-5/-2.5$ and Glu -10 ; AUCs 0.546 – 0.709) — a degeneracy of the vertex, not of the biology. At-optimum flux variability gives phosphoglucose isomerase a range of 45.2 at NH_4 -10 and 177.2 at -2.5 , against 0.0 at the baseline and 4.3 under oxygen limitation; the parsimonious wild type takes up $9.58, 2.49, 4.15$ mmol/gDW/h of glucose where the returned vertices take $10.00, 7.14, 10.00$; the non-essential median κ_V^{flux} inflates from 50 to $1,846$ and its ranking decorrelates from the baseline ranking (Spearman -0.04). Canonical vertex selection — parsimonious FBA for the wild type and every knockout, labels unchanged by construction ($\kappa = 1.000$; the objective is preserved to 10^{-13}) — restores the association everywhere: $r = +0.940$ at NH_4 -10 , $+0.915$ at -2.5 , $+0.872$ at Glu -10 (AUCs $1.000, 0.988, 0.979$; MCC $0.972, 0.965, 0.938$), the canonical ranking correlated $+0.975, +0.956, +0.735$ with the baseline canonical ranking; and it sharpens the baseline itself (r $+0.603 \rightarrow +0.945$, AUC $0.977 \rightarrow 1.000$).*

Remark 13.18 (What the third axis adds: the two-sided boundary). With the nitrogen axis, the invariance claim of Remarks 13.14 and 13.16 extends the bracket to a third axis — carbon, electron acceptor, nitrogen — and its boundary acquires both sides. On the supply side the labels are unconditionally invariant: a 76% reduction of the wild-type optimum moves no gene across the 5% line in either reconstruction, exactly as the bimodal-ratio mechanism of Remark 13.14 predicts. On the substitution side the re-stratification is confined to the module the perturbation

rewires: replacing ammonium by an organic donor shuts down the assimilation arm (the channel *amtB*, the α -ketoglutarate-supply TCA steps, GOGAT) and, when the donor is arginine, the biosynthetic route to the donor itself — losses only, every one substitution-mediated, the exact mirror of the anaerobic gains in the energy-strategy module. The statistic acquires its own boundary condition: κ_V^{flux} under plain FBA is conditioned on the vertex, and nitrogen-limited optima are carbon-degenerate — the surplus glucose may be burned or left on the table, the overflow may take any of several secretion patterns — so the collapse of the plain-FBA association is a property of the solver, not of the biology; under canonical selection the association is the invariant object on all three axes (AUC ≥ 0.979 ; strengthened at the baseline). Where the nitrogen enters is itself legible in the optimum: under glutamate supply the released ammonia comes from D-amino-acid dehydrogenase (*dadA*, 1.95) on iJO1366 and from GLUDy (2.08) on iML1515; under arginine supply the succinylarginine dihydrolase step of the AST pathway (*astB*, 19.3/16.0) dominates, with GLUDy and aspartase (*aspA*) secondary. The medium-robustness statement of the battery is therefore threefold: essentiality labels invariant under any supply perturbation that preserves the growth regime, on all three axes; re-stratification confined to the rewired module at a regime switch; and the association invariant under canonical selection, with plain-FBA association readings reported only where the optimum is unique (carbon- or oxygen-limited).

Proposition 13.19 (Fourth perturbation axis: phosphate limitation). *Limiting the inorganic-phosphate supply — the fourth classical macronutrient supply axis, and the purest test of the supply side: phosphate enters the biomass directly (as the ester phosphate of nucleic acids, phospholipids, and the energy currency), with no assimilation pathway to rewire, so a limitation gradient scales one biomass component while leaving every route topology intact — with the glucose-only medium otherwise unchanged and labels declared at 5% of each level’s own wild type, gives on iJO1366 EX_pi_e $\in \{-0.5, -0.25, -0.1\}$ (wild-type optima 0.518, 0.259, 0.104 — 47%, 74%, 89% reductions; the unlimited uptake is 0.948, fixing the informative range; sulfur, the alternative fourth axis, compresses to a single informative level at its 0.25 uptake and was rejected by the same pre-screen) and on iML1515 $\{-0.25, -0.1\}$ (0.259, 0.104). The essentiality labels are invariant at every level on both reconstructions: 289/1,367 on iJO1366 and 286/1,516 on iML1515, zero flips, Cohen’s $\kappa = 1.000$ throughout, the label-based strata unchanged. The association behaves exactly as the nitrogen axis predicted: every phosphate-limited optimum is carbon-degenerate (at-optimum phosphoglucose-isomerase FVA width 123/172/201 on iJO1366 against 0.0 at the baseline and 4.3 under oxygen limitation; the parsimonious wild type takes up 5.4/2.8/1.2 glucose where the returned plain vertices take 10.0/7.9/3.7), and the plain-FBA statistics collapse with the degeneracy depth ($r = +0.323, +0.054, -0.067$ on iJO1366; $+0.025, +0.090$ on iML1515; AUCs down to 0.506), while canonical vertex selection restores them everywhere: $r = +0.950, +0.916, +0.914$ on iJO1366 and $+0.910, +0.900$ on iML1515 (AUCs 0.988/0.986; MCC 0.965/0.904; canonical ranking correlated $+0.765$ to $+0.968$ with the baseline canonical ranking; labels $\kappa = 1.000$ by construction). Three plain calls and one canonical call at the deepest level (Pi -0.1 , wild type 0.104) are solver-tolerance artifacts of the biotin trace-quota family — bioC, bioF, bioH, and the fabZ side-chain arm, all essential by independent-engine adjudication (§13.6) — and are excluded from the label counts reported above.*

Remark 13.20 (What the fourth axis closes). With the phosphate axis the macronutrient side of the supply claim closes: the four classical macronutrient supplies — carbon source, electron acceptor, nitrogen, phosphate — have now each been limited by up to 89% of the wild-type optimum, and not one supply perturbation moves a gene across the 5% relative line in either reconstruction (289/1,367 and 286/1,516 at every level of every axis; zero flips, $\kappa = 1.000$). Re-stratification appears only where the perturbation changes the growth mode, not the supply level: the anaerobic switch (six glycolysis gains on iML1515, no losses after the tolerance correction) and the organic-nitrogen substitutions (losses confined to the rewired assimilation module). The phosphate axis adds the quota-scaling isolation: because phosphate cannot be rerouted, the axis background-

variation is pure biomass-component scaling — which is precisely what exposed the trace-quota boundary of §13.6 (the biotin and quinone side-chain drains scale with growth and fall toward the solver’s feasibility tolerance at low wild-type optima). The medium-robustness statement of the battery is therefore fourfold: essentiality labels invariant under any supply perturbation that preserves the growth regime, on all five axes (iron: Proposition 13.21); re-stratification confined to the rewired module at a regime switch or source substitution; the association invariant under canonical selection at every level of every axis (and across the anaerobic switch itself); and plain-FBA readings meaningful only where the optimum is unique (carbon- or oxygen-limited).

Proposition 13.21 (Fifth perturbation axis: iron limitation). *Limiting the ferrous-iron supply — the trace-metal class, the last classical nutrient supply after carbon source, electron acceptor, nitrogen, and phosphate; iron pinned to the single ferrous channel (the bound reaction EX_fe2_e, the ferric channel closed, the closure leaving the wild-type optimum unchanged to solver noise) — with the glucose-only medium otherwise unchanged and labels declared at 5% of each level’s own wild type, gives on iJO1366 $\text{EX_fe2_e} \in \{-0.01, -0.005, -0.0025\}$ (wild-type optima 0.623, 0.311, 0.156 — 37%, 68%, 84% reductions; the parsimonious unlimited requirement is 0.0158 mmol/gDW/h, fixing the informative range; zinc and manganese, the alternative trace metals, compress their informative bands to the 10^{-4} bound scale and were rejected by the same pre-screen) and on iML1515 $\{-0.005, -0.0025\}$ (0.311, 0.156 — 62% and 81% reductions). The pre-screen also exposes a structural fact: the iron dose response is identical on the two reconstructions to six decimals — both biomass equations carry the same iron quota, so $b = \text{fe2}/0.0161$: a pure linear biomass-quota scaling, every iron demand scaling with growth, with no assimilation pathway to reroute (the trace-metal counterpart of the quota purity that made phosphate the cleanest macronutrient axis). The essentiality labels are invariant at every level on both reconstructions: 289/1,367 on iJO1366 and 286/1,516 on iML1515, zero flips, Cohen’s $\kappa = 1.000$ throughout, the label-based strata unchanged. The association replicates the phosphate pattern in full: every iron-limited optimum is carbon-degenerate (at-optimum phosphoglucose-isomerase FVA width 103/162/192 on iJO1366 and 139/176 on iML1515 against 0.0 at the baseline; the parsimonious wild type takes up 6.5/3.3/1.7 glucose where the returned plain vertices take 10.0/9.3/5.0), the plain-FBA statistics collapse with the degeneracy depth ($r = +0.518, +0.303, -0.011$ on iJO1366; $+0.127, -0.037$ on iML1515; AUCs down to 0.522), and canonical vertex selection restores them everywhere: $r = +0.957, +0.800, +0.913$ on iJO1366 and $+0.843, +0.905$ on iML1515 (AUCs 0.988/0.986; MCC 0.965/0.904; canonical ranking correlated $+0.63$ to $+0.85$ with the baseline canonical ranking; labels $\kappa = 1.000$ by construction). One plain LP at the deepest level (the cysteine/glutathione ABC-exporter ATPase b0887, non-essential) hangs under GLPK simplex cycling; it was settled by the independent engine under both arm conventions, the row recorded as an engine substitution (§13.6). The deepest levels (wild type 0.156, above the 0.14 tolerance boundary) were integrity-scanned by cross-arm agreement: 0 discrepancies in 7,133 gene-level biomass comparisons across the five levels, no gene in the false-viability band on either arm, zero corrupted calls.*

Remark 13.22 (What the fifth axis closes). With the trace-metal axis the supply side of the invariance claim closes across all five classical nutrient classes — carbon source, electron acceptor, nitrogen, phosphate, iron: no supply perturbation that preserves the growth regime moves one gene across the 5% relative line in either reconstruction (289/1,367 and 286/1,516 at every level of every axis; zero flips, $\kappa = 1.000$), down to the deepest levels probed (89% wild-type reduction on phosphate, 84% on iron). Beyond the count, the iron axis adds two structural facts. First, quota purity: because the two reconstructions share the iron quota constant, the axis collapses both media onto one identical linear dose response — even phosphate, whose unlimited uptake differs (0.948 against 0.793), does not do that — making iron the purest supply perturbation of the battery. Second, the gradient’s floor is set by arithmetic, not biology: the deepest iron level sits at wild type 0.156, deliberately above the 0.14 trace-quota tolerance boundary of §13.6, and the cross-arm scan verifies it clean (0 discrepancies in 7,133 comparisons); pushing to the next pre-screen level (94% reduction, wild type 0.062) would place every trace-quota call inside the

tolerance zone and convert the label invariance claim into an adjudication exercise, so the axis stops where the instrument is still exact.

Proposition 13.23 (Sixth perturbation axis: non-medium maintenance stress). *Genome-scale flux balance has no temperature or pH state variable; the standard constraint-based surrogate for temperature-style stress is escalation of the non-growth-associated ATP maintenance demand (the ATPM lower bound; defaults 3.15 iJO1366 and 6.86 iML1515 mmol ATP/gDW/h). A pre-screen compared this encoding head-to-head with proton-leak forcing (the pH-style surrogate): the maintenance axis spans 29–90% wild-type reduction with four-to-five usable levels in both models, while the proton-leak axis compresses to two usable levels in iJO1366 (the proton leaves partly for free with organic-acid secretion) before hard infeasibility — ATPM is selected, and the axis carries a structural property no supply axis has: zero wild-type carbon-sector degeneracy, phosphoglucose-isomerase at-optimum variability 0.0 at every level in both models against 45–201 at every nitrogen-, phosphate-, and iron-limited level. Levels {40, 60, 80, 100} (iJO1366) and {60, 80, 100} (iML1515) drop the wild-type optima from 0.982 to 0.695, 0.496, 0.298, 0.099 and from 0.822 to 0.398, 0.239, 0.080. The labels re-stratify in one direction only: the nine ATP-synthase genes of the atp operon are gained already at the mildest level (289 → 298 and 286 → 295; $\kappa = 0.981$ on both), and at the –90% endpoints the gain set is 42/43 genes — the atp operon, the cytochrome-bo₃ operon (cyoA–cyoD), the thirteen-gene NDH-I operon, lower glycolysis (pgk, eno), ackA, lpd, and outer-membrane porins (ompG, ompL), oxidative phosphorylation enriched eight-fold over the genome share (26 against 3.0 expected) — the mirror image of the anaerobic glycolysis enrichment of Proposition 13.15; after the independent-engine adjudication below there are no losses at any level ($\kappa = 0.912$ at the endpoints, gains only). The association splits by arm exactly as the degeneracy reading predicts, with one refinement. Plain FBA is strong at moderate stress — $r = +0.949, +0.925, +0.821$ on iJO1366, all above the baseline $+0.603$ — and collapses only at the –90% endpoints: $r = +0.444$ (iJO1366), $+0.244$ with held-out AUC 0.550 (iML1515). Canonical selection restores the endpoint fully on iJO1366 ($r = +0.937$, AUC 0.991, MCC 0.974; moderate levels $+0.969, +0.949, +0.964$) and on iML1515 restores the ranking separation at every level (AUC 0.992, 0.992, 0.978; labels $\kappa = 1.000$) while the transitive correlation on the deposited simplex path reads $r = +0.703, +0.655, +0.475$ — the near-tie boundary of Remark 13.25, where a stateless second engine resolves the near-tie consistently and restores the same statistic to $+0.952, +0.968, +0.943$ (labels unchanged at $\kappa = 1.000$; maximum biomass discrepancy 9×10^{-8}); the symmetric re-run of the iJO1366 levels confirms the restoration itself is engine-invariant — labels $\kappa = 1.000$ at all four levels, canonical r within 0.002 at the three deeper levels and $+0.9685$ to $+0.9496$ at the mildest, whose 961-of-971 floor is the deposited path’s realization — and the deterministic tie-break of the selection subsection closes the near-tie under a declared rule, the iML1515 maintenance levels reading $+0.953, +0.969, +0.944$. Since the wild-type optimum is unique at every level on this axis, the plain endpoint collapse establishes that wild-level flux-variability width is a sufficient but not necessary indicator: the operative mechanism is per-knockout vertex instability at near-zero-biomass optima, which canonical selection repairs wherever the parsimony stage is discriminating.*

Remark 13.24 (What the sixth axis adds). The five medium-side supply axes — carbon source, electron acceptor, nitrogen, phosphate, iron — and the non-medium maintenance-stress axis bracket the claim from both sides. The genuine re-stratifications are both regime switches confined to the module the perturbation rewires: the anaerobic switch gains the substrate-level-phosphorylation module; the maintenance-demand switch gains the energy-transduction module, and does so already at mild stress — a 40 mmol/gDW/h ATP demand makes the ATP synthase itself load-bearing. The nitrogen-substitution losses remain the only re-wiring-driven losses. The association is the invariant object under canonical selection on all five supply axes in both models ($r \geq +0.843$, held-out AUC ≥ 0.979), is restored on the maintenance axis in iJO1366, and acquires — then closes, by the deterministic tie-break — its measured boundary on the maintenance axis in iML1515 (§13.6).

13.6 Canonical Vertex Selection and Solver-Tolerance Integrity

The plain-FBA association readings collapse on every level of the nitrogen, phosphate, and iron axes whose optimum is carbon-degenerate. The statistics of Table 5 are computed under one homogeneous construction across all six axes — canonical vertex selection at every level, the oxygen re-run and the arginine-substitution levels included — and the sixth, non-medium maintenance-stress axis (Proposition 13.23) extends the same construction — with the refinement that its wild-type optima are carbon-unique (phosphoglucose-isomerase width 0.0 at every level) and the plain arm still collapses at the extreme endpoints. The finding is a property of the statistic, not the biology: κ_V^{flux} aggregates absolute squared flux changes, so wherever the optimum is non-unique in the carbon sector (how much surplus glucose is taken up and how the overflow is split are optimum-equivalent), the LP solver’s arbitrary vertex choice injects noise into every knock-out’s comparison. The at-optimum FVA signature separates the regimes cleanly: phosphoglucose-isomerase range 0.0 at the carbon-limited baseline and 4.3 under oxygen limitation, against 45–201 at every nitrogen-, phosphate-, and iron-limited level. Canonical vertex selection — parsimonious FBA for the wild type and every knockout, the objective preserved exactly (verified per level) and labels therefore unchanged by construction — is the declared selection rule under which the statistic is well posed.

Table 5 reports the canonical-selection statistics across the six axes. Five features matter. First, canonical selection sharpens even the unique-optimum levels: the baselines rise from $r = +0.603$ to $+0.945$ (iJO1366) and $+0.875$ to $+0.937$ (iML1515), and the deepest oxygen limitation from $+0.519$ to $+0.945$ — the plain baseline vertex itself sits on a free acetate-kinase cycle (at-optimum ACKr range $\approx 10^3$) whose arbitrary resolution the canonical rule removes. Second, the association is restored at every degenerate level of the nitrogen, phosphate, and iron axes ($r \in [+0.80, +0.96]$, AUC ≥ 0.979 , MCC ≥ 0.904); the iron axis replicates the full collapse-and-restore pattern at the deepest degeneracy in the battery (plain r down to -0.011 (iJO1366) and -0.037 (iML1515) against phosphoglucose-isomerase ranges of 192/176, canonical $+0.913/+0.905$). Third — and this strengthens the regime-switch reading of Remark 13.16 — the iML1515 anaerobic endpoint is restored too: under canonical selection the association survives the regime switch intact ($r = +0.949$, AUC 0.997, MCC 0.979, against the plain $+0.258/0.682$), so the switch moves labels only (six glycolysis gains); the apparent collapse of the association at the switch was vertex-conditioning, not biology. Fourth, the substitution side is symmetric with the limitation side: the arginine levels, computed under canonical selection, confirm that where the substituted optimum is carbon-re-pinned and unique, the plain reading is already well posed (iML1515: plain $+0.915$, canonical $+0.850$, AUC 0.997, MCC 0.993; iJO1366 sharpens $+0.802$ to $+0.953$), so the canonical rule is applied uniformly at every level. Fifth, the non-medium axis refines the degeneracy reading itself: its wild-type optima are unique (widths 0.0), plain FBA is strong at moderate stress (above the baseline), and the collapse appears only at the near-zero-biomass endpoints — so wild-level variability width is a sufficient but not necessary indicator, the operative mechanism being per-knockout vertex instability — and the axis adds the one corner where the canonical vertex itself is only weakly determined: at elevated maintenance demand the iML1515 parsimony stage itself carries *near-tied* optima (vertices ≈ 200 apart in the squared-flux metric whose L_1 objectives differ by 8.6×10^{-4} absolute, 1.2×10^{-6} relative — within the simplex optimality tolerance), so the returned canonical vertex is solver-path dependent: on the deposited simplex path 782 of 1,127 compensable knockouts record the near-tie floor $kV \approx 200.01$ while the rest sit at $kV \approx 0$ (supply axes carry no floor; compensable medians 0.00–0.10), and the floor decorrelates that path’s transitive statistic on iML1515 (r down to $+0.475$) without touching the essentiality ranking (AUC 0.978, labels $\kappa = 1.000$ at every level). A full second-engine re-run of the three iML1515 levels under the stateless cold-start rule (scipy/HiGHS, same two-stage selection) measures the path dependence directly: the near-tie itself is engine-invariant (wild-type L_1 reproduced to 1×10^{-5} ; labels $\kappa = 1.000$ at all three levels, maximum biomass discrepancy 9×10^{-8} , AUC 0.992/0.992/0.984), but the floor is the deposited path’s realization — the stateless

engine collapses the floor band to the one genuinely forced knockout (*lamB*, $kV = 200.0$ at every level) with a single further five-gene rerouting block (*dhaKLM*, *fsaA/fsaB*, kV 162–416 across the levels), and restores the transitive association to +0.952, +0.968, +0.943.

The deterministic tie-break. The near-tie boundary is not only measurable but closable, and the closure is the selection rule’s own completion: a third lexicographic stage minimizing a fixed functional $w^T v$ over the wild-type-pinned, parsimony-pinned face (weights $w \sim U(0.5, 1.5)$ under a fixed seed, the optimum generically unique) selects a unique vertex by construction, so the returned canonical vertex becomes a property of the declared protocol rather than of the solver’s path. A pre-registered pilot at the two worst levels verified exactly this. On a stratified sample of 105 genes plus the wild types (the random 40-gene floor draw at each level, the near-tie gene *ltaE*, *lamB*, the stateless top-rerouting blocks, and compensable controls), the GLPK warm-start path and the stateless cold-start engine return the *same* tie-broken vertex — maximum cross-engine squared distance 6×10^{-11} , against two-stage vertices some $kV \approx 200$ apart at the same genes — and every one of the 80 randomly drawn floor genes collapses to $kV \leq 0.1$ under the declared rule in both engines, with labels unchanged (maximum biomass discrepancy 6×10^{-8}). The pilot also exposes the limit of engine substitution alone: the stateless two-stage reading had kept *lamB* at $kV = 200$ on iJO1366 — its one-gene floor — while the declared rule returns $kV = 0$ there; even a stateless engine’s vertex is a path, and only a declared rule makes the statistic well posed. Full sweeps under the declared rule at the four floor-affected levels close the boundary in the instrument’s own terms: the transitive association reads $r = +0.953, +0.969, +0.944$ on the iML1515 maintenance levels and +0.951 at the iJO1366 floor level (labels $\kappa = 1.000$ against the deposit at all four; held-out AUC 0.992, 0.992, 0.984, 1.000), and the floor census collapses to the rule-determined rerouting sets: *lamB* at $kV = 200.0$ at every iML1515 level with the *dhaKLM/fsaA/fsaB* block deepening from 194 to 427 as maintenance demand grows, and the iJO1366 floor level carrying instead a parallel-routing block (*pfkB*, *fbaB*, *fsaA/fsaB*, *dhaKLM*, and the putative receptor *ydjI*) at kV 69–90. Canonical selection is necessary everywhere, and with the deterministic tie-break it is sufficient at every level measured, including the near-tied maintenance levels.

Gene-by-gene comparison of every plain sweep level against its canonical counterpart (24,282 biomass-and-label comparisons across seventeen levels) delimits a solver-integrity boundary: exactly eight corrupted calls, every one a false-viability call of a trace-quota gene at a low-growth level: the *fabZ* ubiquinone-side-chain row at the iML1515 anaerobic endpoint, and at the Pi -0.1 levels (wild type ≈ 0.104) the plain *bioC/fabZ/bioH* rows and the canonical *bioF/bioH* rows. The mechanism is arithmetic, not stochastic: the biomass drains of biotin (stoichiometric coefficient 2×10^{-6}) and of the ubiquinone side chain scale with the growth rate, so at wild-type optima below ≈ 0.14 they fall to $2\text{--}3 \times 10^{-7}$ mmol/gDW/h — within a factor of two of the simplex’s primal feasibility tolerance (10^{-7}) — and the solver can then accept a “solution” that violates the hard zero bounds of the knocked-out steps. The failure is one-directional (false viability only) and confined to genes whose sole essential role is such a trace quota, which is why every level with wild-type optimum above ≈ 0.23 (quota $\geq 4.6 \times 10^{-7}$) is clean. All eight calls were re-adjudicated with an independent LP engine (HiGHS, two-stage split-variable parsimonious solves: every biotin-pathway knockout has biomass maximum exactly 0) and corrected; after correction the label sets are the invariant ones reported in Propositions 13.13–13.21, the cross-check is 0 discrepancies in 24,282 comparisons, and the corrected calls reproduce across engines to the reported precision (the three iJO1366 biotin knockouts share the identical canonical curvature 164.975; the three iML1515 ones 253.551). The deepest iron-restriction levels sit at wild type 0.156 — between the 0.14 corruption boundary and the 0.23 sufficiency margin, the one band not otherwise sampled — and were verified directly: cross-arm agreement over all five iron levels gives 0 discrepancies in 7,133 gene-level biomass comparisons (maximum $|\Delta b| = 6 \times 10^{-8}$), no gene lands in the false-viability band on either arm, and the trace-quota family reads exactly zero on iJO1366. A further solver pathology arises at a degenerate optimum: one plain LP (the

cysteine/glutathione ABC-exporter ATPase b0887, non-essential) hangs under GLPK simplex cycling and was settled under the independent HiGHS rule in both arms, the row recorded as an engine substitution in the deposited sweeps.

Remark 13.25 (What the selection rule requires). The methodological content is twofold. A flux-distance statistic computed at non-unique optima is conditioned on the vertex, so it requires a declared selection rule — ours is parsimonious FBA for the wild type and every knockout, with the objective preserved and labels unchanged by construction; plain readings remain meaningful only where the at-optimum FVA width certifies uniqueness. And essentiality calls for genes whose blocking quota sits within an order of magnitude of the solver’s feasibility tolerance require independent adjudication at low growth rates — the trace-quota family (biotin, quinone side chain, lipoate) is exactly the at-risk set, and a second engine settling the LP is sufficient — as the iron levels at wild type 0.156 verify directly (0 discrepancies in 7,133 cross-arm comparisons). The same second engine is the fallback for solver pathology at degenerate optima: a plain LP that cycles indefinitely under one simplex implementation is settled by the other, with the row marked as an engine substitution in the deposited sweep. The sixth axis meets both requirements and one new boundary. Its sweeps run under a simplex time limit (a handful of at-optimum L_1 LPs cycle indefinitely at the deepest maintenance level; timed-out solves fall back to the plain optimum vertex, label-preserving and flagged per gene), and its endpoints sit at wild type 0.099 and 0.080 — inside the false-viability band — so the cross-arm scan was run there: six corrupted calls, all trace-quota genes (the plain *fabZ/bioH* rows on iJO1366, the plain *bioD/fabZ* and the canonical *bioF/fabI* rows on iML1515), every one settled at biomass exactly zero by the independent engine and corrected under the same conventions; after correction the maintenance-stress re-stratification is purely one-directional (energy-transduction gains only, no losses) and the plain-versus-canonical arm agreement is $\kappa = 1.000$ at all seven levels of the axis. The new boundary is the near-tie floor above: unlike the trace-quota calls, it is not a corrupted reading but a property of the parsimony stage itself, and it bounds the rule rather than violating it. Both requirements are properties of the instrument, and the battery’s reported statistics satisfy both. A third requirement is measured, not imposed: where the parsimony stage itself carries near-tied optima — the maintenance axis in iML1515, ΔL_1 at 1.2×10^{-6} relative — the canonical vertex is only weakly determined and the transitive statistic is path-dependent, the measured engine bracket at the deepest level being $[+0.475, +0.943]$ (the deposited simplex path against the stateless second engine of the re-run above); the essentiality ranking is engine-invariant (labels $\kappa = 1.000$, held-out AUC ≥ 0.978 under both engines), which is why the table’s iML1515 maintenance rows are reported with their AUC and MCC intact and their r read as the deposited path’s realization. The requirement is also closable by construction: the deterministic tie-break — a third lexicographic stage minimizing the fixed functional $w^T v$ over the parsimony-pinned face, with fixed-seed weights whose optimum is generically unique — returns the same vertex under both engines (cross-engine squared distance at most 6×10^{-11} on the stratified pilot; labels unchanged), collapses the floor to the rule-determined rerouting set, and reads the maintenance levels at +0.953, +0.969, +0.944 (iML1515) and +0.951 (iJO1366, floor level) under the single declared rule; the engine bracket above records the two-stage spread closed by the rule.

13.7 Benchmark against the structural closure instruments

The closure criteria of Definition 3.20 are the dynamical counterpart of chemical organization theory [24], and the designed progression sits in the network-expansion territory of Handorf and Ebenhöf [29]. The dynamical closure test is benchmarked against both structural instruments on the same iJO1366 model.

Proposition 13.26 (Structural-instrument benchmark (E14)). *With the network-expansion (NE) scope computed from the glucose-minimal-medium seed (18 extracellular metabolites; iterative expansion converges in 3 rounds to a scope of 45 metabolites, expansion factor $2.50\times$) and the largest*

*chemical organization (COT) computed on the central-carbon subnetwork (28 cytosolic metabolites, 14 reactions; the largest closed set is the full 28 and is self-maintaining by LP feasibility), the 50-metabolite dynamical closure-test record (28 causally internal, 22 homeostatic) compares as follows. Of the 28 dynamically internal metabolites, 0 are in the NE scope and 19 are outside the COT organization (the discriminative cases: versus NE, all 28, e.g. *g6p_c*, *pep_c*, *pyr_c*, *accoa_c*, *cit_c*; versus COT, 19, e.g. *25drapp_c*, *2ahbut_c*, *h2mb4p_c*, *skm_c*, *xmp_c*). Agreement with the dynamical verdict: 0.440 (NE), 0.600 (COT). The dynamical closure test identifies as causally internal metabolites that both structural instruments miss — the discriminative cases in which the dynamical test separates systems the structural tests cannot.*

Proof. Direct computation (deposited artifacts): the NE scope and the COT largest organization are computed on the deposited model files; the dynamical verdicts are the deposited closure-test records; the cross-tabulations, agreement statistics, and discriminative-case lists are the deposited benchmark outputs. $\square$

Remark 13.27 (What the three arms establish together). The designed progression establishes sensitivity: each flagged failure is converted by exactly the modification that repairs the named cascade, so the test localizes the missing closure rather than merely scoring networks. The fixed-model arm establishes internal cross-validity: on an unmodified model, the dynamical dependency structure predicts independently computed essentiality at $\kappa = 0.835$ (with the threshold provenance of Remark 13.8). The external anchor carries the outside information: transitive validation against the Keio collection at held-out ROC AUC 0.953, a significant direct rank-level association against the raw screen, a non-transfer verdict across rebuilds reported as a negative result, and the structural-instrument benchmark. No arm alone would carry the conclusion; together they anchor the closure-test language of Sections 3 and 12 in executed computations.

14 Conclusion

The framework assembled here gives a constructive and falsifiable account of path-dependent viability risk. Four statements carry the weight of the claim:

1. *Constraint switches carry exact geometric data.* The second derivatives of active-set systems are not intractable singularities but discrete, matrix-valued Radon measures concentrated on the switching boundaries; the gluing theorem assembles the smooth strata along those boundaries into one stratified connection, and the application paper [1] realizes the measure measure-theoretically.
2. *Path-dependent risk has a scaling law.* Loops that cross a constraint-switching wall accumulate holonomy linearly in loop diameter ($O(\varepsilon)$; machine-verified across five loop geometries and three structure groups), where smooth systems accumulate quadratically — the quantitative form of sequential-adaptation memory. The small-loop theorem accordingly bounds the margin cost of such loops by the viability-weighted curvature κ_V .
3. *Heterogeneous domains compose.* Autocatalysis, self-prediction, contraction, symmetries, compression, learning, and transport type as optics; their composed update is a Banach contraction with a unique operational fixed point (product bound $0.697 < 1$; verified across the 375-configuration grid).
4. *The biological reading is explicit.* The application paper’s genome-scale validation — transcriptional association with rerouting ($r = +0.395$) against a protein-layer null ($r = -0.083$) — supplies the empirical reading: cells keep rerouting capacity ready at the transcript level while buffering its translation.

The open problems that remain — the order- ε^2 boundary expansion, endofunctor semantics, optic-level colimits, the ∞ -optic composition, the sharpened contraction bound, the ratio-form discretization bridge, and external benchmarks for the closure test — are collected in Section 15.

15 Future directions and open problems

The following problems are open.

1. *Closed-loop boundary term at order ε^2 .* Theorem 3.6 controls the residual boundary contribution of a small closed loop at its true order ($O(\varepsilon)$, the tangential variation of the transition function), and the abelian case exhibits an exact leading-order cancellation. A complete expansion of the *boundary-interaction* contribution at $O(\varepsilon^2)$ (commutators of the resets with the adjacent stratum transports, and the second-order variation of the transition function) is not derived here; for non-abelian G_C the coefficient of the $O(\varepsilon)$ term under general matching conditions is likewise not fully tracked.
2. *Endofunctor semantics.* The full functorial construction of the composed update on a category of periodic typed systems $\mathbf{Sys}_7$ (object map, morphism map, identity and composition preservation) remains open (Remark 6.8); the present fixed-point claims use only the realized endomap.
3. *Filtered colimits beyond $\mathbf{Set}$, and the optic-level colimit.* The componentwise statements of Theorem 10.3 use that $\mathbf{Set}$ is locally finitely presentable and complete. Two extensions are open: (a) the general lfp monoidal case of the adapter statement; and (b) the componentwise colimit for diagrams with nontrivial feedback residuals — the backward carriers are contravariant along the diagram’s arrows, so a componentwise cocone requires declared backward structure, which a general diagram does not supply (Remark 10.4); the residual-level colimit over the comma category $(i \downarrow \mathbb{I})$ likewise requires declared residual structure maps and is not constructed here.
4. *An explicit ∞ -optic composition.* The symmetric monoidal structure of $\mathbf{Optic}(\mathcal{C}_\infty)$ is cited in the proof of Theorem 12.2, not constructed. An explicit construction of ∞ -categorical optics with its coherence data, or a proof that no such construction exists under the stated hypotheses, would close the most significant gap in Section 12 (Remark 12.4).
5. *Sharpening the contraction bound.* The analytic bound $\prod_i \text{Lip}(f_i) \leq 0.697$ is sufficient but not tight: the measured contraction factors of the Krasnoselskii–Mann-averaged iteration across the 375-configuration grid are systematically smaller (the averaging projection contracts each iterate toward $\mathcal{K}$, and the instantiation’s iterates are better behaved than the worst case the bound tracks). Recovering the empirical rates analytically is open.
6. *The ratio-form domination (discretization bridge).* The envelope curvature κ_V^{alg} of Theorem 9.6(e2) is a raw directional derivative of the envelope; Definition 2.11 defines the ratio form $\kappa_\alpha = [-Dh_\alpha(F(u, v))]^+ / h_\alpha$. Establishing (or refuting) a domination or identity between the two — the precise discretization correspondence behind the active-atom bridge stated in Definition 2.11 and instantiated measure-theoretically in the application paper [1] — is open.
7. *Operational instantiations of κ_V .* Definition 2.11 is geometric; the application paper [1] instantiates the idea measure-theoretically on parametric FBA. Other faithful operationalizations — ODE-level viability models, control benchmarks, experimentally measured channels — would test the framework’s reach; each should carry the selection-rule robustness check that the application paper introduces.
8. *Benchmarks for the closure test.* The three-phase autopoiesis closure test (Definition 12.5) has been exercised on a series of designed catalytic maintenance networks deposited in the project repository. Calibration on benchmarks *not* designed by the framework’s authors — and a theoretical account of when Phase III’s contractibility criterion is strictly weaker than Phase I in a way that matters — is open.

A Proof of the Terminal-Coalgebra Theorem

Proof of Theorem 11.2. (i) Deflationarity is immediate: $D(S) = S \cap \Phi(S) \subseteq S$. For monotonicity, let $S \subseteq T$. The closure is monotone in the reaction set: $\text{cl}_S(F) \subseteq \text{cl}_T(F)$ (a reaction applicable under S is applicable under T , so the iteration (3) from F can only add molecules); both conditions in (54) are therefore satisfied by at least as many reactions for T as for S , so $\Phi(S) \subseteq \Phi(T)$ and $D(S) = S \cap \Phi(S) \subseteq T \cap \Phi(T) = D(T)$.

(ii) For $S \subseteq U$, $D(S) = S$ iff $S \subseteq \Phi(S)$ iff every $r \in S$ has a catalyst in $\text{cl}_S(F)$ and all reactants of r in $\text{cl}_S(F)$. The first clause is reflexive autocatalysis (each reaction is catalyzed by a molecule reachable from the food through S itself); the second is food-generatedness (each reaction’s inputs are reachable from the food set through the reactions of S). These are exactly the RAF conditions of Definition 2.7 and of 33, 53, restricted to subsets of U .

(iii) Existence of the greatest fixed point: the sequence $U \supseteq D(U) \supseteq D^2(U) \supseteq \dots$ is decreasing by deflationarity, and each strict inclusion removes at least one reaction, so by finiteness of U the sequence stabilizes at some $D^m(U)$ with $m \leq |U|$; the stabilized value is a fixed point, $D(D^m(U)) = D^{m+1}(U) = D^m(U)$. It is the greatest fixed point: if $S = D(S)$ then by monotonicity and induction $S = D^n(S) \subseteq D^n(U)$ for every n , in particular $S \subseteq D^m(U)$. Identification with $R_{\max}$: the union of any family of RAFs is again a RAF — the catalyst pool and the reactant closure only grow under unions — so $R_{\max} = \bigcup \{S : S \text{ a RAF}, S \subseteq U\}$ is itself a fixed point by (ii), hence a subset of $D^m(U)$; conversely $D^m(U)$ is a RAF, hence a subset of $R_{\max}$. Therefore $D^m(U) = R_{\max}$, and since the sequence is stable from m on, $\bigcap_{n < \omega} D^n(U) = D^m(U) = R_{\max}$. Terminal coalgebra: regard 2^U as a posetal category (objects the subsets of U , a morphism $S \rightarrow T$ the inclusion $S \subseteq T$). A monotone map D is an endofunctor of 2^U ; a D -coalgebra is an object S with a morphism $S \rightarrow D(S)$, i.e. a post-fixed point $S \subseteq D(S)$; a morphism of coalgebras $S \rightarrow S'$ is an inclusion compatible with the structure, i.e. $S \subseteq S'$. The terminal coalgebra is therefore the greatest post-fixed point of D . By Tarski’s fixed-point theorem for monotone endomaps of complete lattices, the greatest post-fixed point equals the greatest fixed point, so the terminal D -coalgebra is $\nu D = R_{\max}$. This is the posetal instance of the general terminal-coalgebra theory [2, 57].

(iv) The Hordijk–Steel iterative-removal algorithm initializes with the full reaction set and repeatedly deletes every reaction that is neither catalyzed by an element of the current food-closure nor has all its reactants in that closure, until no further deletion is possible. The retained set after one deletion round is exactly $S \cap \Phi(S) = D(S)$: the reactions of S whose catalysts and reactants both lie in $\text{cl}_S(F)$. The algorithm’s iterates are therefore $D^n(U)$, step for step, and its output is the stabilized value $D^m(U) = R_{\max}$.

(v) One application of D scans the reaction set once; for each reaction $r \in U$ it tests membership of r ’s catalysts in the pool (constant time per catalyst against a precomputed catalysis table) and queries the reactant closure of $F \cup \text{prod}(S)$ (bounded by the metabolite count, and incrementally maintainable between rounds since $\text{prod}(S)$ only shrinks). With at most $|U|$ strict rounds the total work is polynomial in the network size, the same order as the published iterative-removal algorithm, which it coincides with by (iv). $\square$

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Data and code availability. All machine-verification scripts, numerical artifacts, and their SHA-256 manifests are available in the public repository [github.com/ MIKEAA2020/metabolic-curvature-measure](https://github.com/MIKEAA2020/metabolic-curvature-measure); the compile-ready submission package of this paper (LaTeX source, figures, and the compiled PDF) is archived on Zenodo at DOI [10.5281/zenodo. 22940820](https://doi.org/10.5281/zenodo.22940820) (CC-BY 4.0);

the application paper [1] and its repository carry the biological data sources used there. No new experiments were performed.

Declaration on generative AI. GLM (Z.ai) and DeepSeek AI assisted with development and documentation of the machine-verification code. All AI-assisted content was reviewed, verified, and edited by the author, who takes responsibility for the final content.

Author contributions. A.A. conceived the study design, developed the methodology and analysis code, performed the data analysis, drafted and revised the manuscript.

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condition	plain r	canonical r	canonical AUC	canonical MCC
<i>iJO1366</i>				
baseline (glucose-only)	+0.603	+0.945	1.000	0.972
oxygen -10	+0.775	+0.895	1.000	0.972
oxygen -5	+0.607	+0.939	1.000	1.000
oxygen -2.5	+0.519	+0.945	1.000	0.958
ammonium -10	+0.350	+0.940	1.000	0.972
ammonium -2.5	+0.079	+0.915	0.988	0.965
glutamate -10	-0.122	+0.872	0.979	0.938
phosphate -0.5	+0.323	+0.950	0.988	0.965
phosphate -0.25	+0.054	+0.916	0.988	0.965
phosphate -0.1	-0.067	+0.914	0.988	0.952
iron -0.01	+0.518	+0.957	0.988	0.965
iron -0.005	+0.303	+0.800	0.988	0.965
iron -0.0025	-0.011	+0.913	0.988	0.952
arginine -10	+0.802	+0.953	0.991	0.978
ATPM ≥ 40	+0.949	+0.969	1.000	0.972
ATPM ≥ 60	+0.925	+0.949	0.988	0.959
ATPM ≥ 80	+0.821	+0.964	0.982	0.934
ATPM ≥ 100	+0.444	+0.937	0.991	0.974
<i>iML1515</i>				
baseline (glucose-only)	+0.875	+0.937	0.987	0.966
oxygen -5	+0.559	+0.941	1.000	1.000
oxygen 0 (anaerobic)	+0.258	+0.949	0.997	0.979
ammonium -2.5	-0.112	+0.909	0.986	0.904
glutamate -10	+0.314	+0.911	0.987	0.965
phosphate -0.25	+0.025	+0.910	0.986	0.904
phosphate -0.1	+0.090	+0.900	0.986	0.904
iron -0.005	+0.127	+0.843	0.986	0.904
iron -0.0025	-0.037	+0.905	0.986	0.904
arginine -10	+0.915	+0.850	0.997	0.993
ATPM ≥ 60	+0.747	+0.703	0.992	0.980
ATPM ≥ 80	+0.435	+0.655	0.992	0.960
ATPM ≥ 100	+0.244	+0.475	0.978	0.868

Table 5 Canonical-selection statistics across the six perturbation axes, medium-side and non-medium (transitive calibration; r is the Pearson coefficient of $\log \kappa_V^{\text{flux}}$ against Δb , AUC/MCC the held-out 70/30 logistic). Labels agree with the plain sweeps at $\kappa = 1.000$ at every level (after the solver-tolerance corrections below); the iML1515 maintenance rows’ canonical r is the deposited simplex path’s realization, the stateless second engine reading +0.952, +0.968, +0.943 and the declared deterministic tie-break +0.953, +0.969, +0.944 (Remark 13.25). Under canonical selection the association is the invariant object at every level of every axis — including across the anaerobic regime switch, where the plain reading collapses (+0.258 $\rightarrow$ +0.949) and only the labels re-stratify.